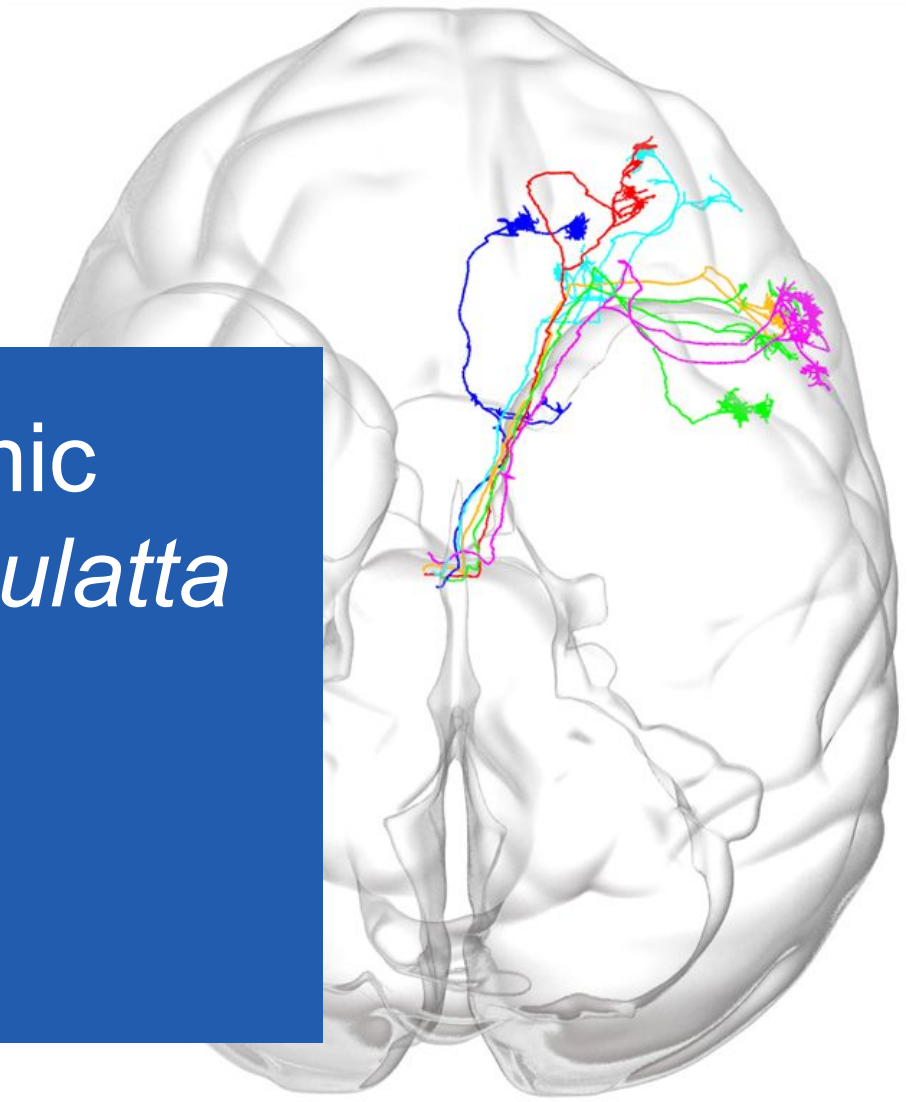
ETH zürich



Developmental Transcriptomic Signatures in the *Macaca mulatta* Frontal Cortex

Déa Karam, Marian Farrera, Rémi Tregan

Project in Systems Genomics

17. December 2025, Basel

Image Credit: Xu, Fang, et al. "High-throughput mapping of a whole rhesus monkey brain at micrometer resolution." *Nature biotechnology* 39.12 (2021): 1521-1528.

The Paper

RESEARCH ARTICLE SUMMARY

PSYCHIATRIC GENOMICS

Spatiotemporal transcriptomic divergence across human and macaque brain development

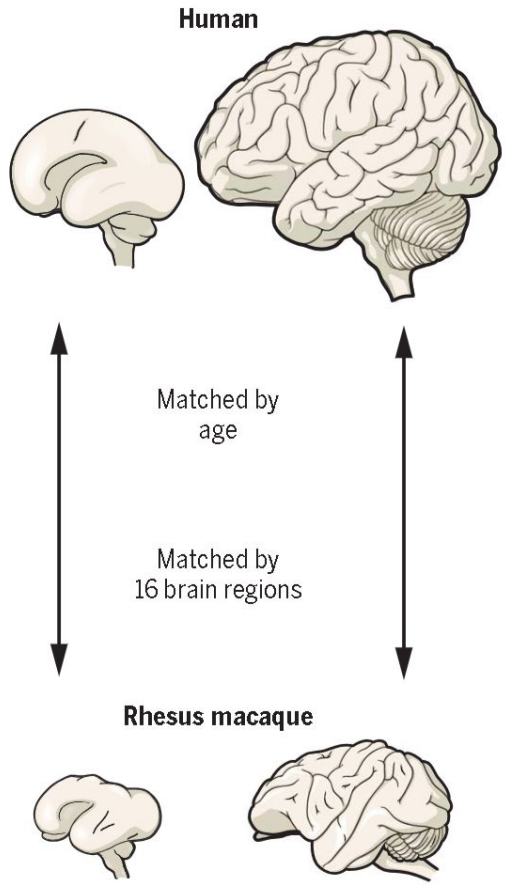
Ying Zhu*, André M. M. Sousa*, Tianliuyun Gao*, Mario Skarica*, Mingfeng Li*, Gabriel Santpere, Paula Esteller-Cucala, David Juan, Luis Ferrández-Peral, Forrest O. Gulden, Mo Yang, Daniel J. Miller, Tomas Marques-Bonet, Yuka Imamura Kawasawa, Hongyu Zhao, Nenad Sestan†

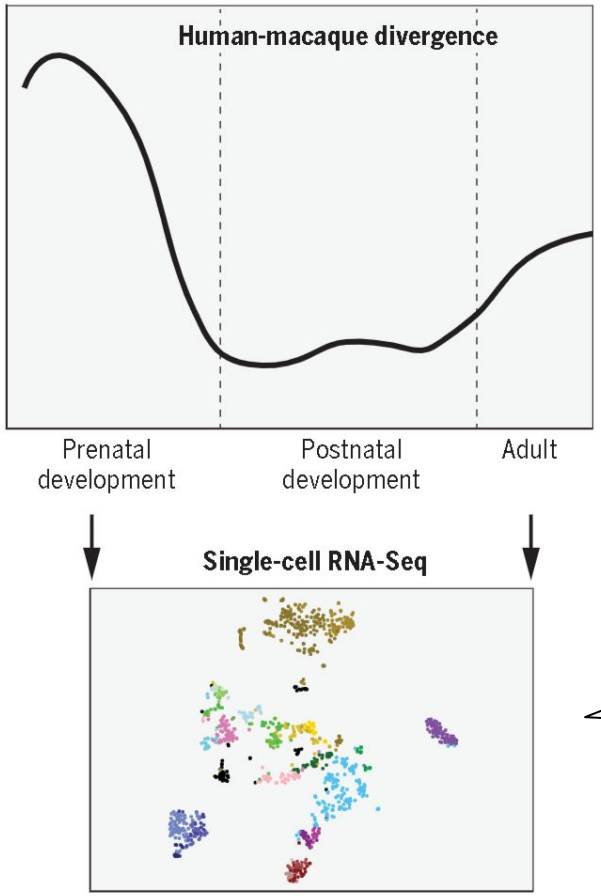
The Paper: Overview of Methods

Human brain transcriptome dataset from (Kang et al. 2011)

Matched by age and by tissue to human dataset

Generated complementary macaque brain RNA-seq dataset



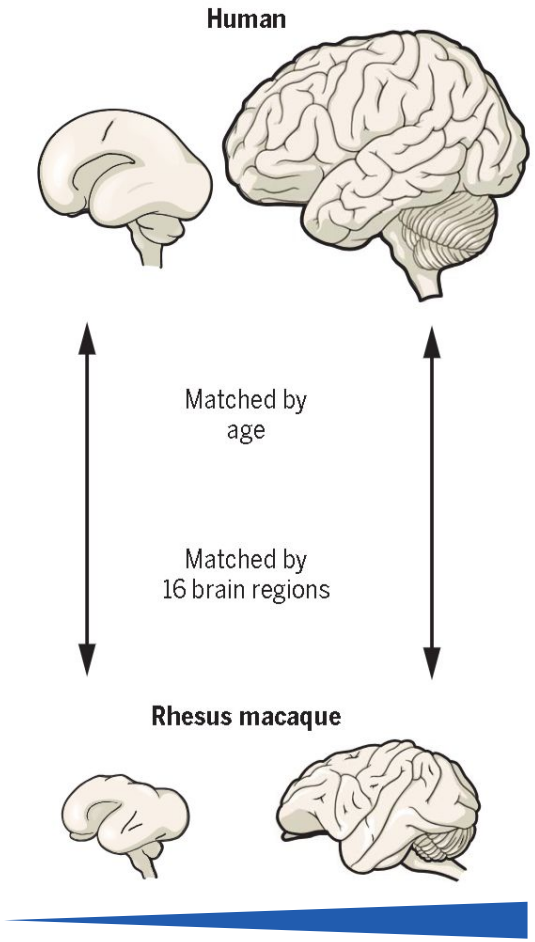


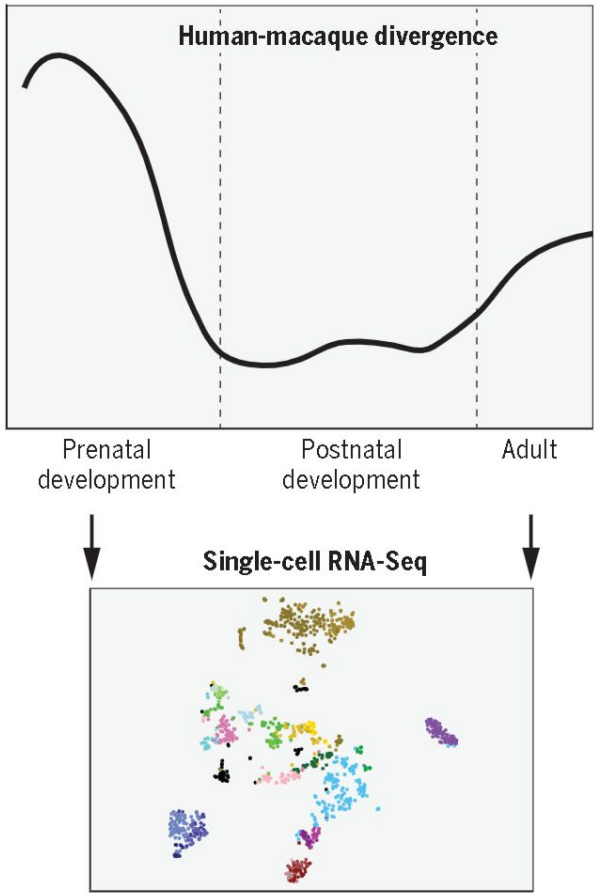
Assessed differences between human and macaque gene expression across development

scRNA-seq revealed divergent transcriptome profiles for distinct cell types (between Hs and Mm)

Zhu, Ying, et al. "Spatiotemporal transcriptomic divergence across human and macaque brain development." *Science* 362.6420 (2018): eaat8077.

The Paper: Our Scope





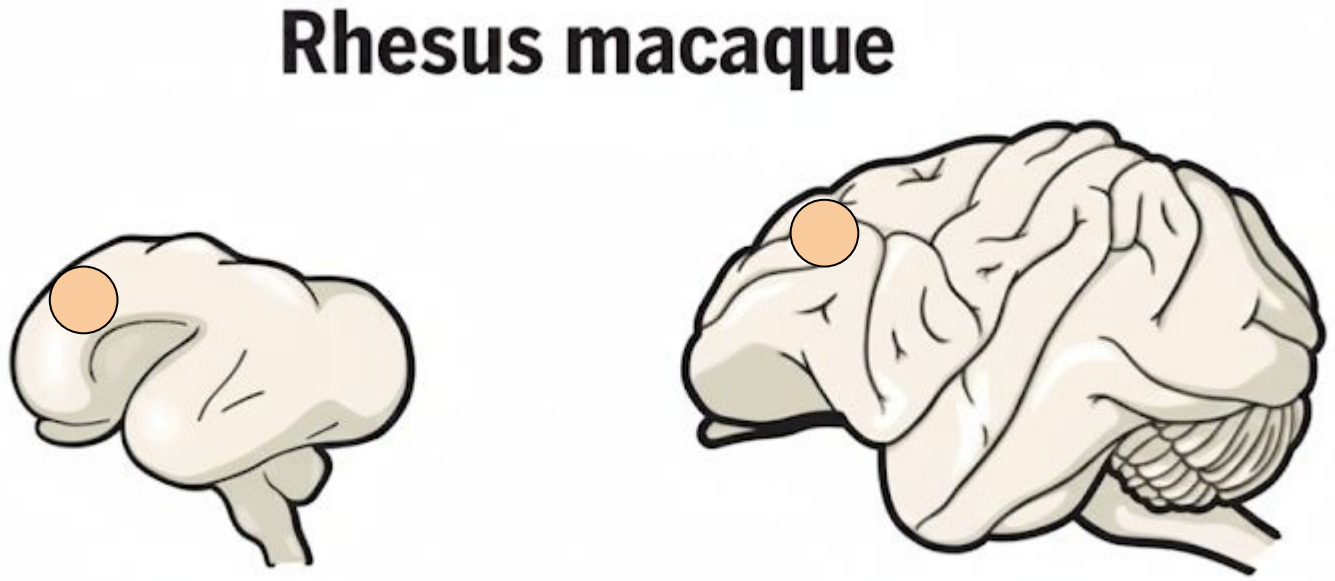
Assess time effects

Zhu, Ying, et al. "Spatiotemporal transcriptomic divergence across human and macaque brain development." *Science* 362.6420 (2018): eaat8077.

Our Samples: Tissue

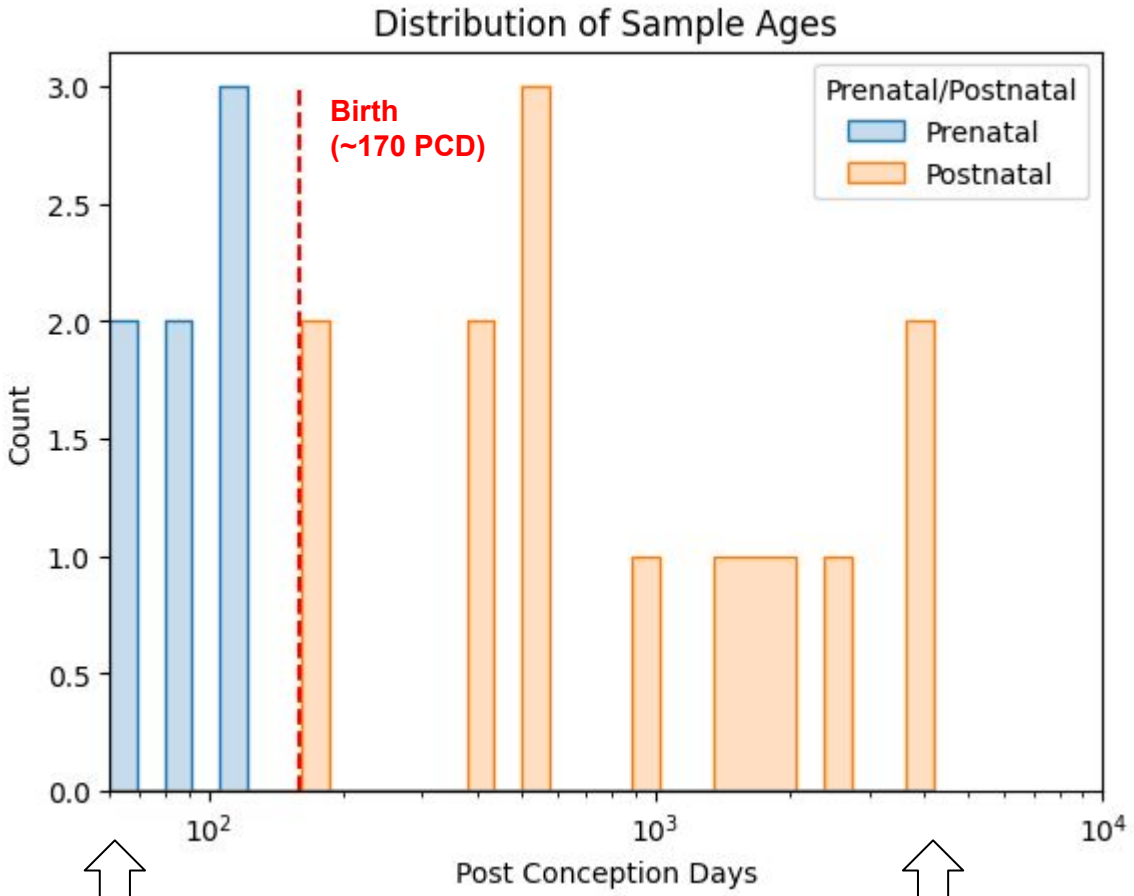
All our samples originate from same tissue: dorsolateral prefrontal neocortex (DFC)

Sequencing mode: Bulk tissue RNA-seq (Illumina)



Adapted from: Zhu, Ying, et al. "Spatiotemporal transcriptomic divergence across human and macaque brain development." *Science* 362.6420 (2018): eaat8077.

Our Samples: Ages



Youngest:
60 PCD

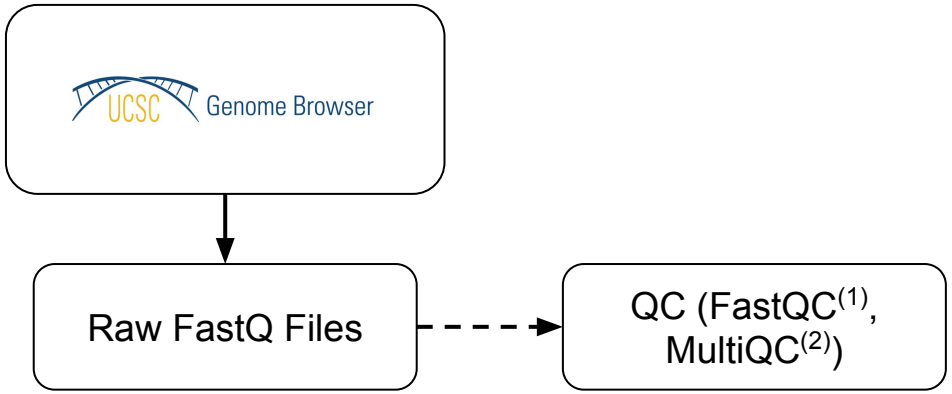
Oldest:
11 Years

For reference, a typical lifespan in the wild is ~15 years(1)

⇒ Samples are representative of macaque development from fetal to adult stages

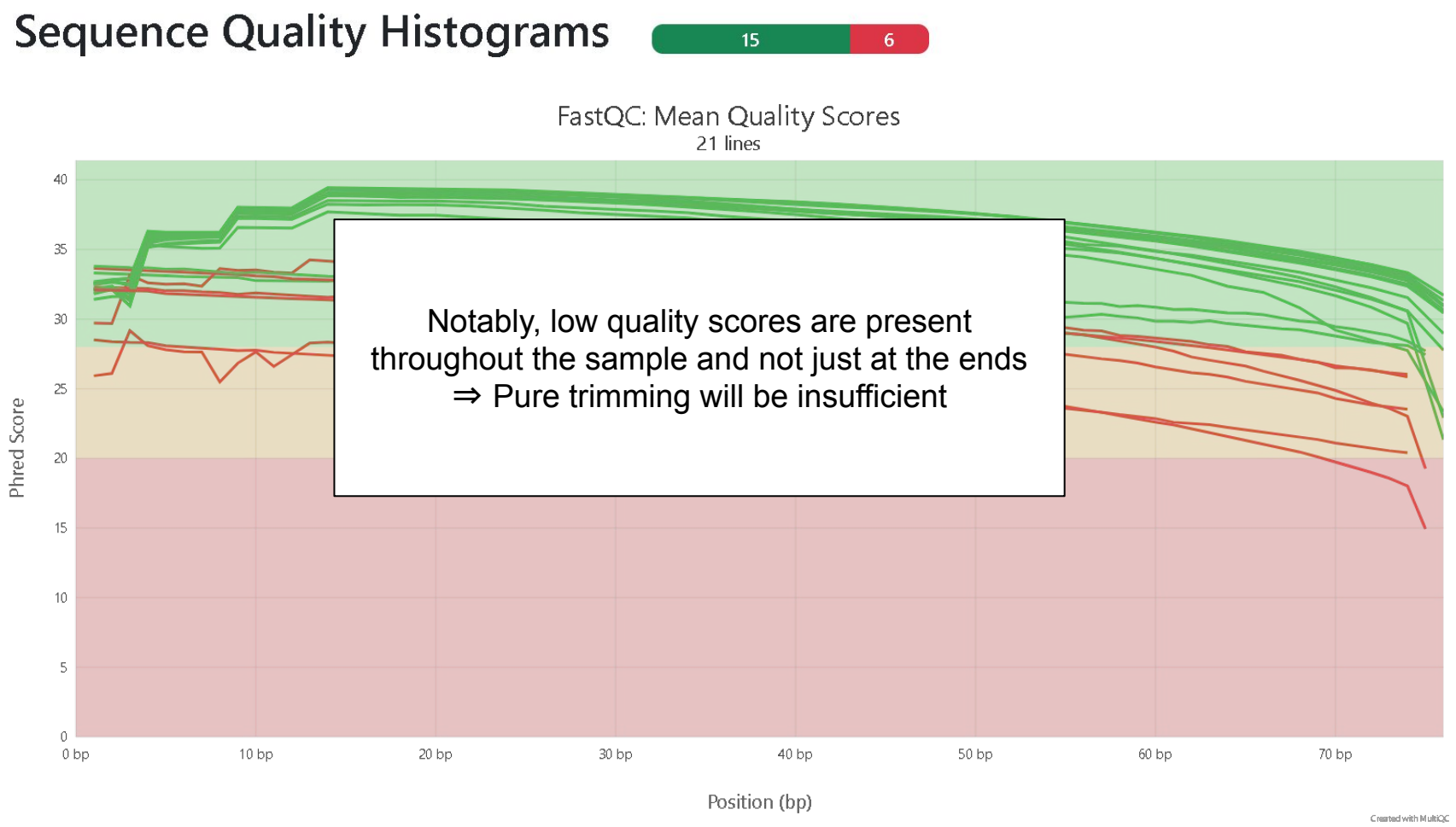
(1) Johnson, R. L., and E. Kapsalis. "Ageing, infecundity and reproductive senescence in free-ranging female rhesus monkeys." *Reproduction* 105.2 (1995): 271-278.

Preprocessing, Alignment, Quantification



(1) Andrews, Simon. *FastQC: A Quality Control Tool for High Throughput Sequence Data*. Version 0.12.1, 2010, www.bioinformatics.babraham.ac.uk/projects/fastqc/.
(2) Chen, Shifu. "Fastp 1.0: An ultra-fast all-round tool for FASTQ data quality control and preprocessing." *Imeta* 4.5 (2025): e70078.
(3) Dobin, Alexander, et al. "STAR: ultrafast universal RNA-seq aligner." *Bioinformatics* 29.1 (2013): 15-21.
(4) Li, Bo, and Colin N. Dewey. "RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome." BMC bioinformatics 12.1 (2011): 323.

Quality Control: Per Base Qualities

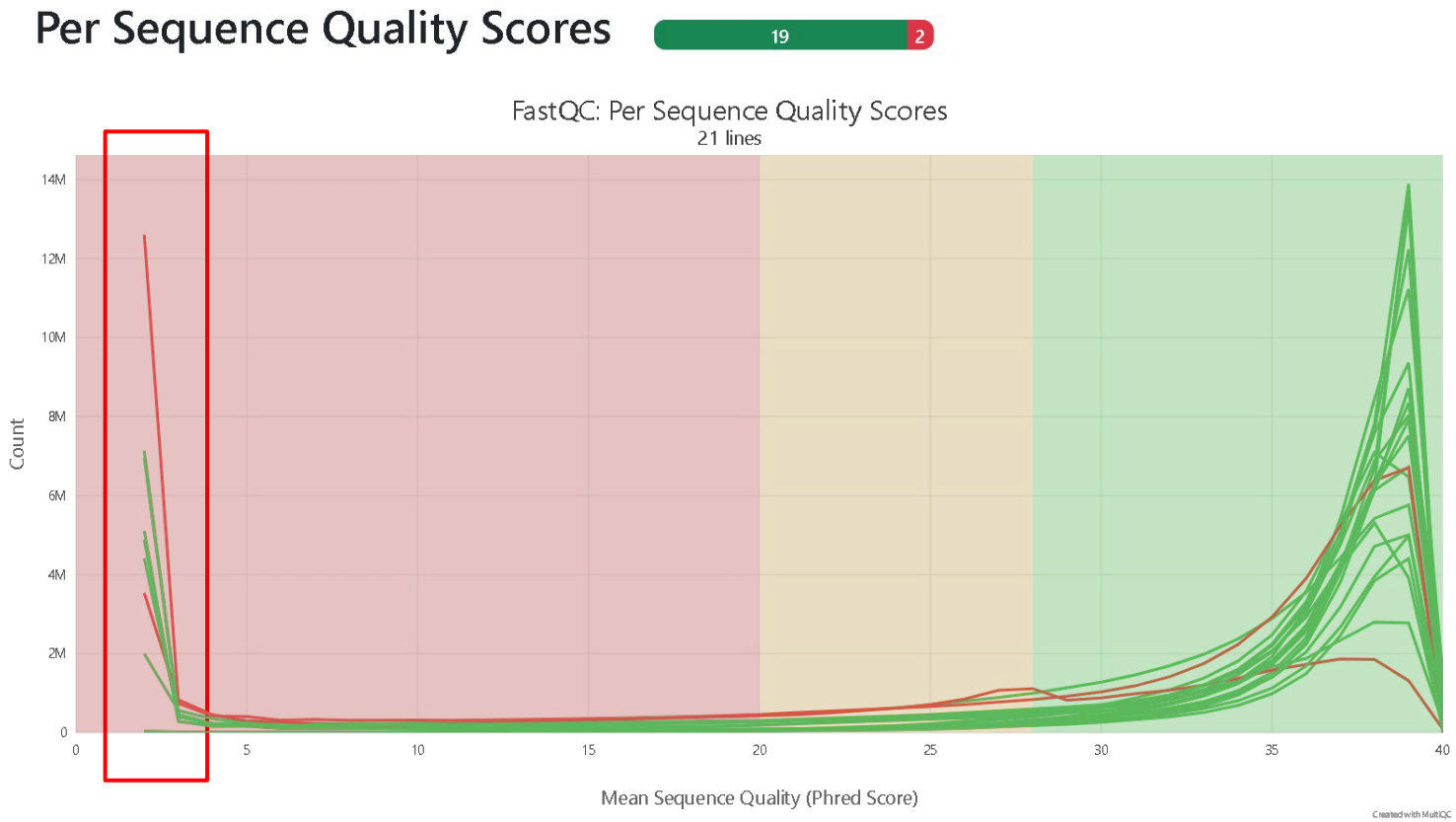


Notably, low quality scores are present throughout the sample and not just at the ends
⇒ Pure trimming will be insufficient

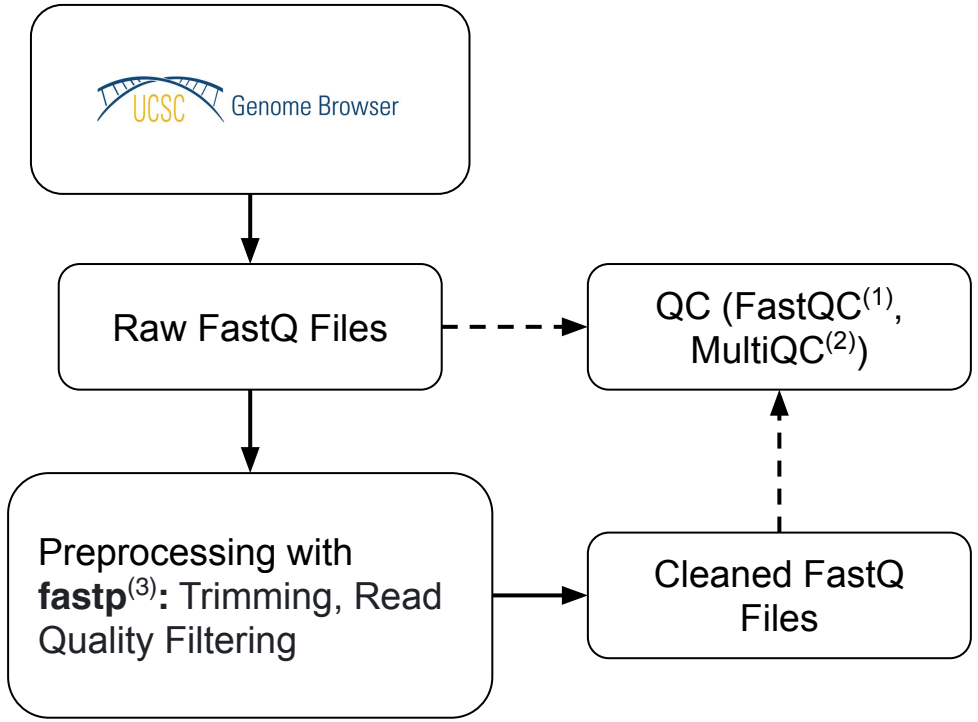
Quality Control: Mean Read Qualities

Main issue seems to be cluster of low-quality reads

⇒ We need to filter



Preprocessing, Alignment, Quantification



(1) Andrews, Simon. *FastQC: A Quality Control Tool for High Throughput Sequence Data*. Version 0.12.1, 2010, www.bioinformatics.babraham.ac.uk/projects/fastqc/.

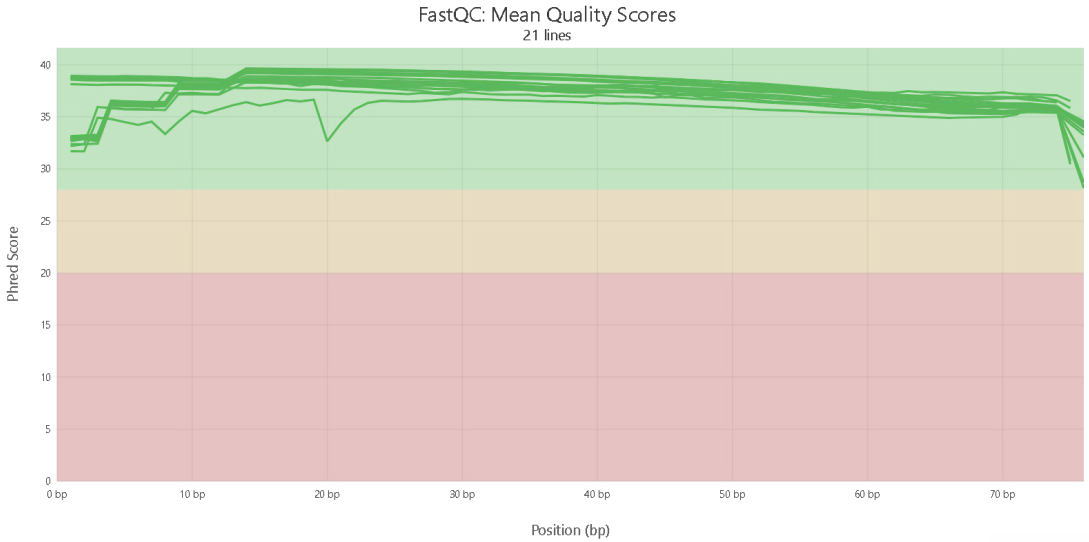
(2) Chen, Shifu. "Fastp 1.0: An ultra-fast all-round tool for FASTQ data quality control and preprocessing." *Imeta* 4.5 (2025): e70078.

(3) Dobin, Alexander, et al. "STAR: ultrafast universal RNA-seq aligner." *Bioinformatics* 29.1 (2013): 15-21.

MultiQC Outputs for Cleaned Samples

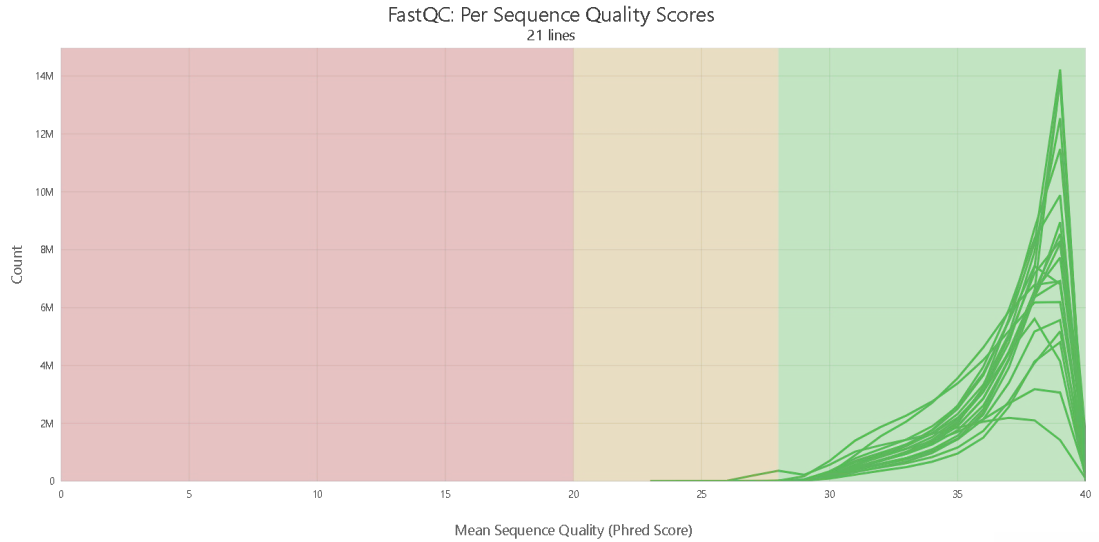
Sequence Quality Histograms

21

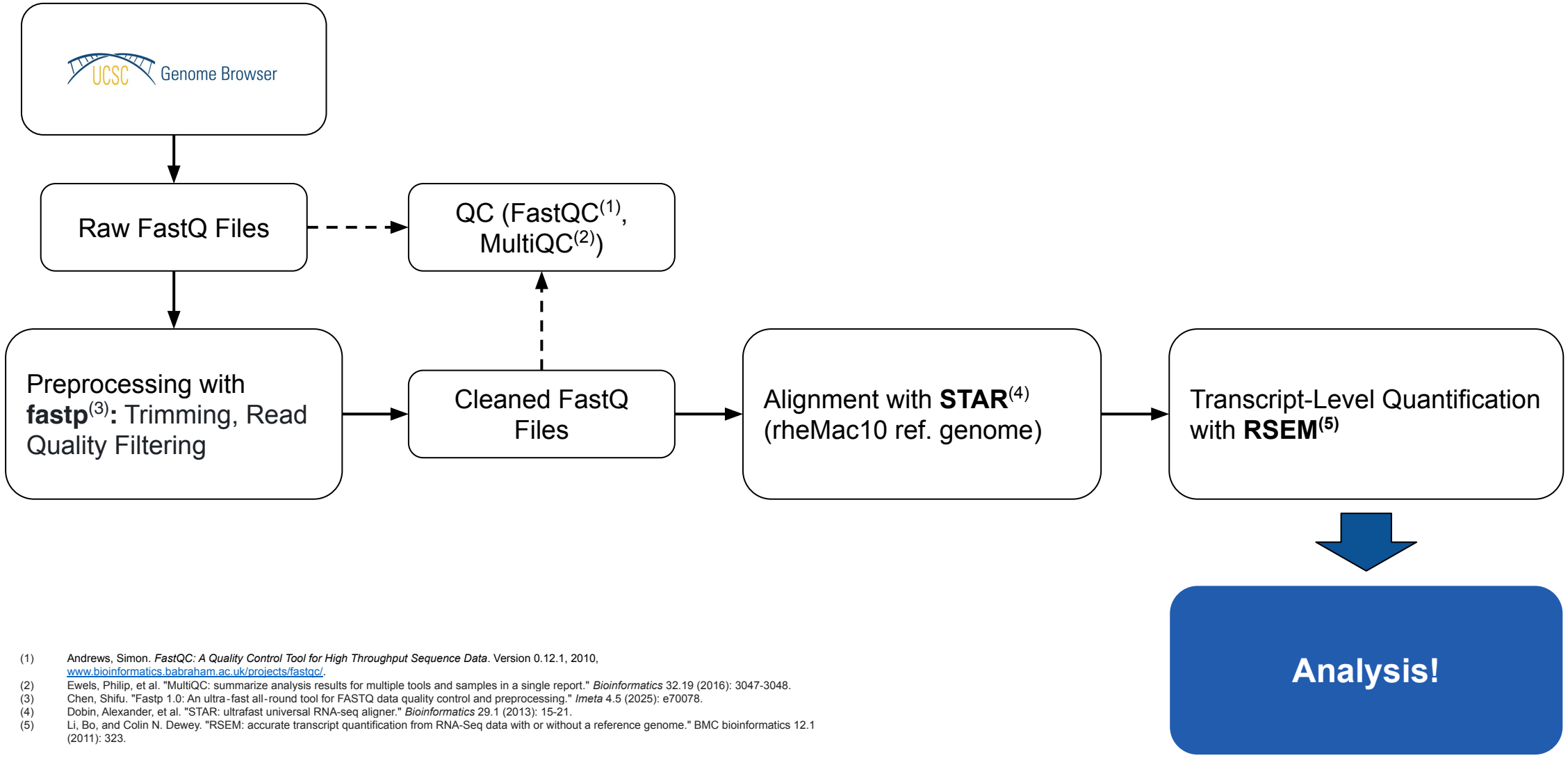


Per Sequence Quality Scores

21



Preprocessing, Alignment, Quantification

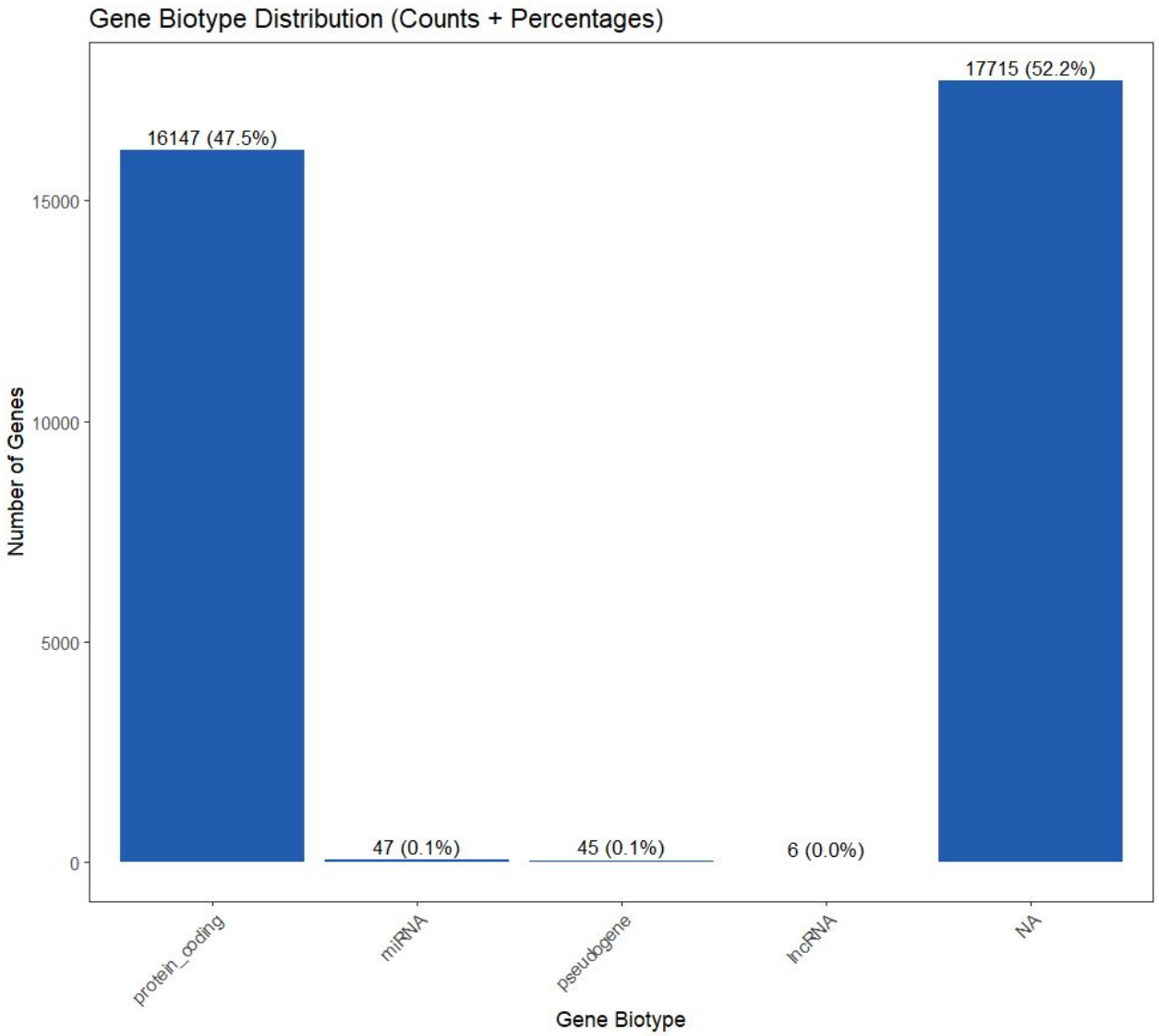


(1) Andrews, Simon. *FastQC: A Quality Control Tool for High Throughput Sequence Data*. Version 0.12.1, 2010, www.bioinformatics.babraham.ac.uk/projects/fastqc/.
(2) Ewels, Philip, et al. "MultiQC: summarize analysis results for multiple tools and samples in a single report." *Bioinformatics* 32.19 (2016): 3047-3048.
(3) Chen, Shifu. "Fastp 1.0: An ultra-fast all-round tool for FASTQ data quality control and preprocessing." *Imeta* 4.5 (2025): e70078.
(4) Dobin, Alexander, et al. "STAR: ultrafast universal RNA-seq aligner." *Bioinformatics* 29.1 (2013): 15-21.
(5) Li, Bo, and Colin N. Dewey. "RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome." BMC bioinformatics 12.1 (2011): 323.

Gene Filtering and Sample Correlations

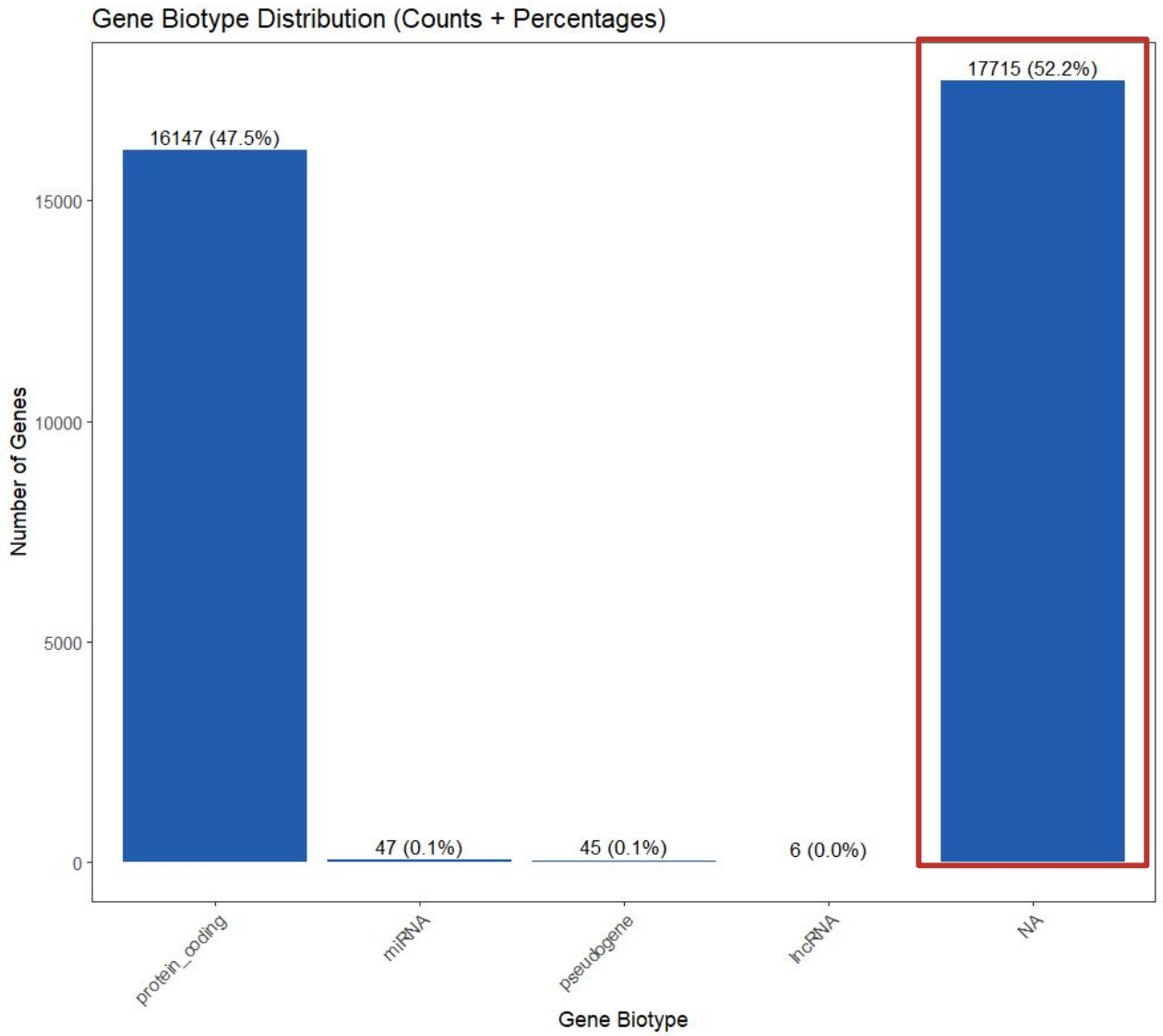
Gene Filtering

- Initially had 33960 genes.
- 17715 genes with Biotype “NA’
- Among those, 15790 were LOC genes.



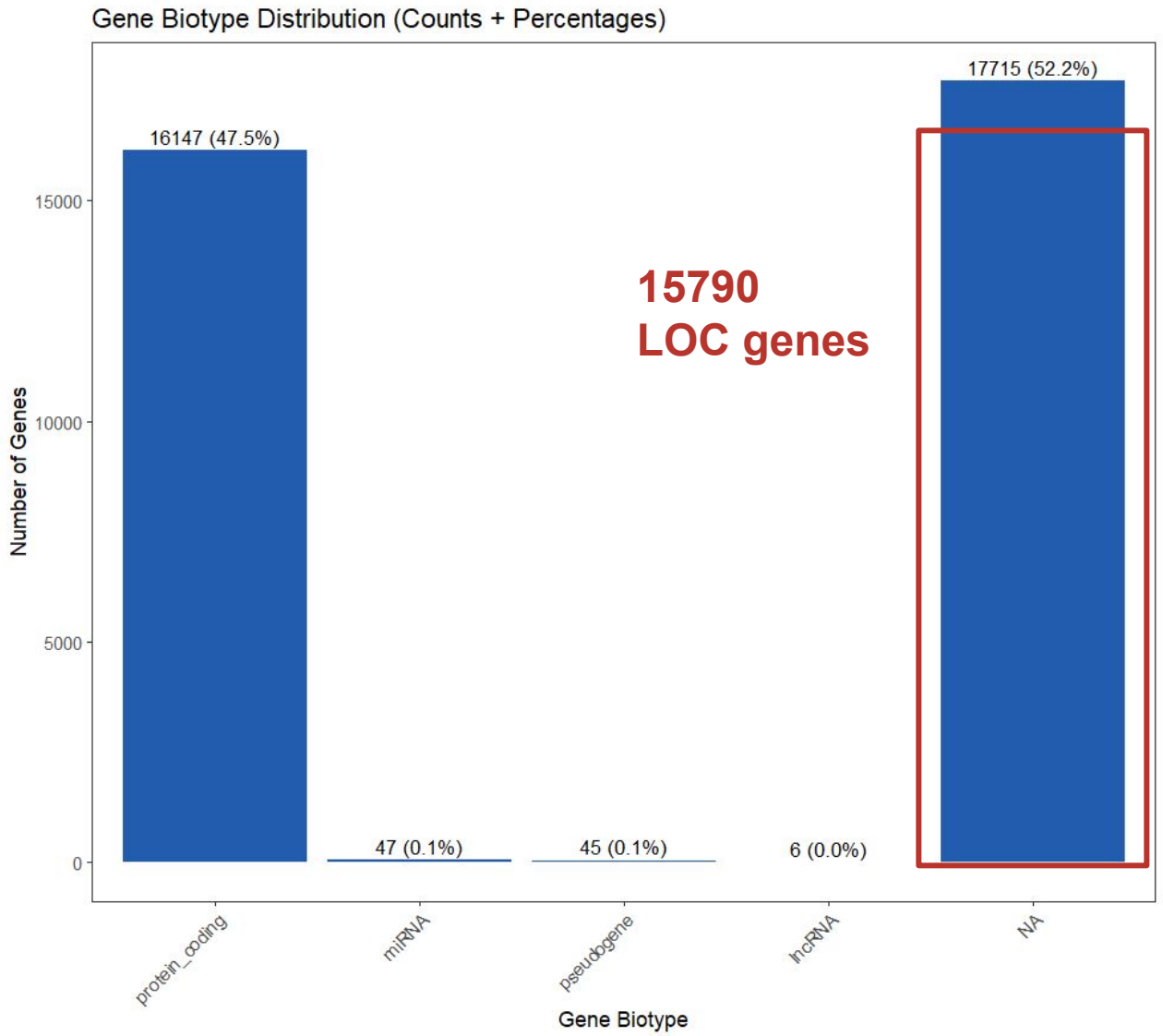
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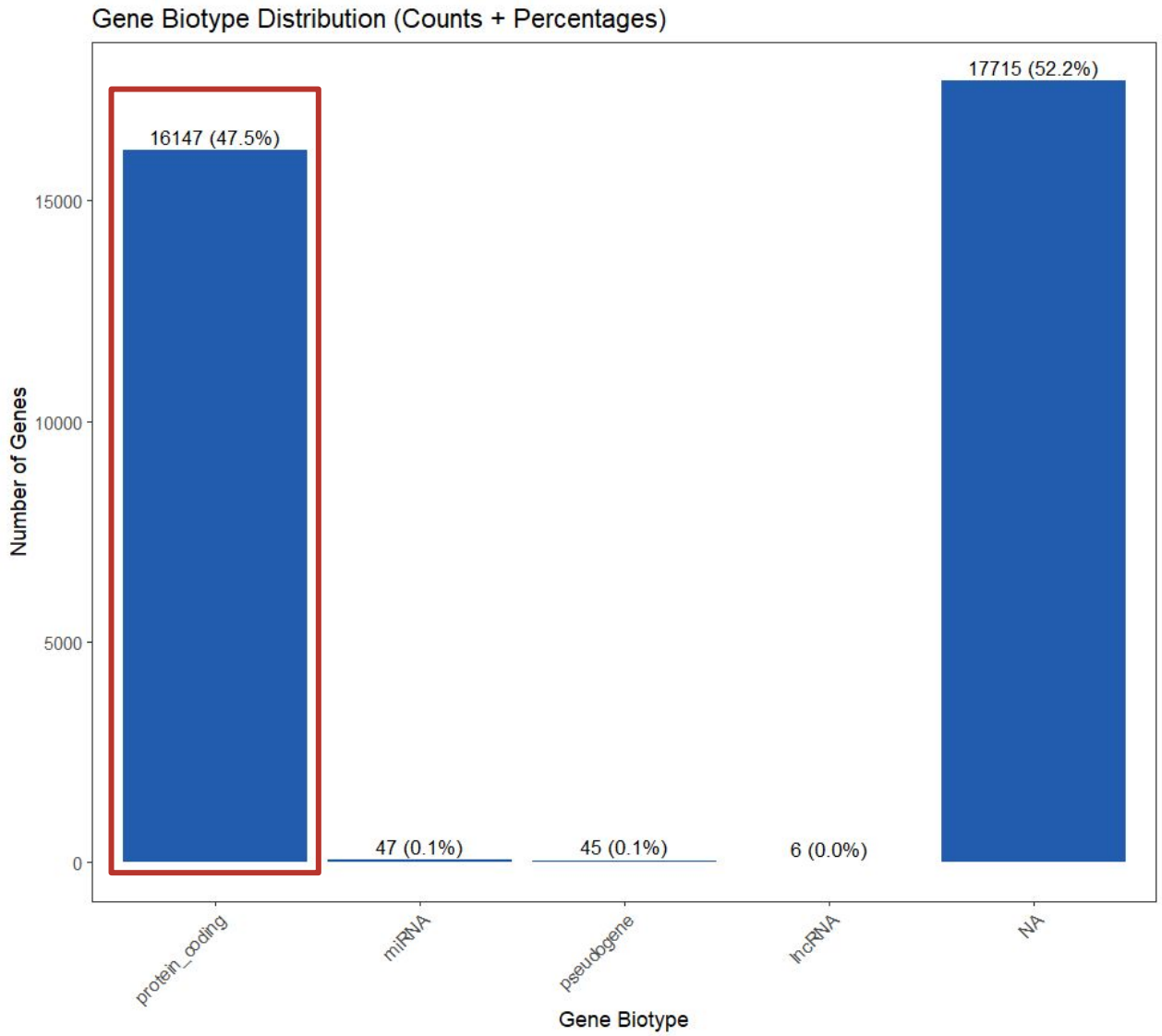
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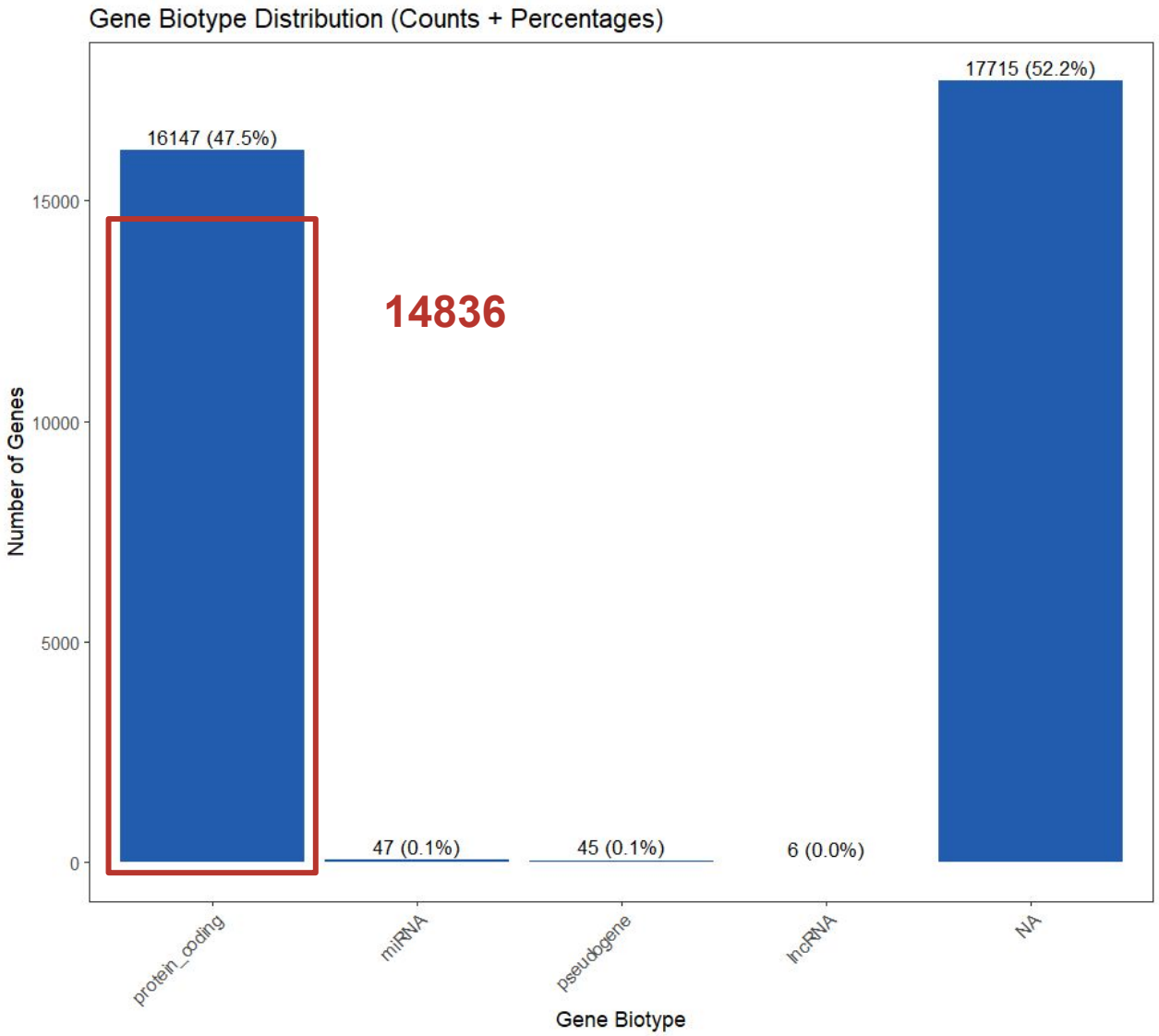
Filtered for protein coding genes



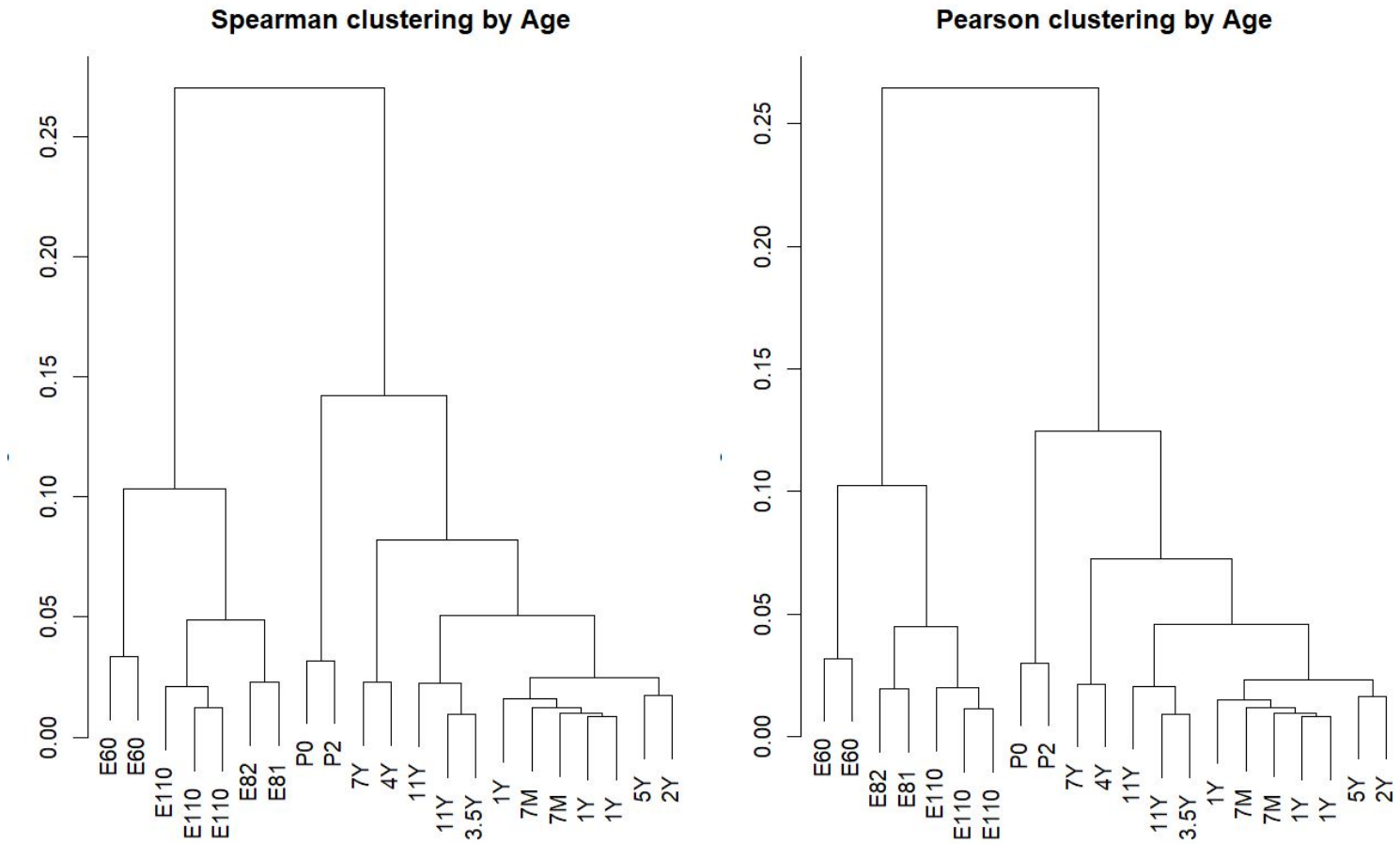
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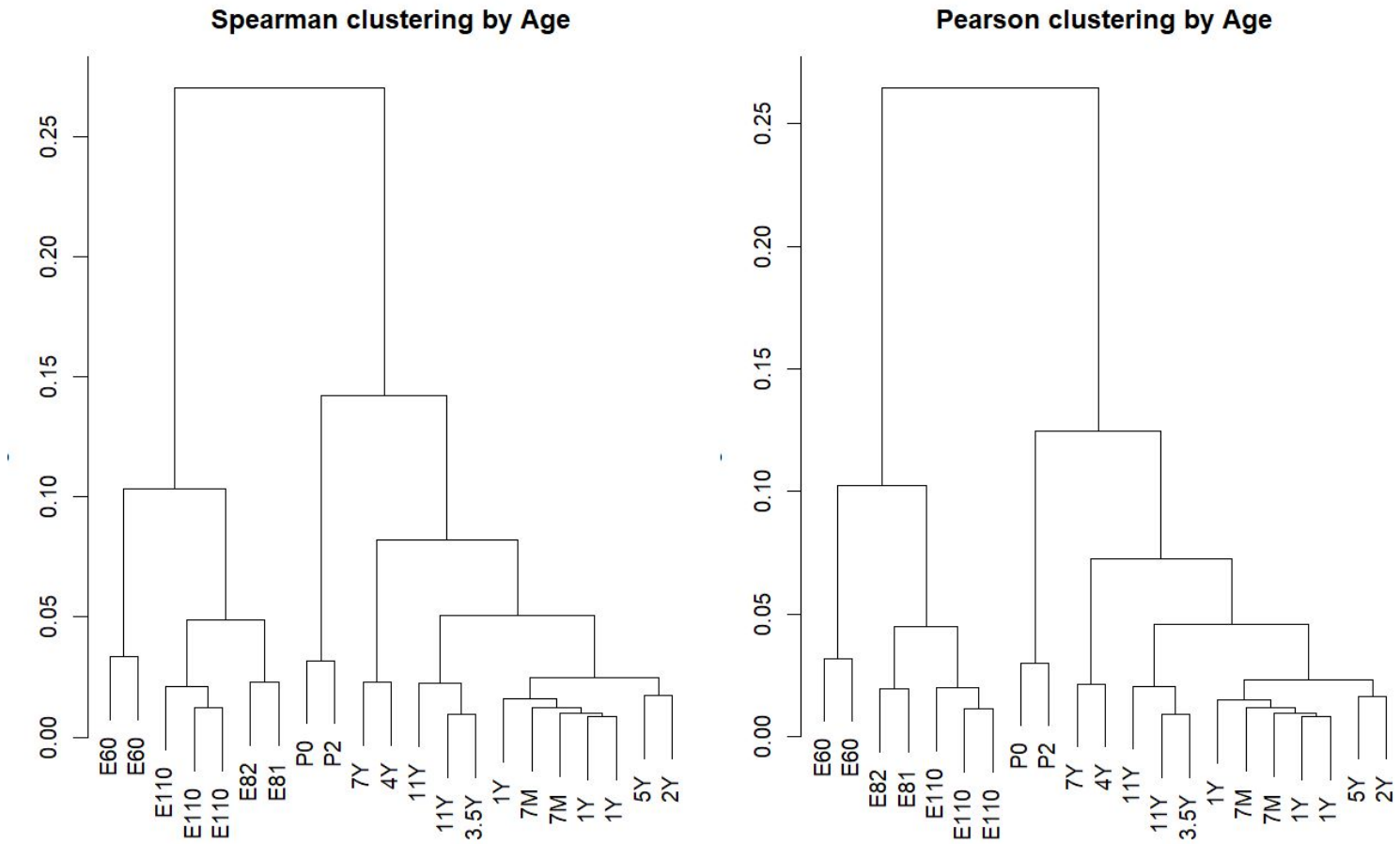
Filtered for genes that were expressed in more than half of the samples or had an average expression >= 1



Sample Correlations and Hierarchical Clustering

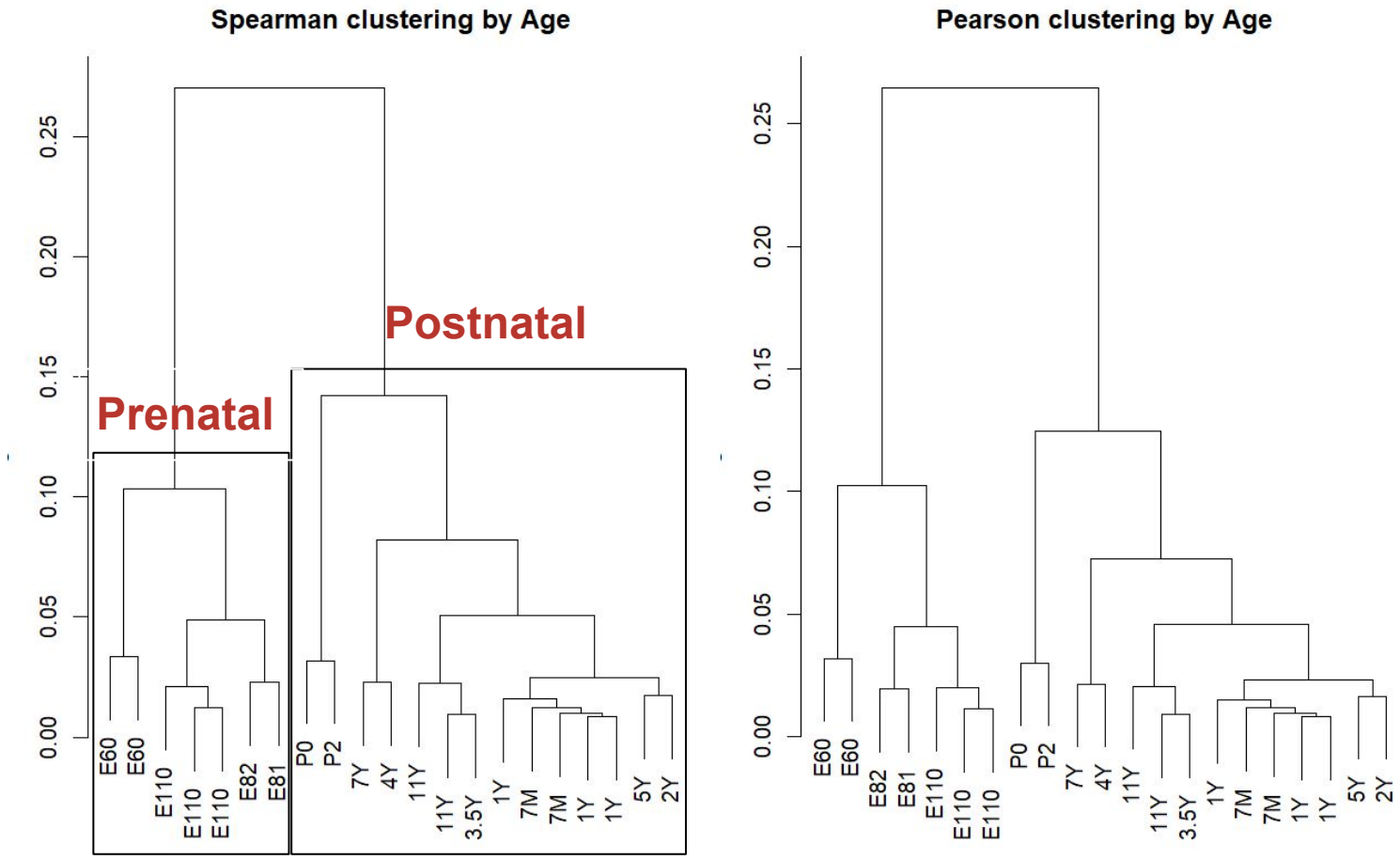


Sample Correlations and Hierarchical Clustering



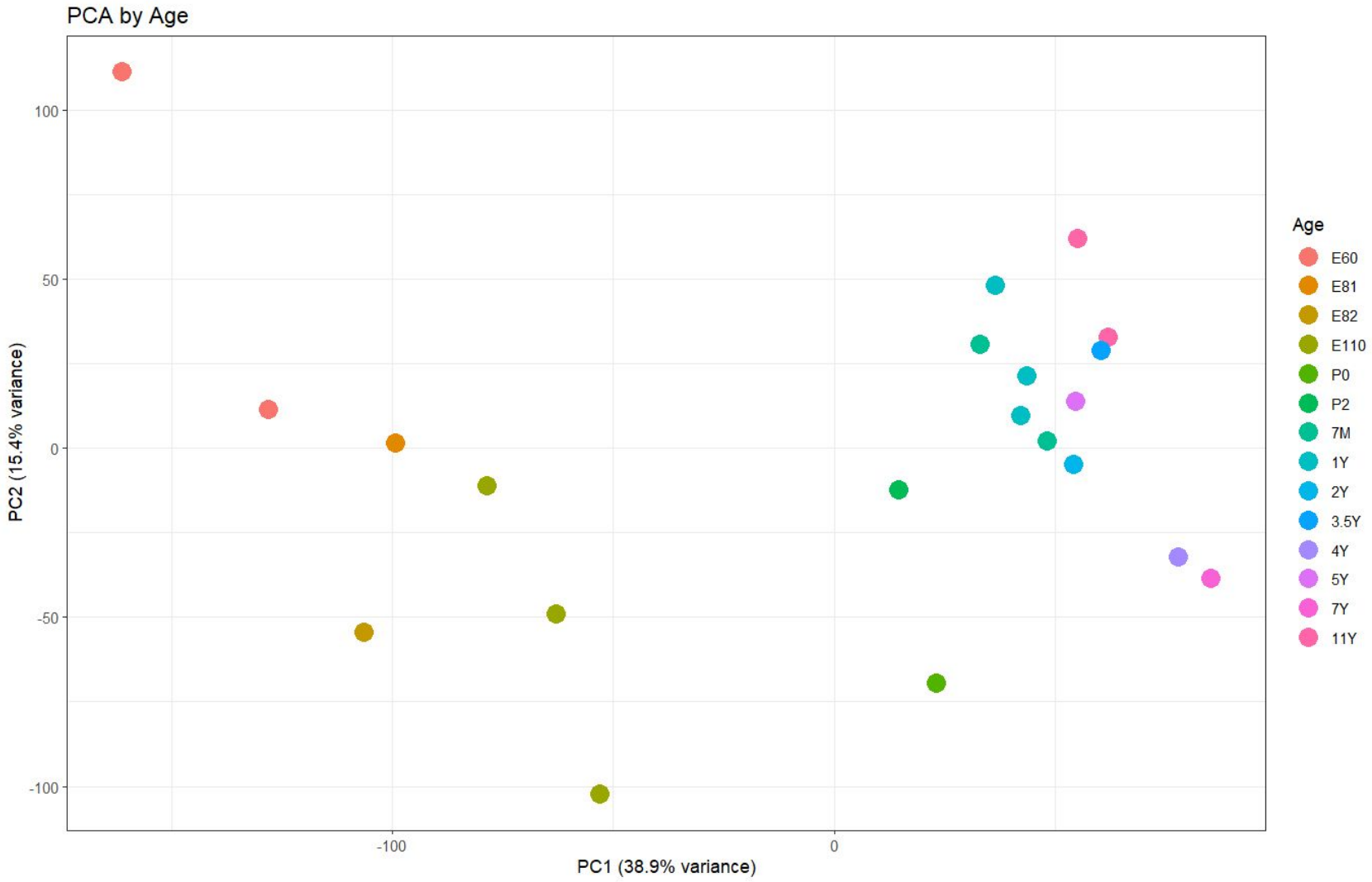
Near identical clustering

Sample Correlations and Hierarchical Clustering

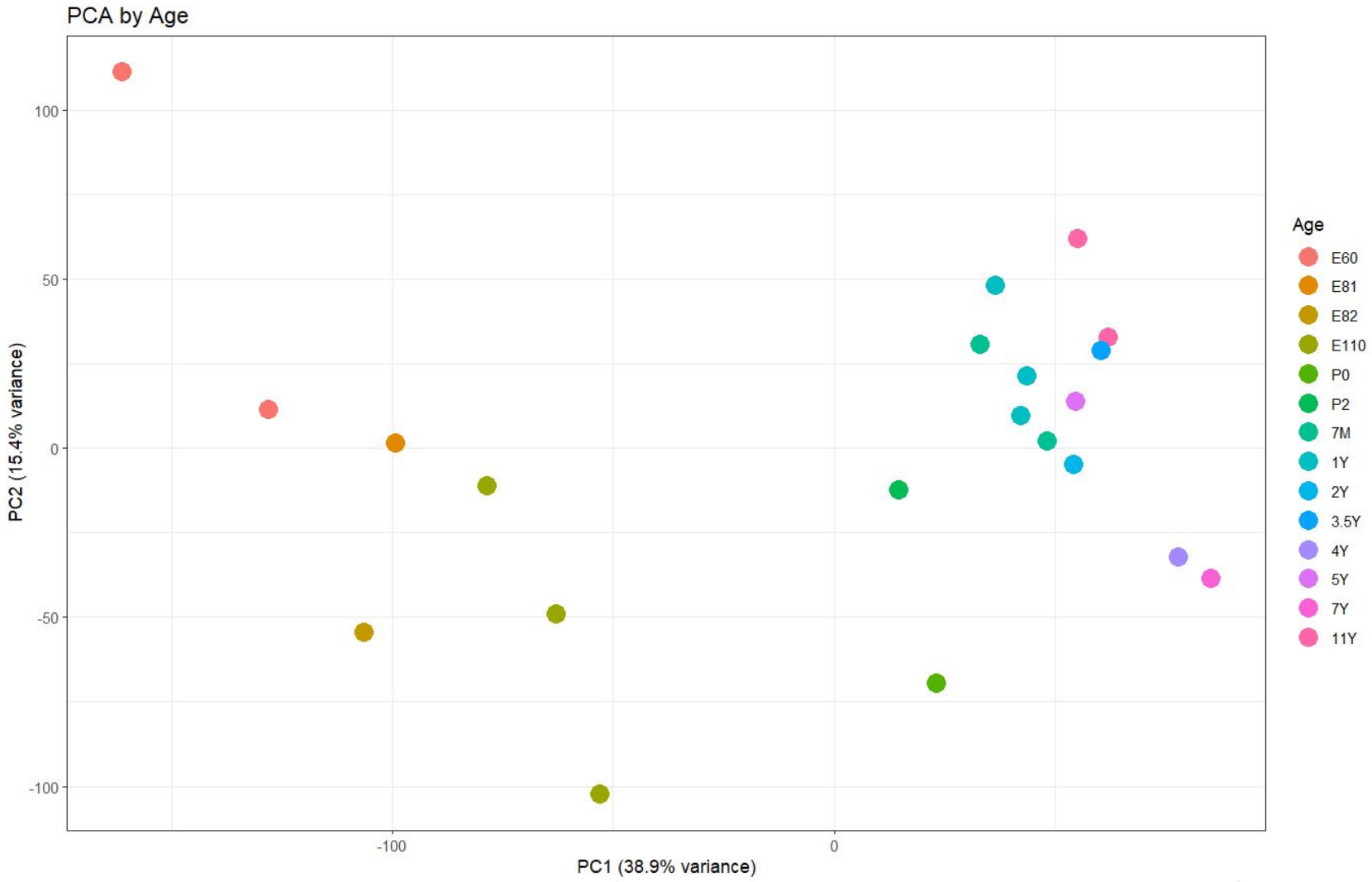


Clear clustering of prenatal and postnatal samples

Principle Component Analysis

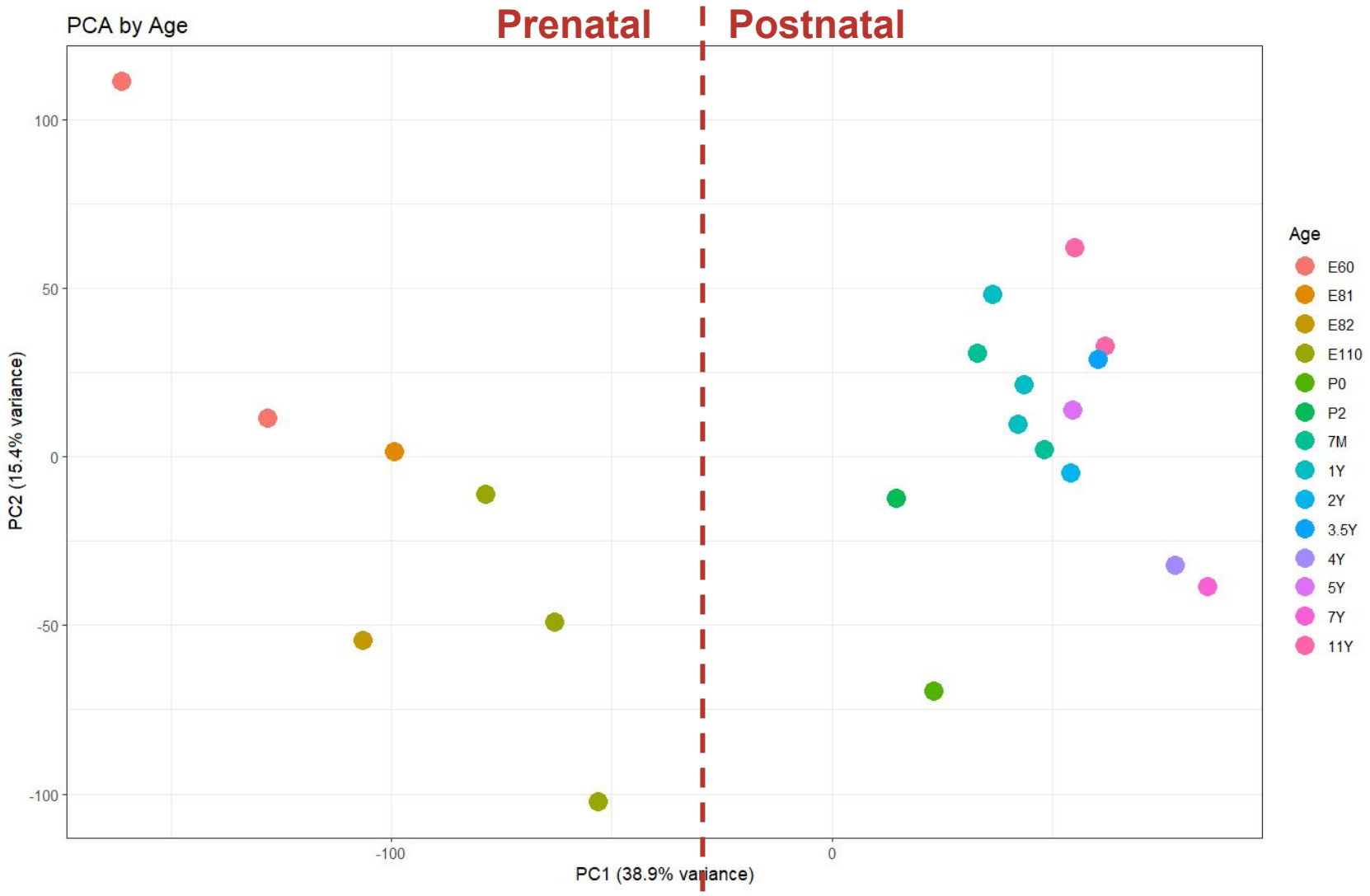


Principle Component Analysis

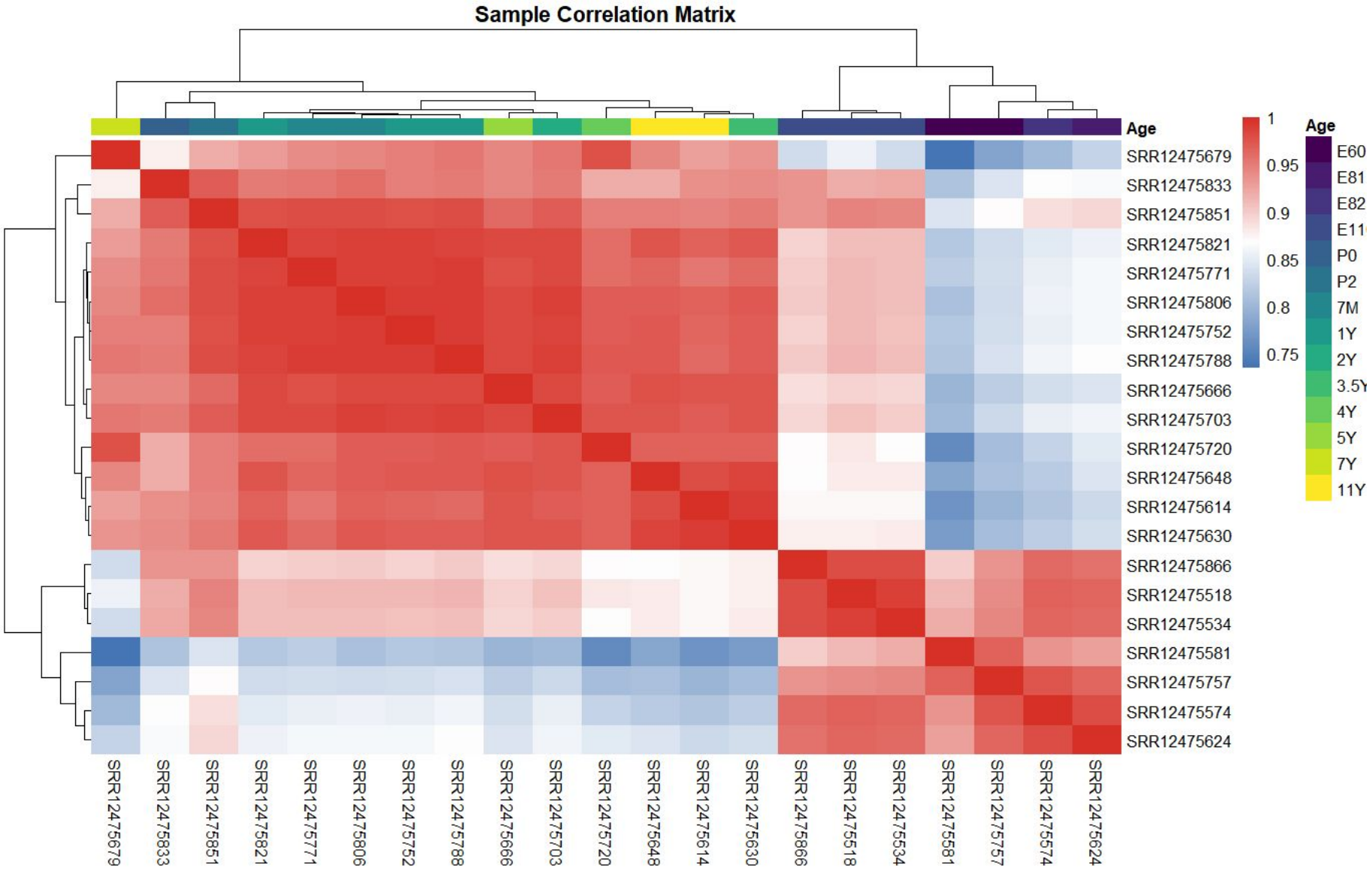


Increasing age

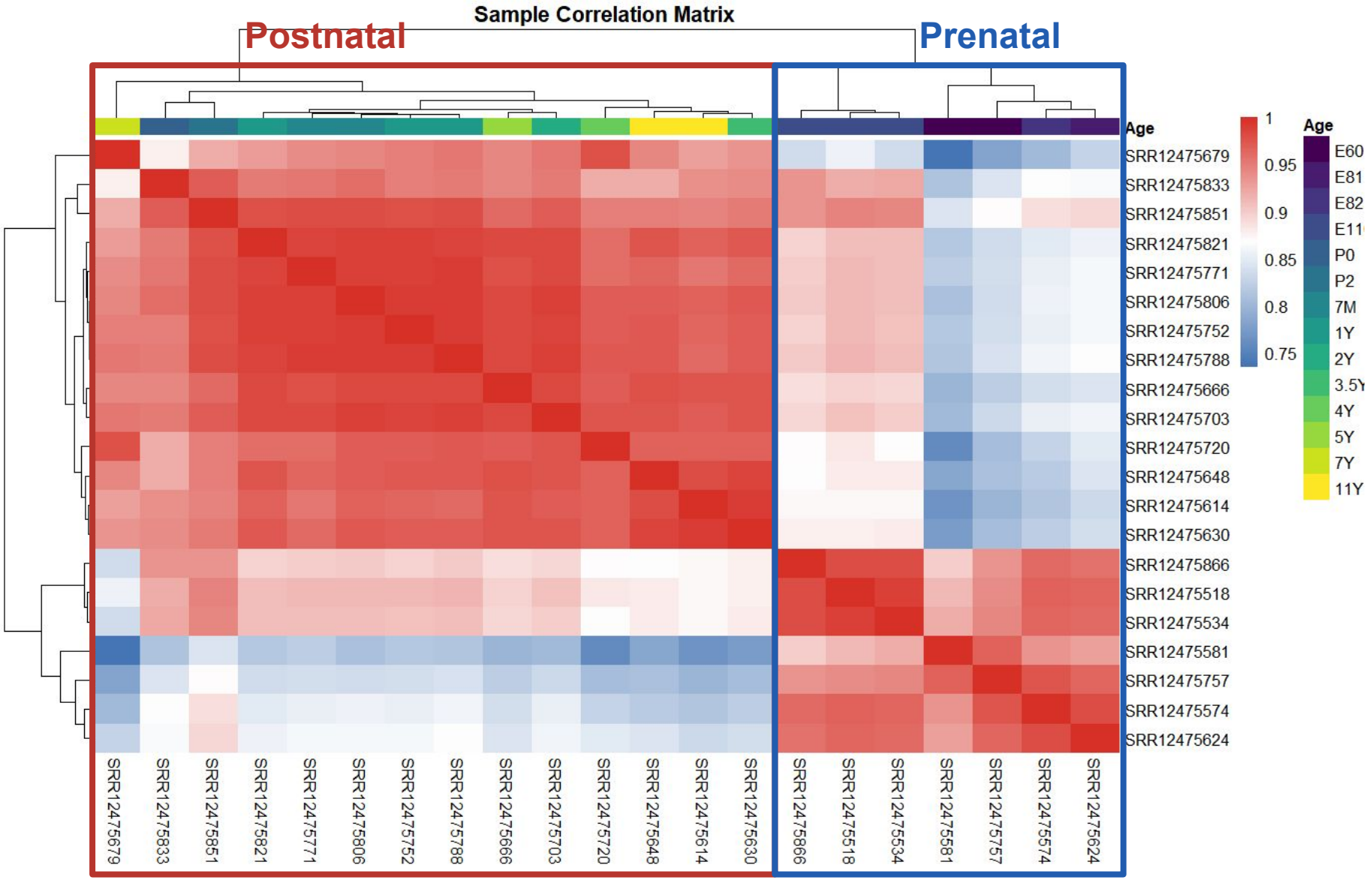
Principle Component Analysis



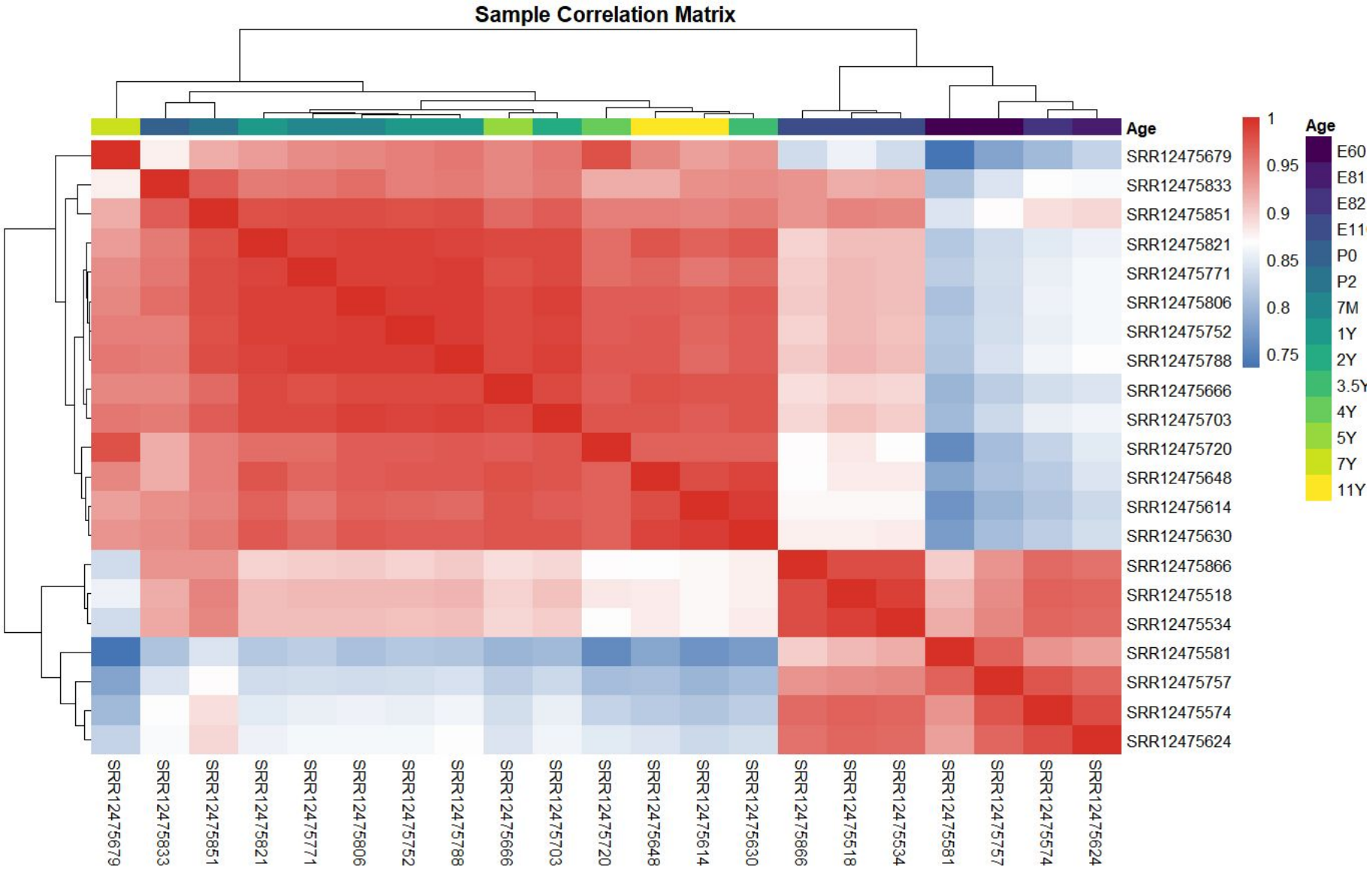
Sample Correlation Matrix



Sample Correlation Matrix



Sample Correlation Matrix

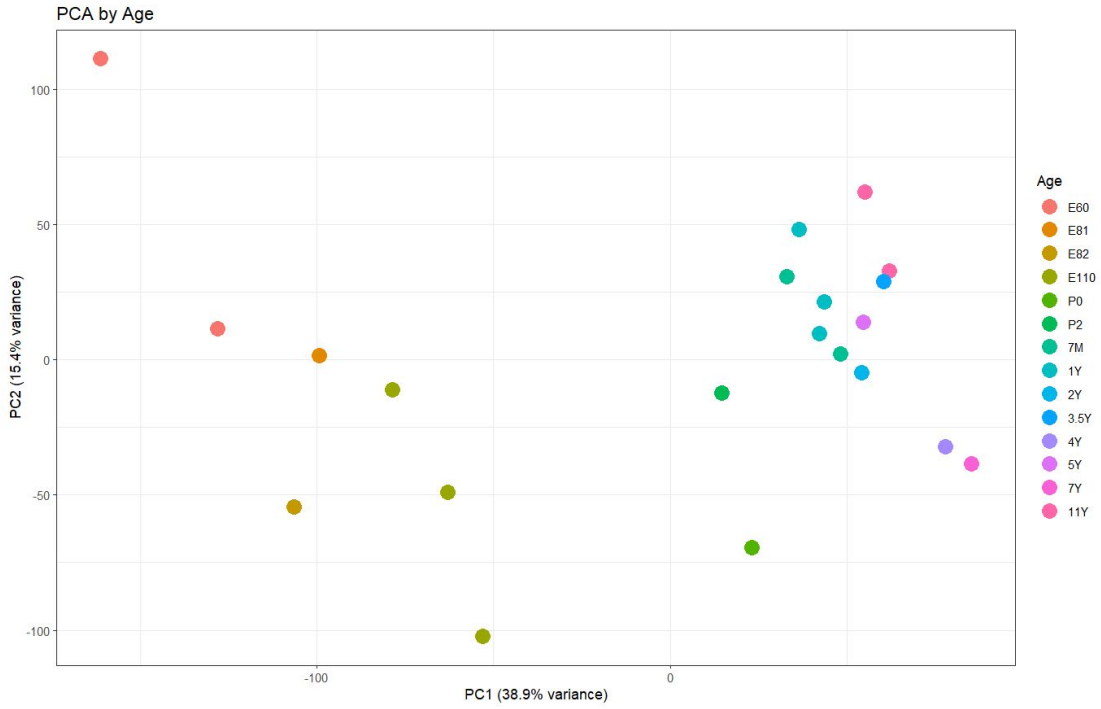


But there are also subtle differences between prenatal and postnatal samples…

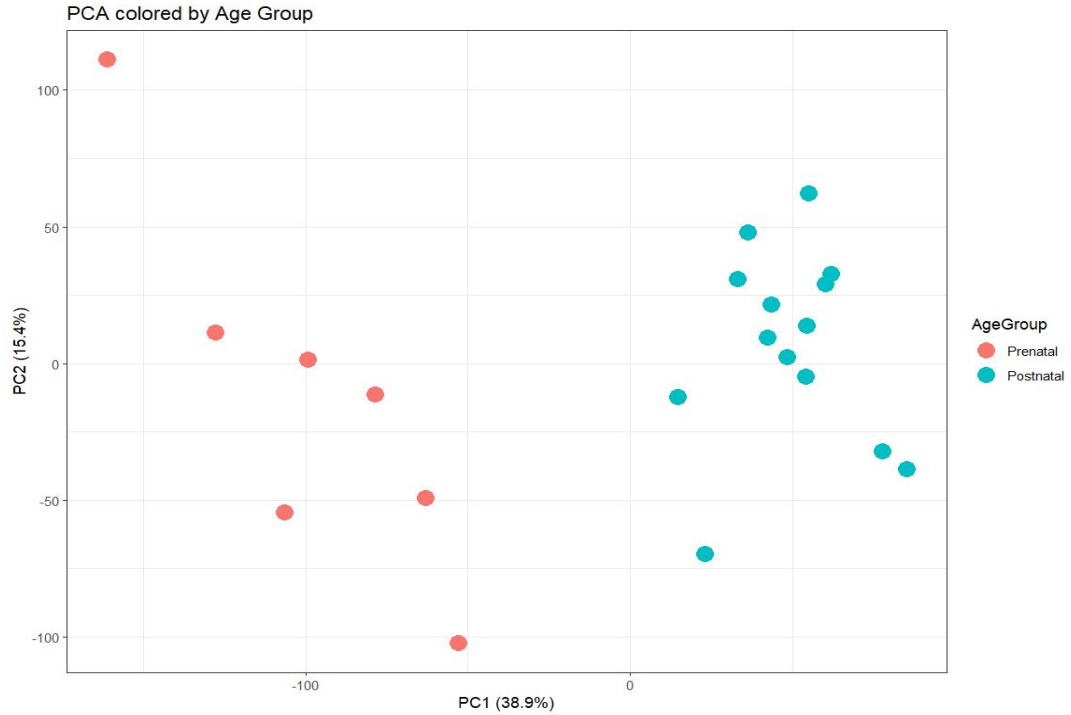
ETH zürich

Should we group, or not?

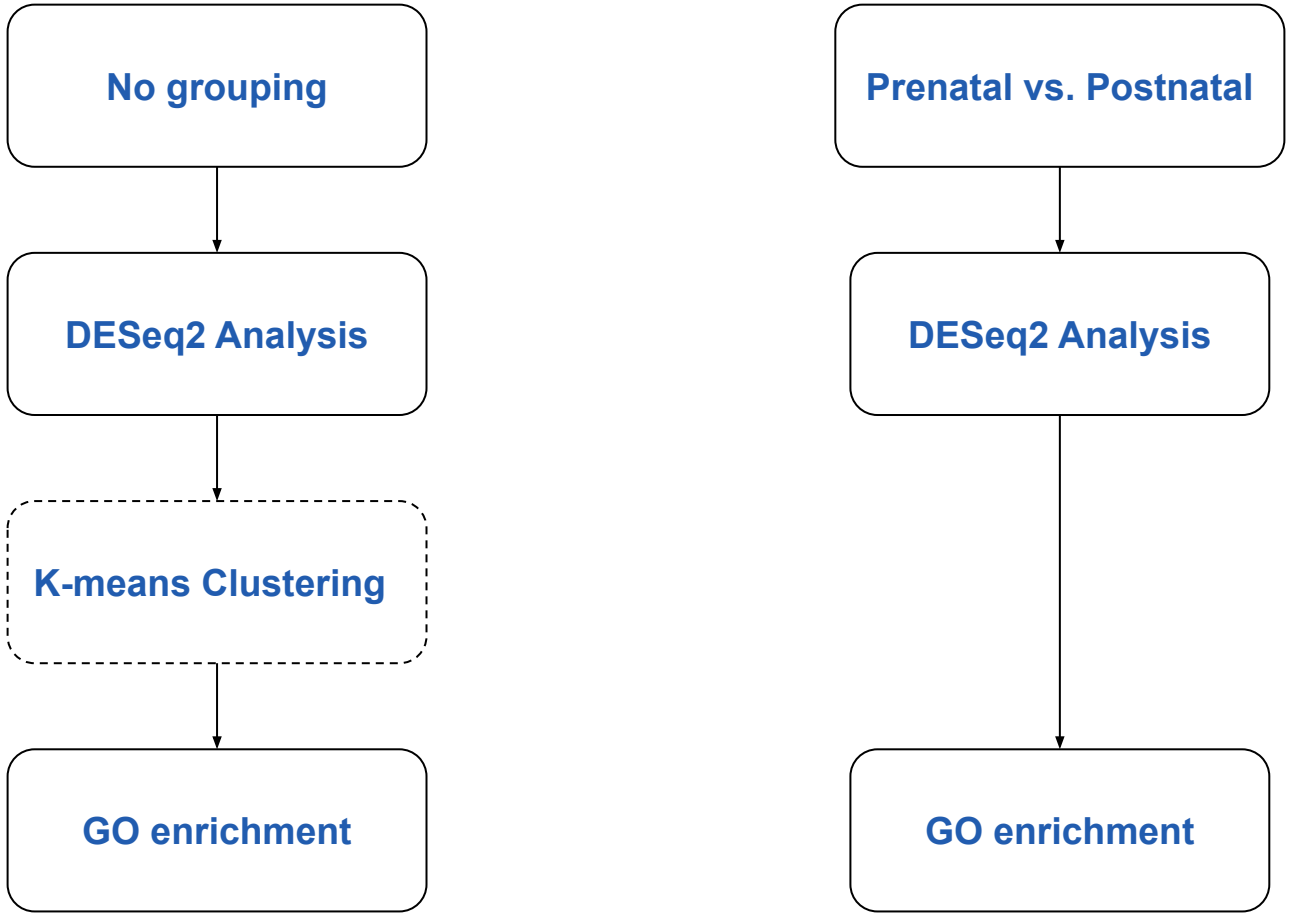
No grouping



Group (Prenatal vs. Postnatal)



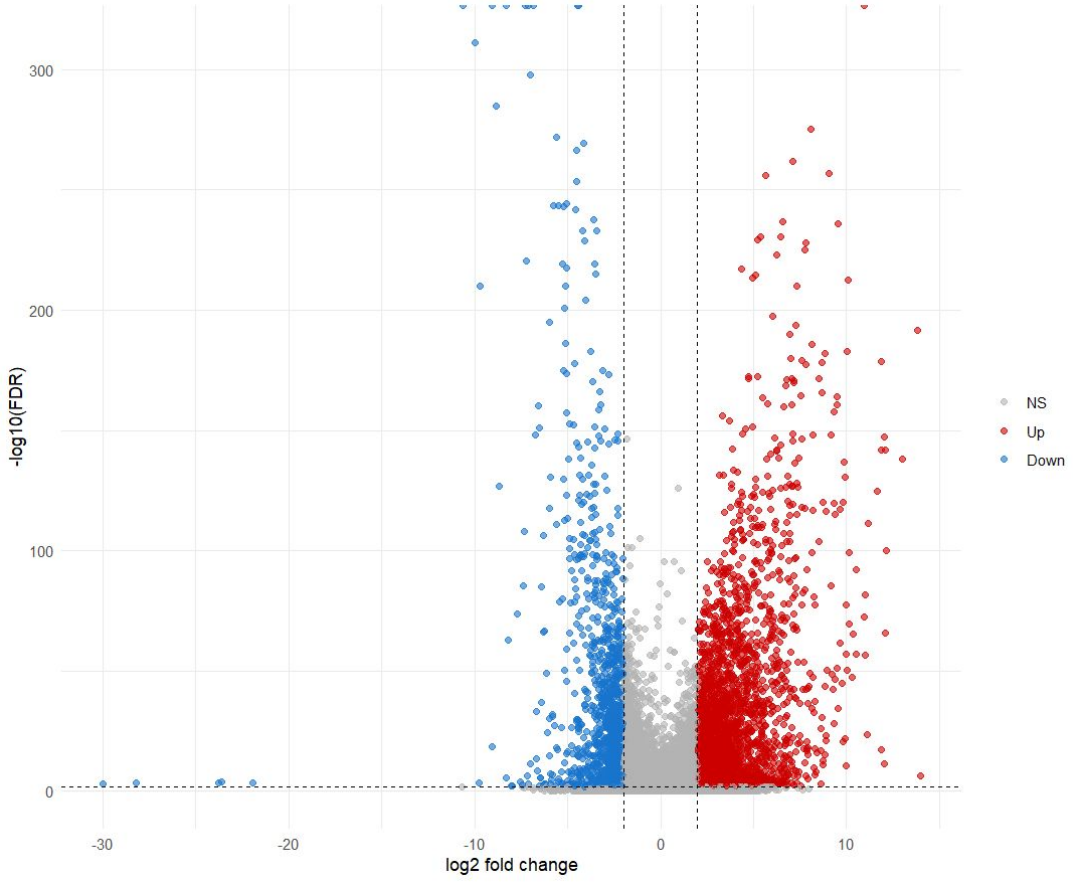
Proceed with two analyses



DESeq2 Analysis

No grouping

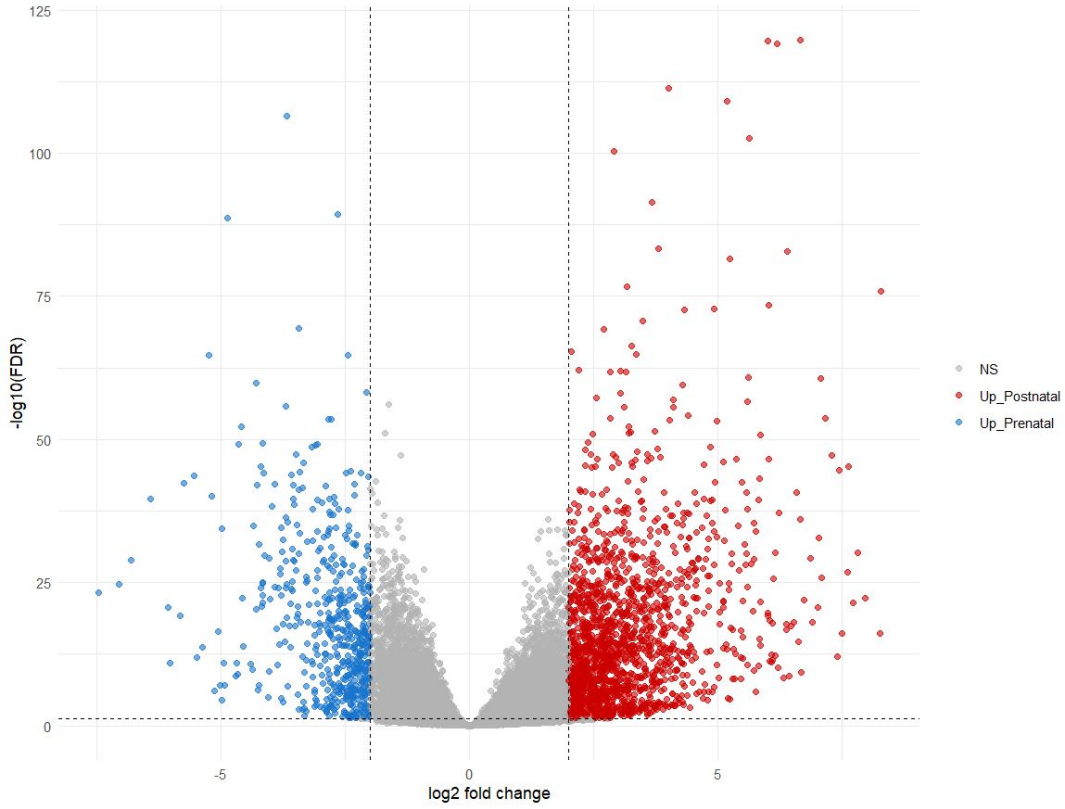
Likelihood Ratio Test



Prenatal vs. Postnatal

Wald Test

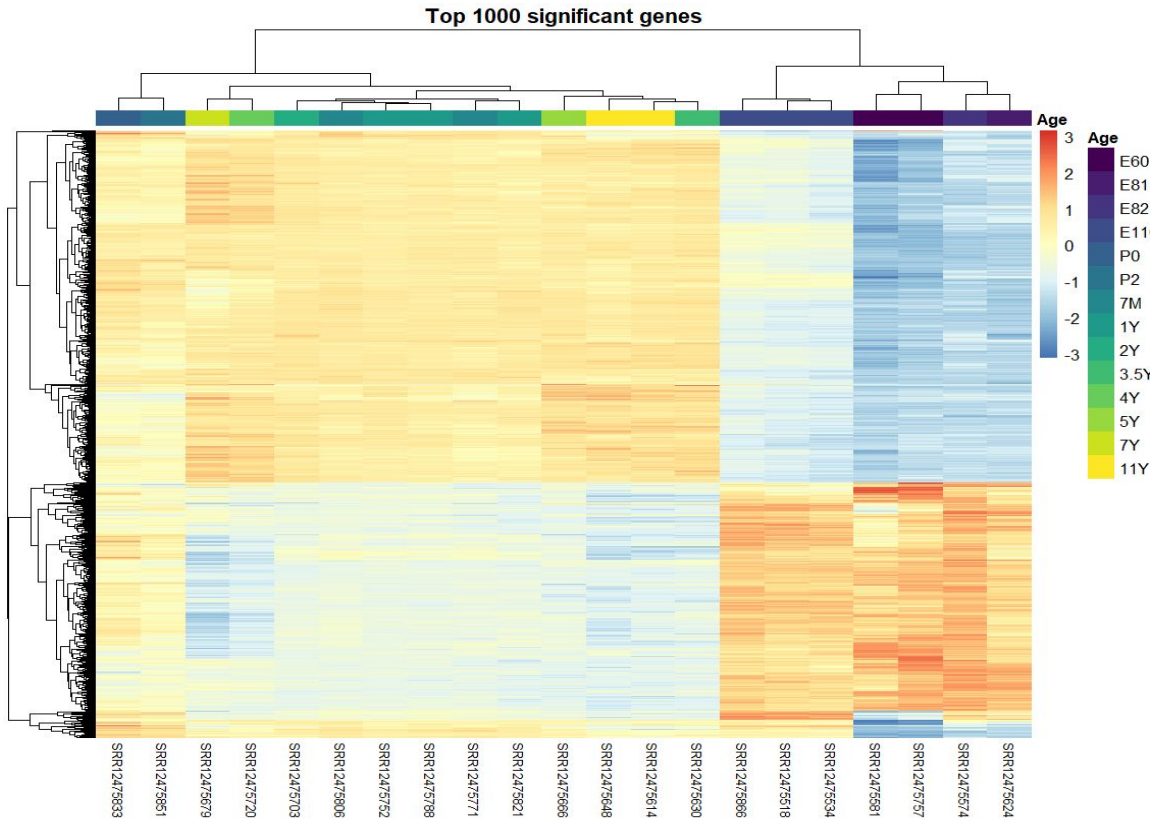
2206 genes with padj <0.05 and abs(Log2FC) > 2



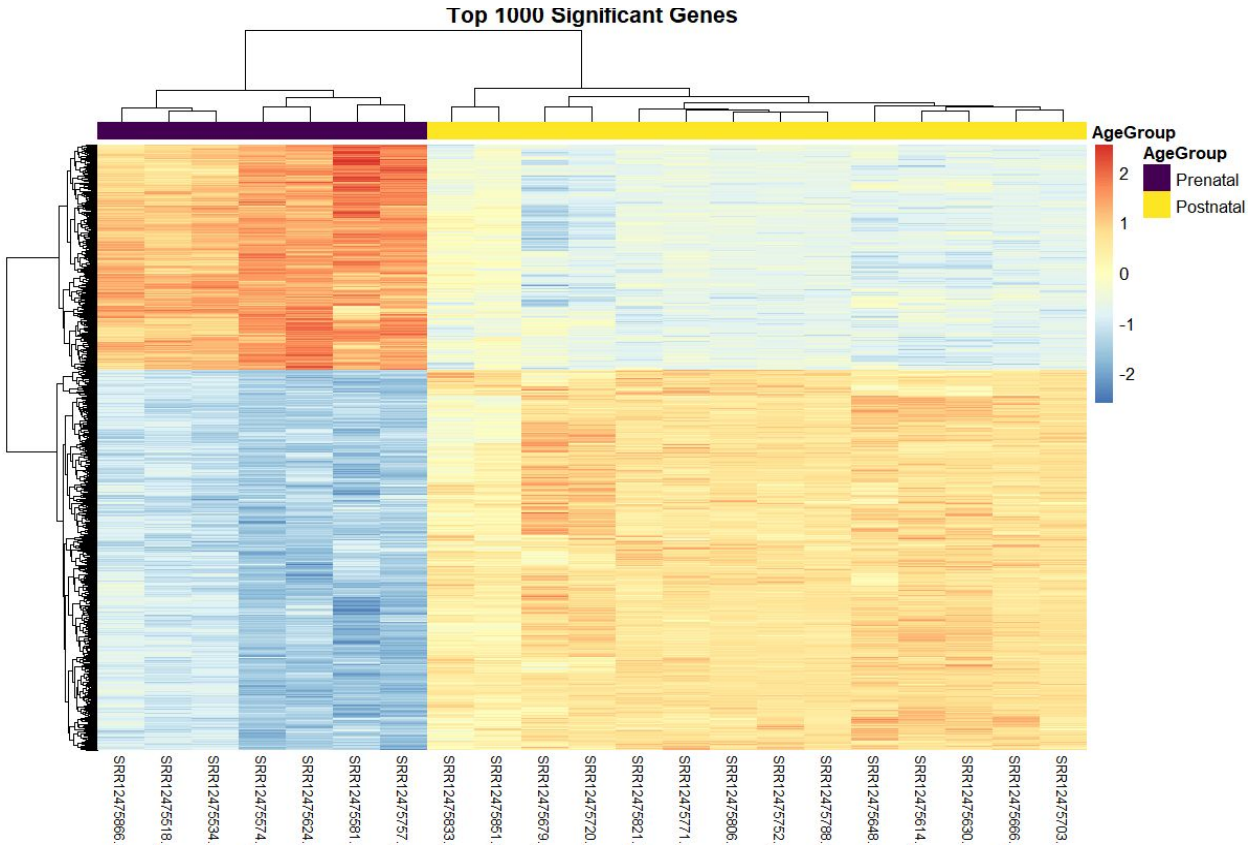
ETH zürich

Top 1000 Significant Genes

No grouping



Grouping



ETH zürich

Comparison with The Paper

Five Major Transcriptional Phases of Primate Cortical Development

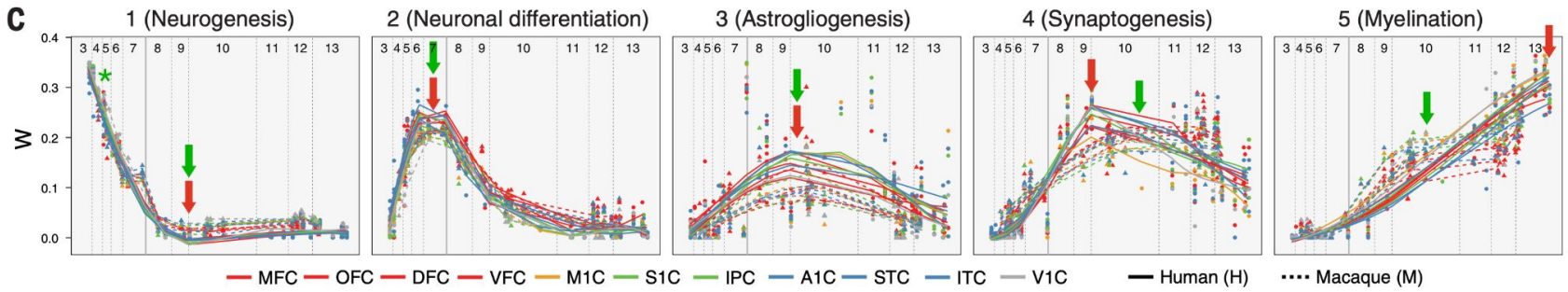
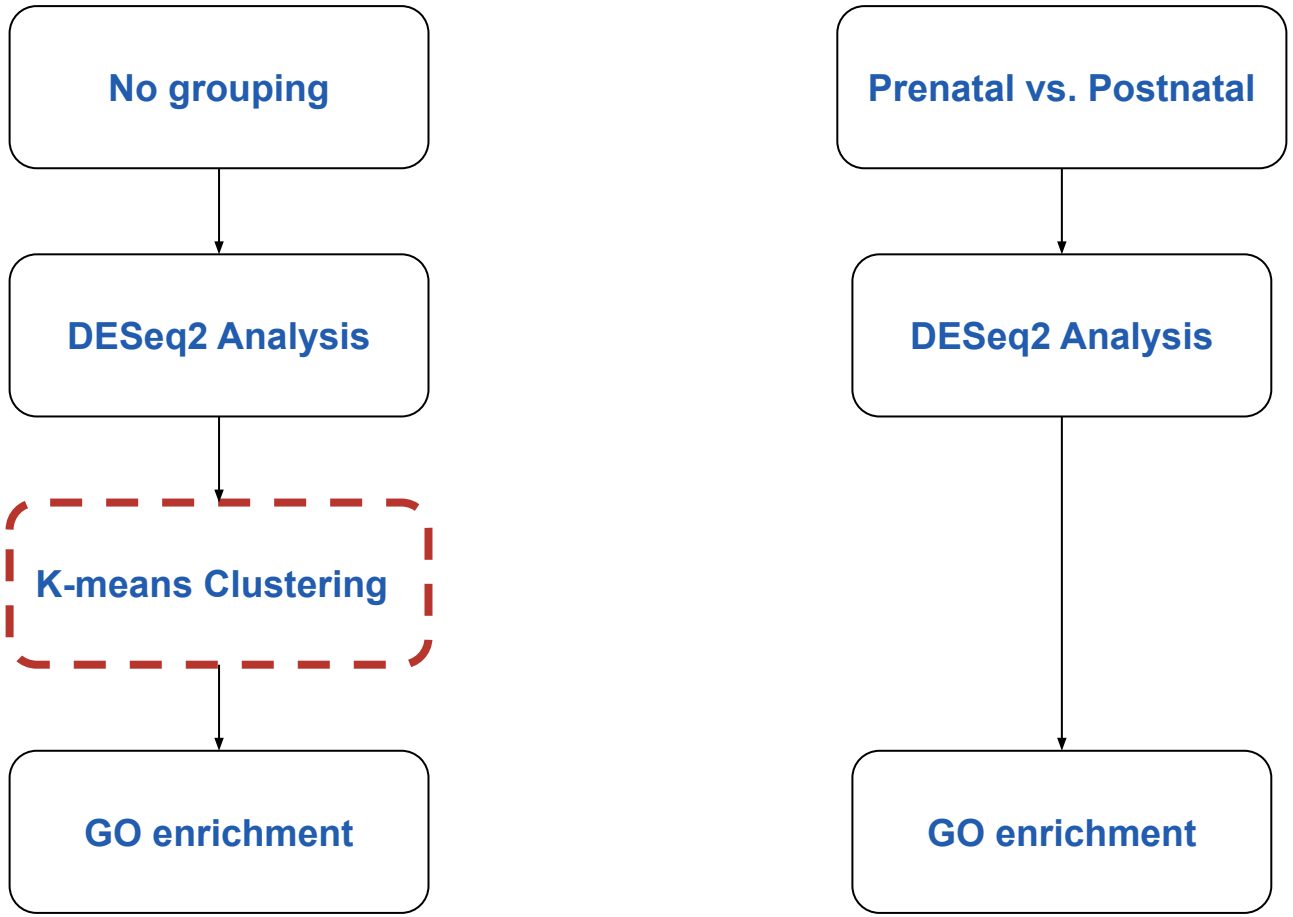


Figure from Zhu et al., 2018. Transcriptomic signatures reveal five conserved developmental processes: neurogenesis, neuronal differentiation, astrogliogenesis, synaptogenesis, and myelination.

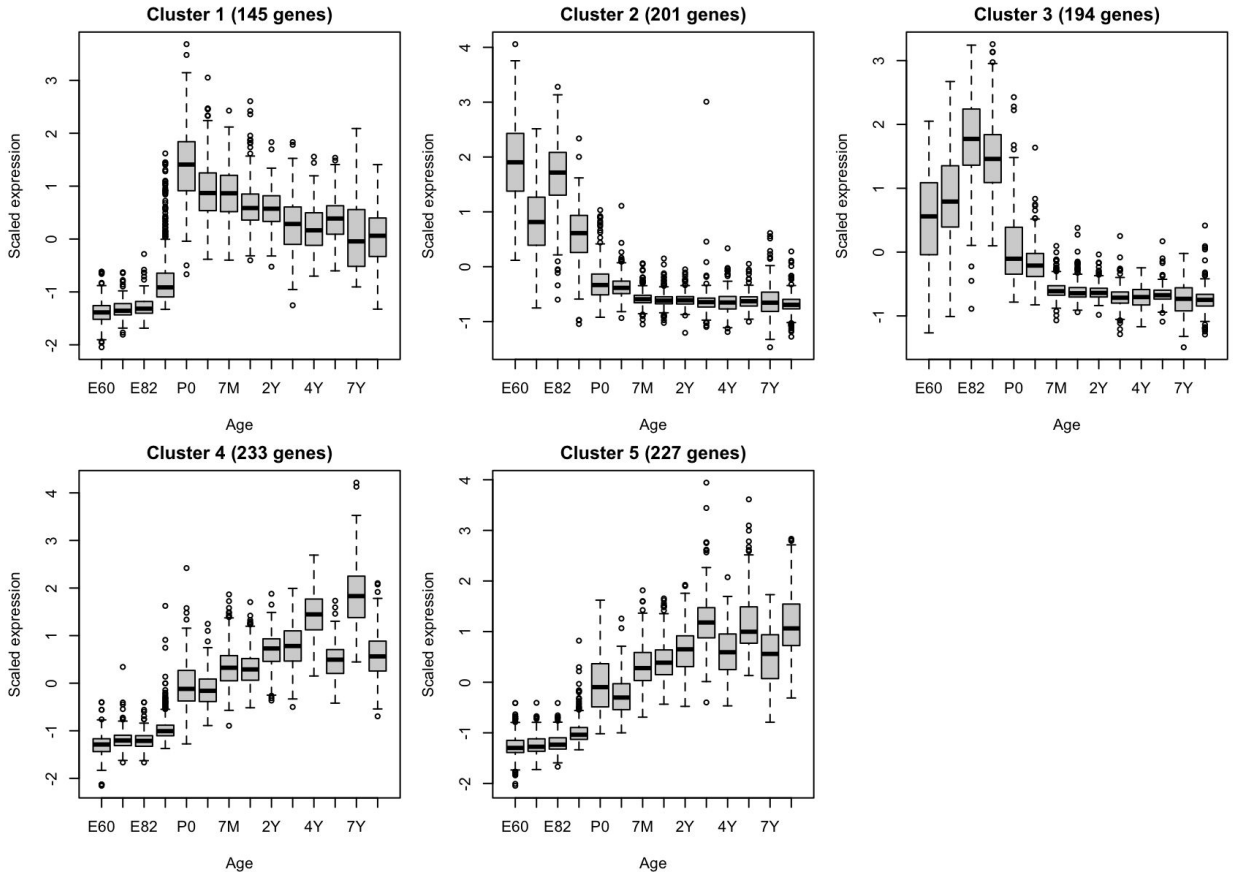
Do our samples reveal similar developmental processes?

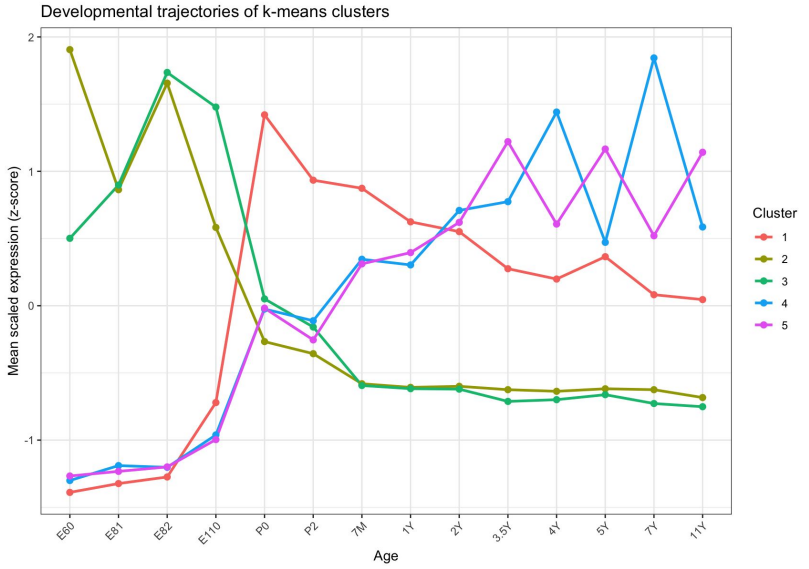
K-means Clustering and Enrichment Analysis



Clustering

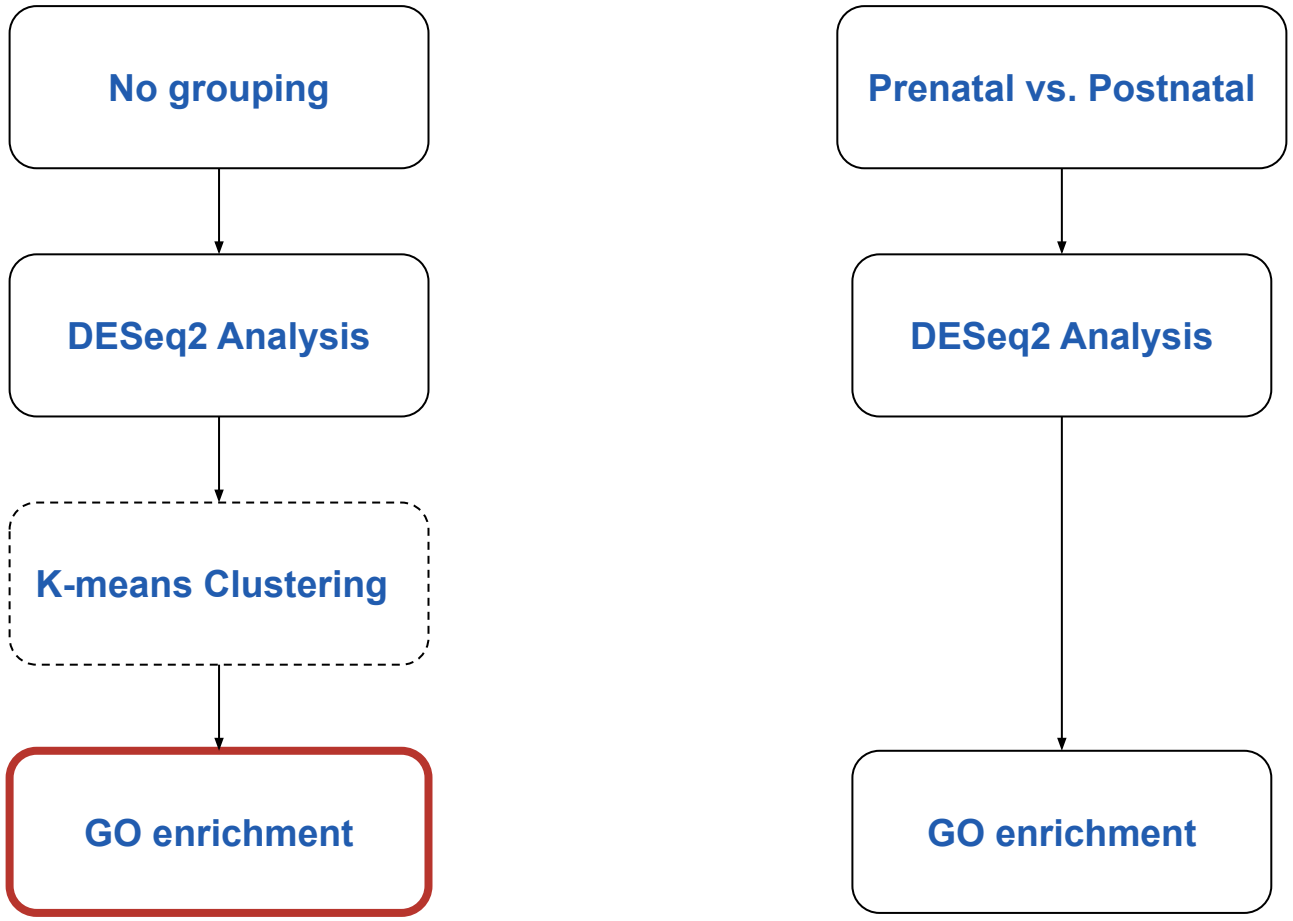
k-means Clusters: Expression Patterns



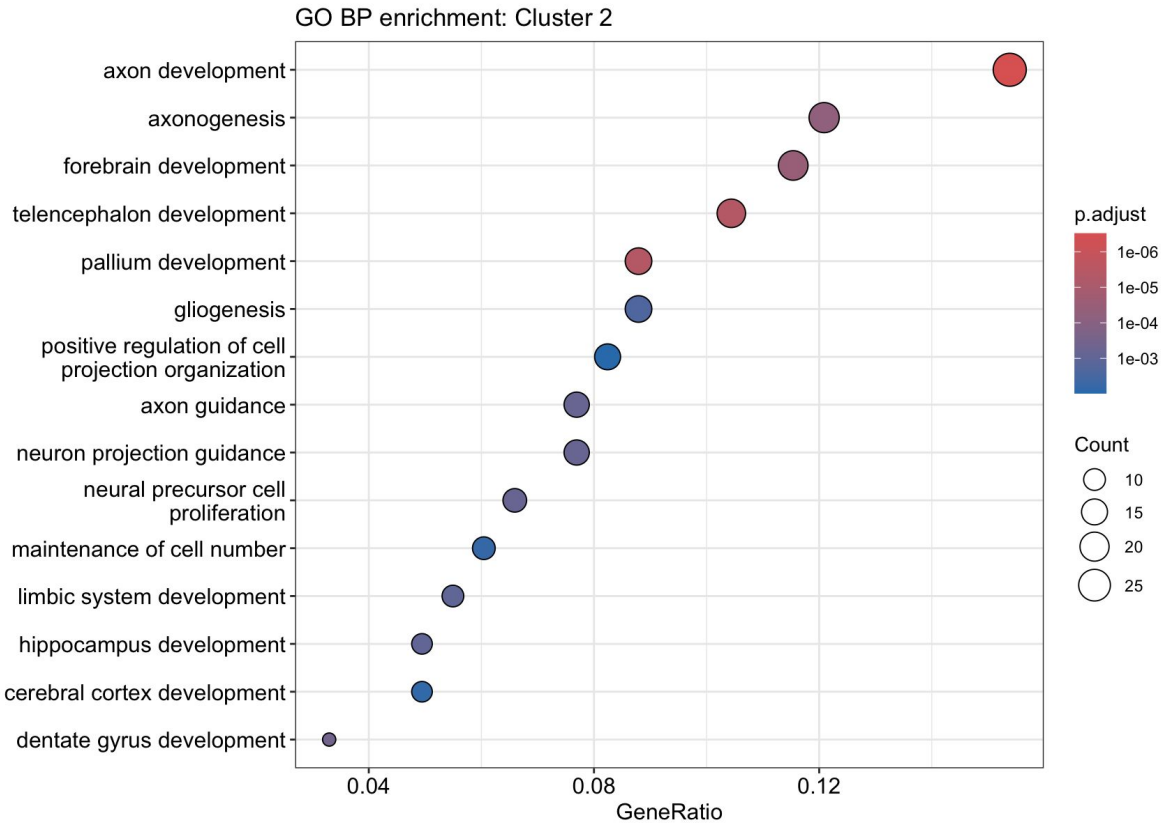


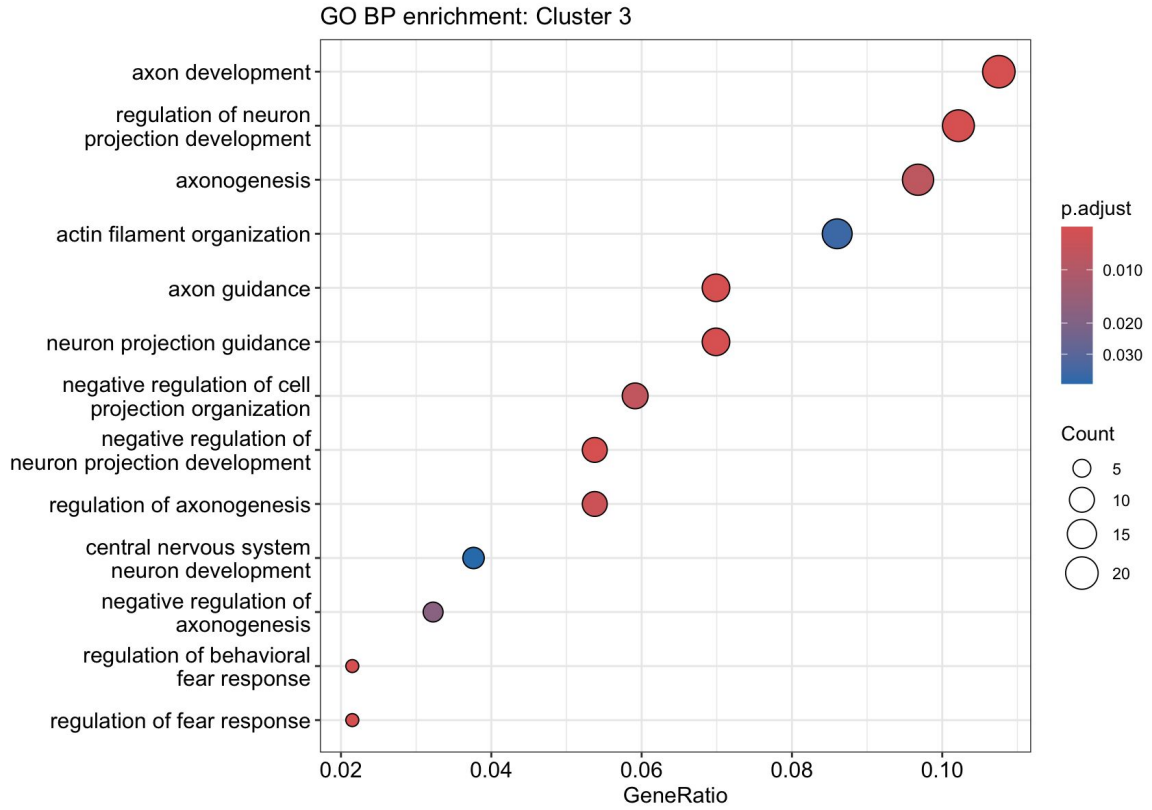
Cluster 2/3 high prenatally
Cluster 1 high postnatally
Cluster 4/5 rises at birth, high in adulthood

K-means Clustering and Enrichment Analysis



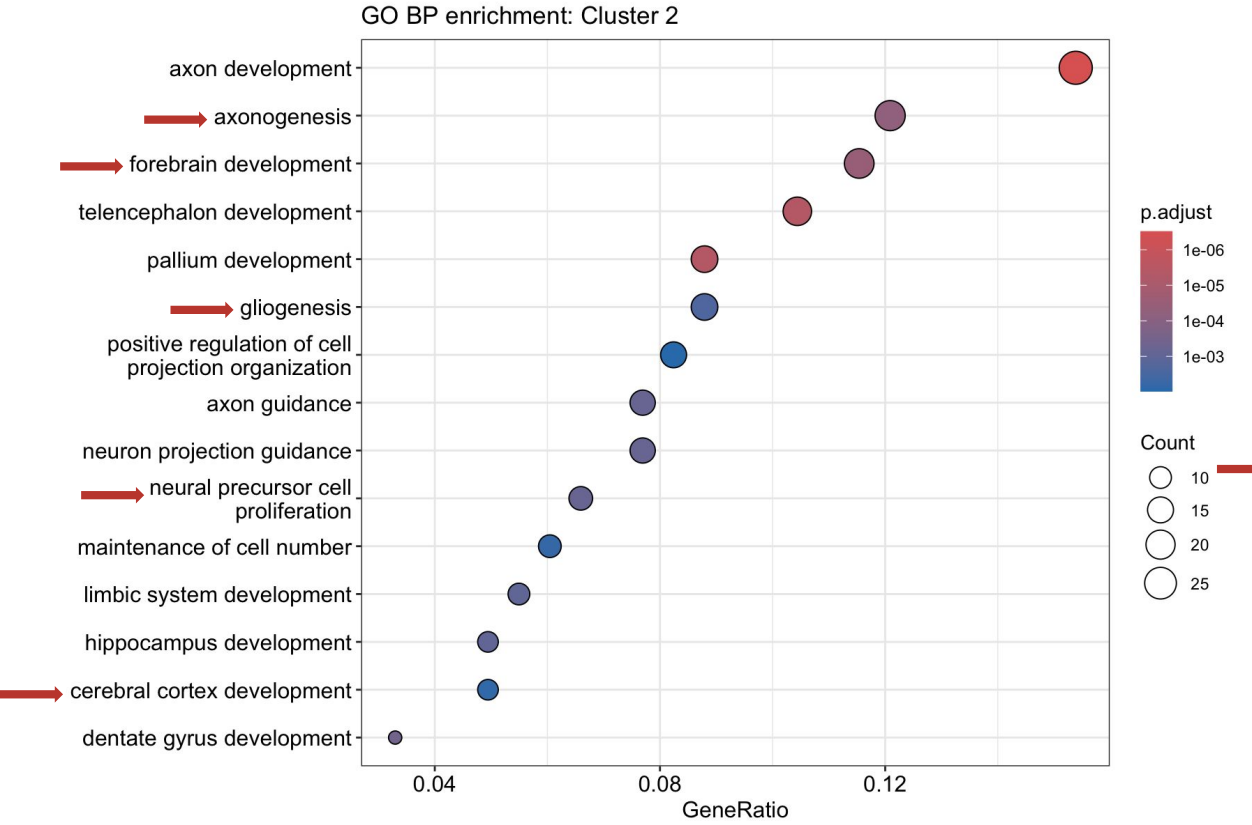
Clusters 2 & 3

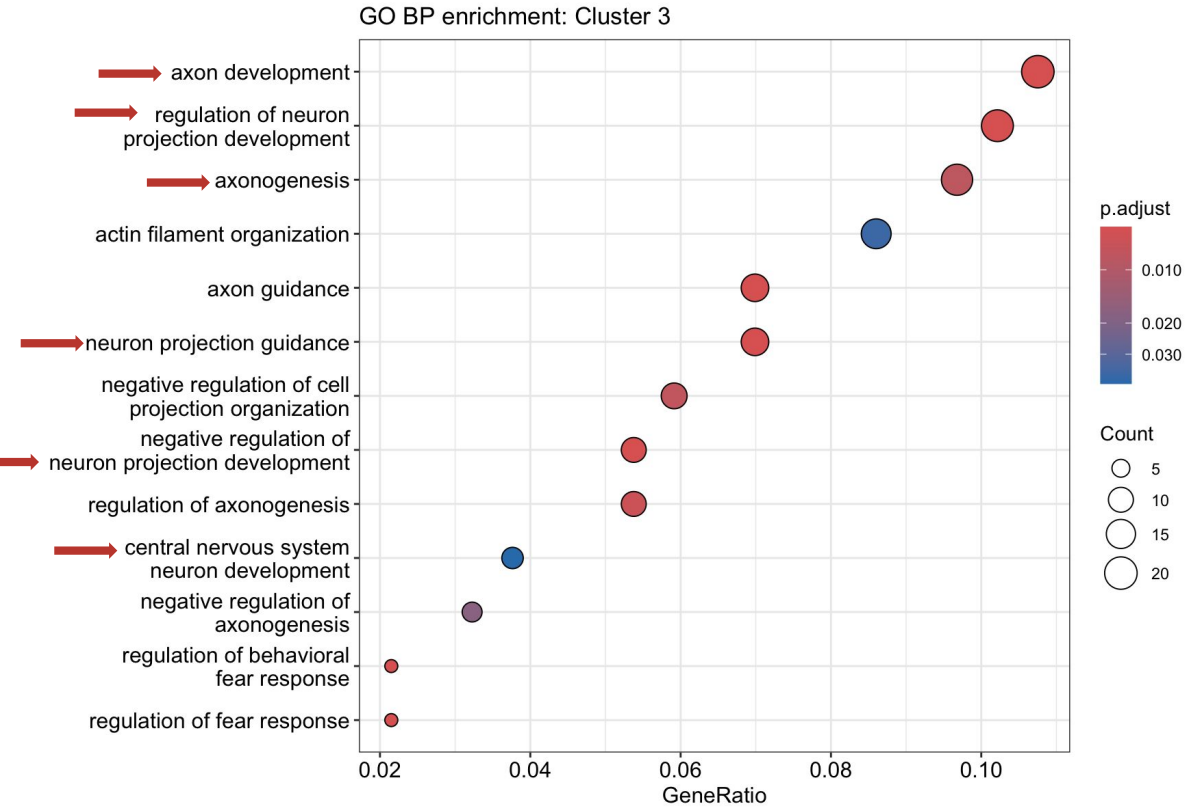

GO BP enrichment: Cluster 2
axon development
axonogenesis
forebrain development
telencephalon development
pallium development
gliogenesis
positive regulation of cell projection organization
axon guidance
neuron projection guidance
neural precursor cell proliferation
maintenance of cell number
limbic system development
hippocampus development
cerebral cortex development
dentate gyrus development
0.04
0.08
0.12
GeneRatio
p.adjust
1e-06
1e-05
1e-04
1e-03
Count
10
15
20
25


GO BP enrichment: Cluster 3
axon development
regulation of neuron projection development
axonogenesis
actin filament organization
axon guidance
neuron projection guidance
negative regulation of cell projection organization
negative regulation of neuron projection development
regulation of axonogenesis
central nervous system neuron development
negative regulation of axonogenesis
regulation of behavioral fear response
regulation of fear response
0.02
0.04
0.06
0.08
0.10
GeneRatio
p.adjust
0.010
0.020
0.030
Count
5
10
15
20

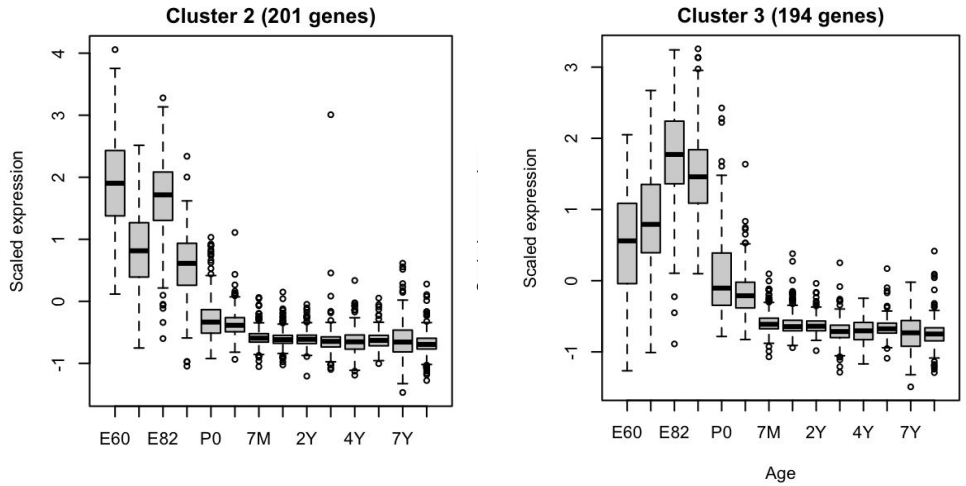
Clusters 2 & 3

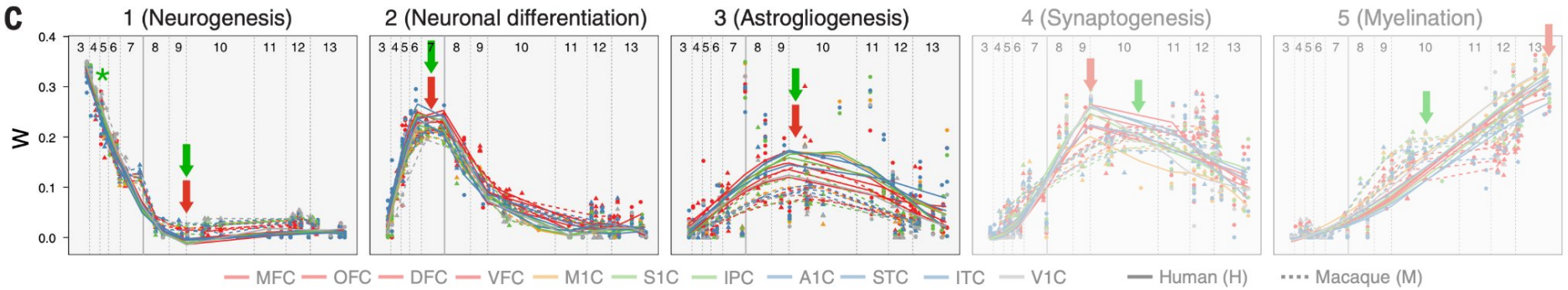
- → Neurogenesis
- → Neuronal differentiation
- → Axonogenesis
- → Gliogenesis



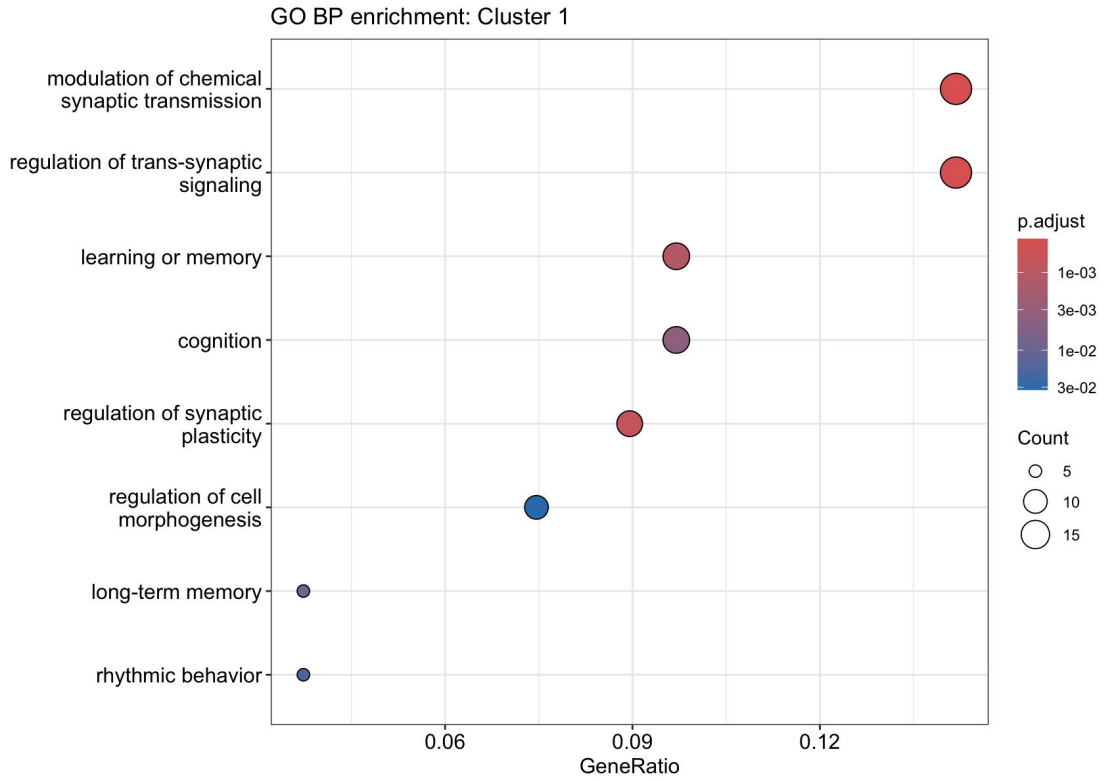


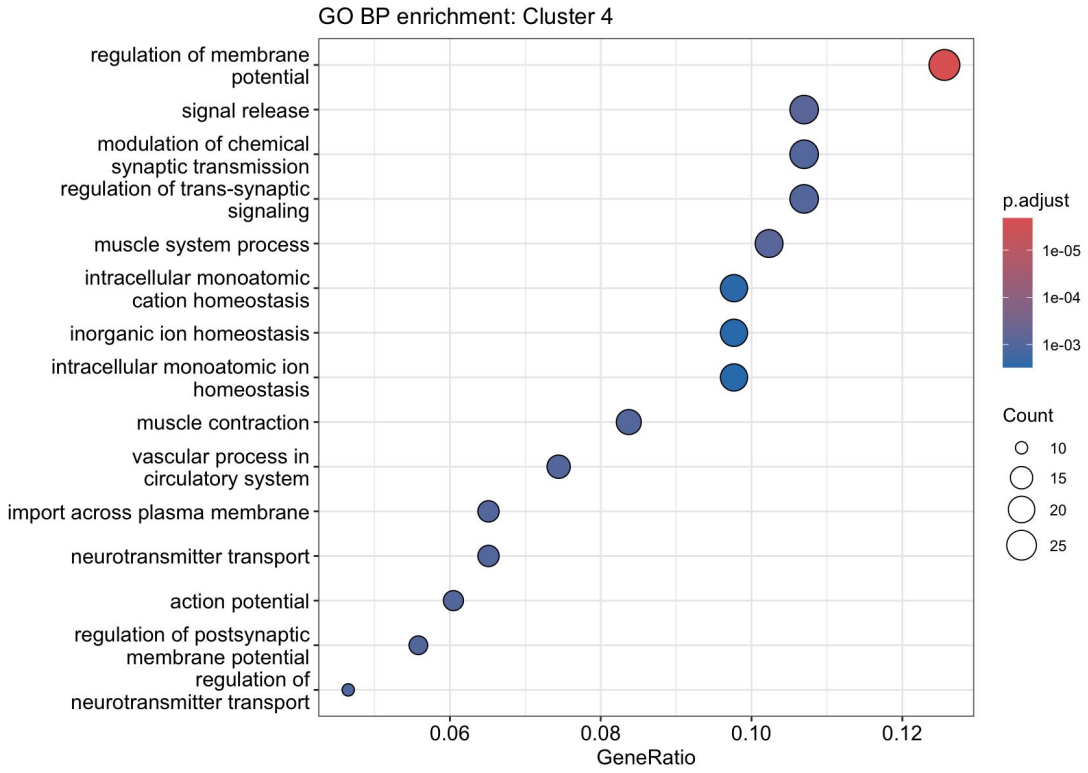
Clusters 2 & 3


Cluster 2 (201 genes)
Scaled expression
E60 E82 P0 7M 2Y 4Y 7Y
Cluster 3 (194 genes)
Scaled expression
E60 E82 P0 7M 2Y 4Y 7Y
Age


C
1 (Neurogenesis)
2 (Neuronal differentiation)
3 (Astrogliogenesis)
4 (Synaptogenesis)
5 (Myelination)
W
MFC OFC DFC VFC M1C S1C IPC A1C STC ITC V1C
Human (H)
Macaque (M)

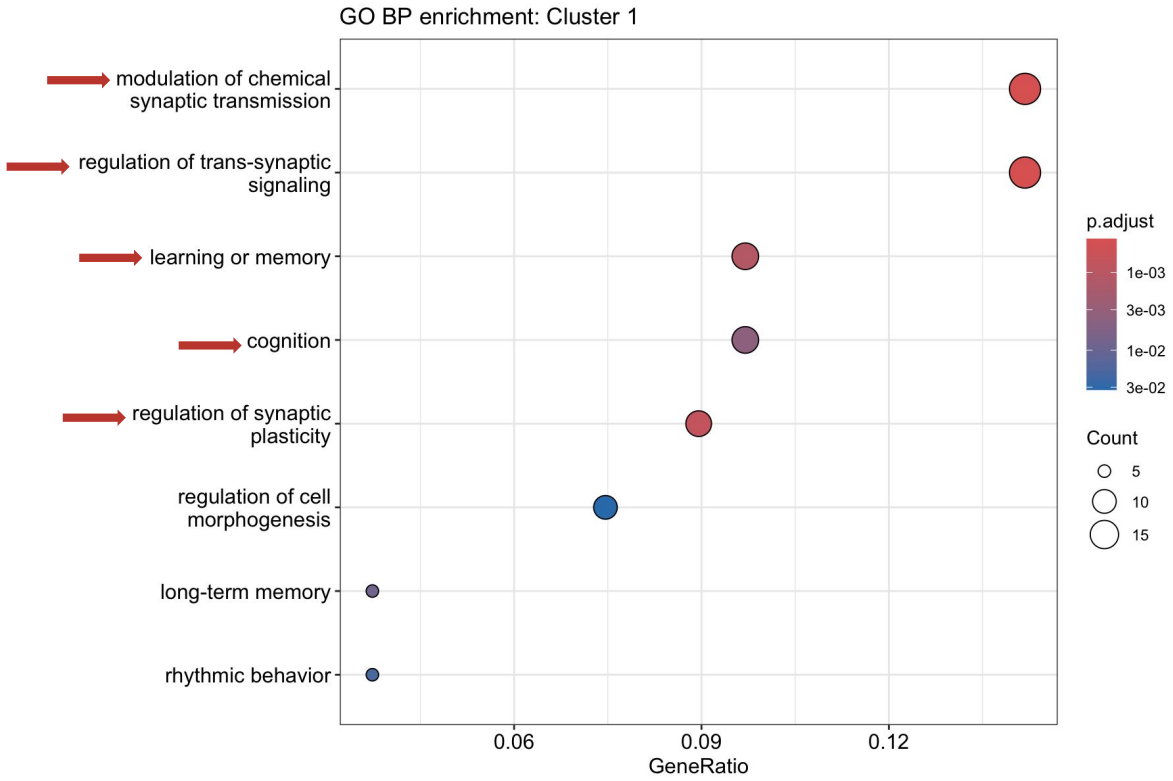
Clusters 1 & 4

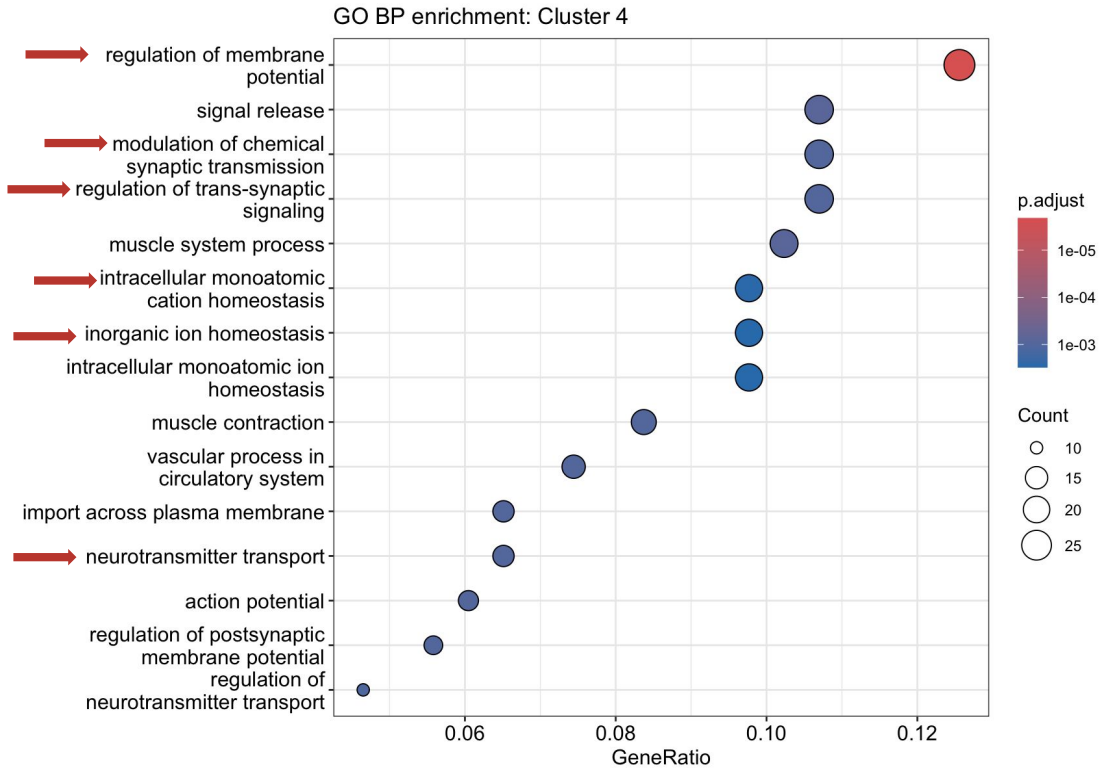

GO BP enrichment: Cluster 1
modulation of chemical synaptic transmission
regulation of trans-synaptic signaling
learning or memory
cognition
regulation of synaptic plasticity
regulation of cell morphogenesis
long-term memory
rhythmic behavior
0.06
0.09
0.12
GeneRatio
p.adjust
1e-03
3e-03
1e-02
3e-02
Count
5
10
15


GO BP enrichment: Cluster 4
regulation of membrane potential
signal release
modulation of chemical synaptic transmission
regulation of trans-synaptic signaling
muscle system process
intracellular monoatomic cation homeostasis
inorganic ion homeostasis
intracellular monoatomic ion homeostasis
muscle contraction
vascular process in circulatory system
import across plasma membrane
neurotransmitter transport
action potential
regulation of postsynaptic membrane potential
regulation of neurotransmitter transport
0.06
0.08
0.10
0.12
GeneRatio
p.adjust
1e-05
1e-04
1e-03
Count
10
15
20
25

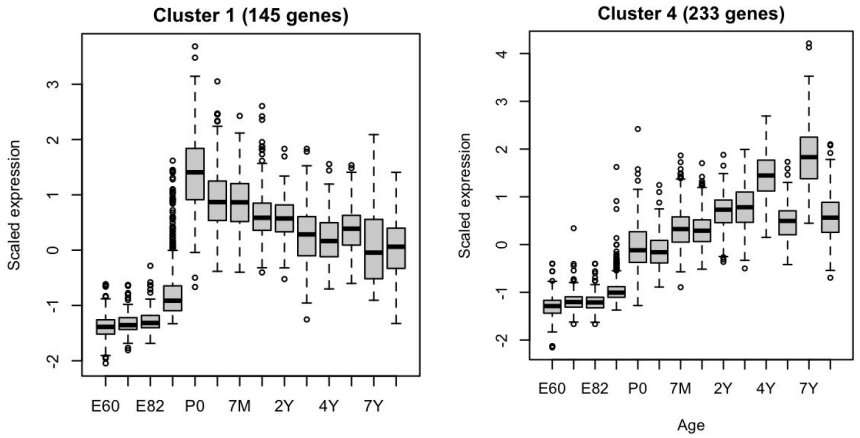
Clusters 1 & 4

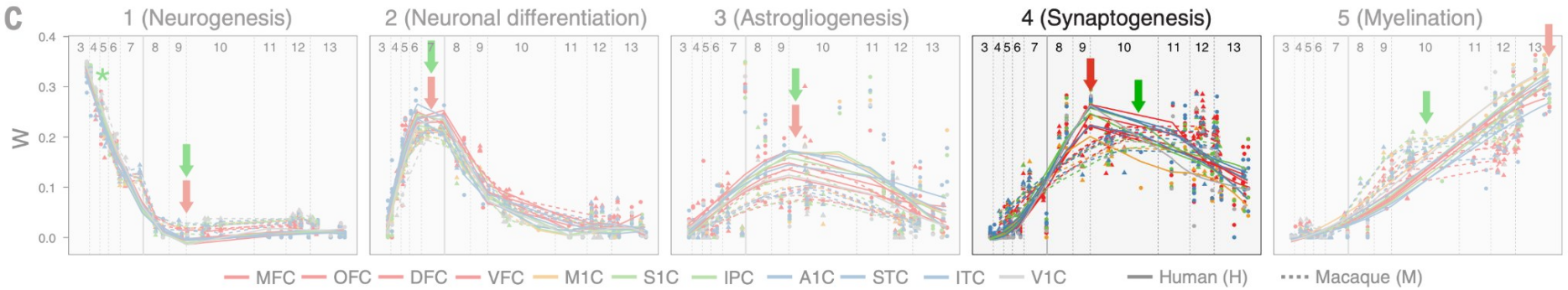
→ Early and late synaptogenesis



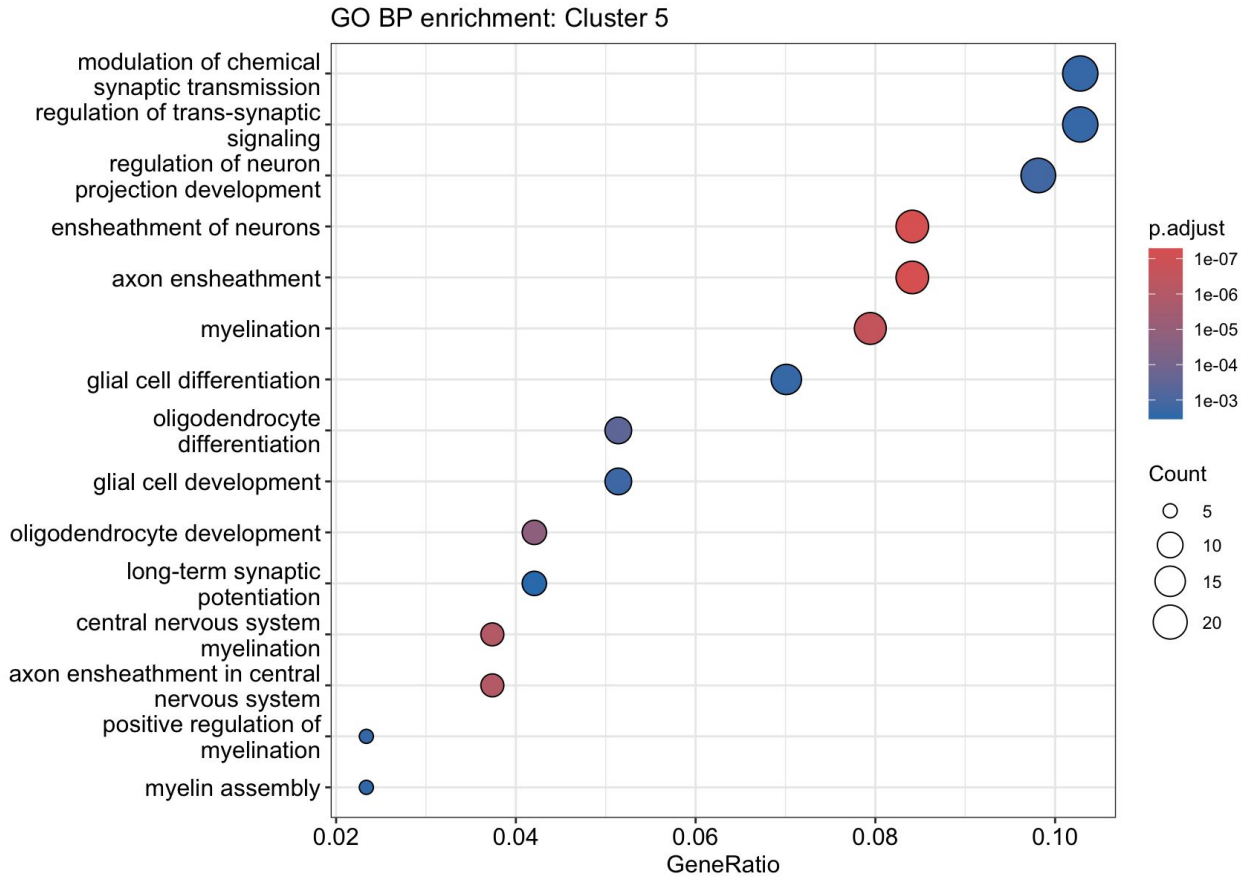


Clusters 1 & 4

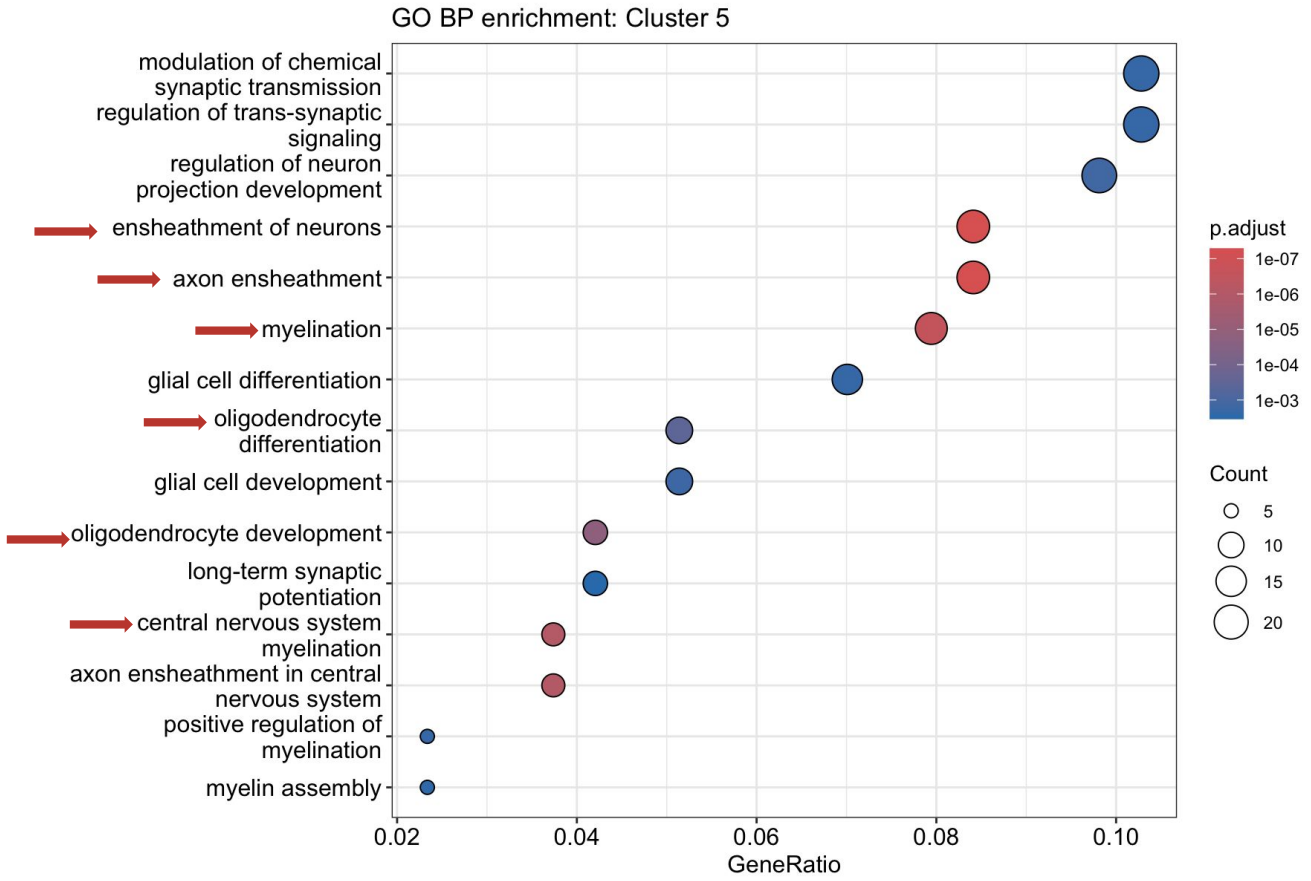

Cluster 1 (145 genes)
Scaled expression
E60 E82 P0 7M 2Y 4Y 7Y
Cluster 4 (233 genes)
Scaled expression
E60 E82 P0 7M 2Y 4Y 7Y
Age


C
1 (Neurogenesis)
2 (Neuronal differentiation)
3 (Astrogliogenesis)
4 (Synaptogenesis)
5 (Myelination)
W
MFC OFC DFC VFC M1C S1C IPC A1C STC ITC V1C
Human (H)
Macaque (M)

Cluster 5


GO BP enrichment: Cluster 5
modulation of chemical synaptic transmission
regulation of trans-synaptic signaling
regulation of neuron projection development
ensheathment of neurons
axon ensheathment
myelination
glial cell differentiation
oligodendrocyte differentiation
glial cell development
oligodendrocyte development
long-term synaptic potentiation
central nervous system myelination
axon ensheathment in central nervous system
positive regulation of myelination
myelin assembly
0.02
0.04
0.06
0.08
0.10
GeneRatio
p.adjust
1e-07
1e-06
1e-05
1e-04
1e-03
Count
5
10
15
20

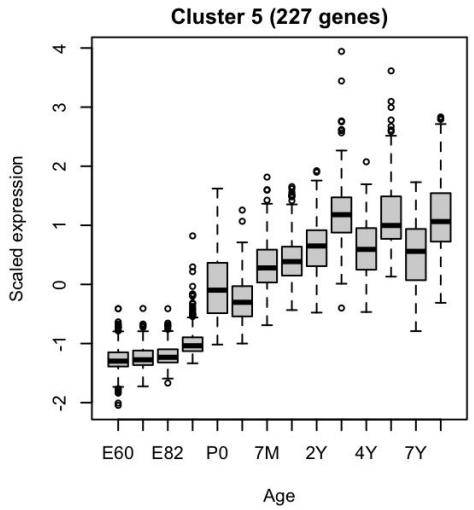
Cluster 5

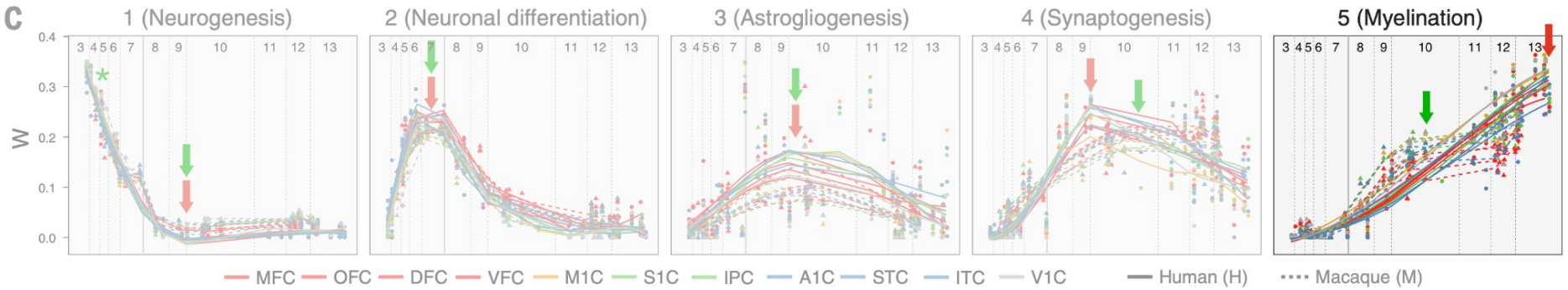


➔ Oligodendrocyte maturation

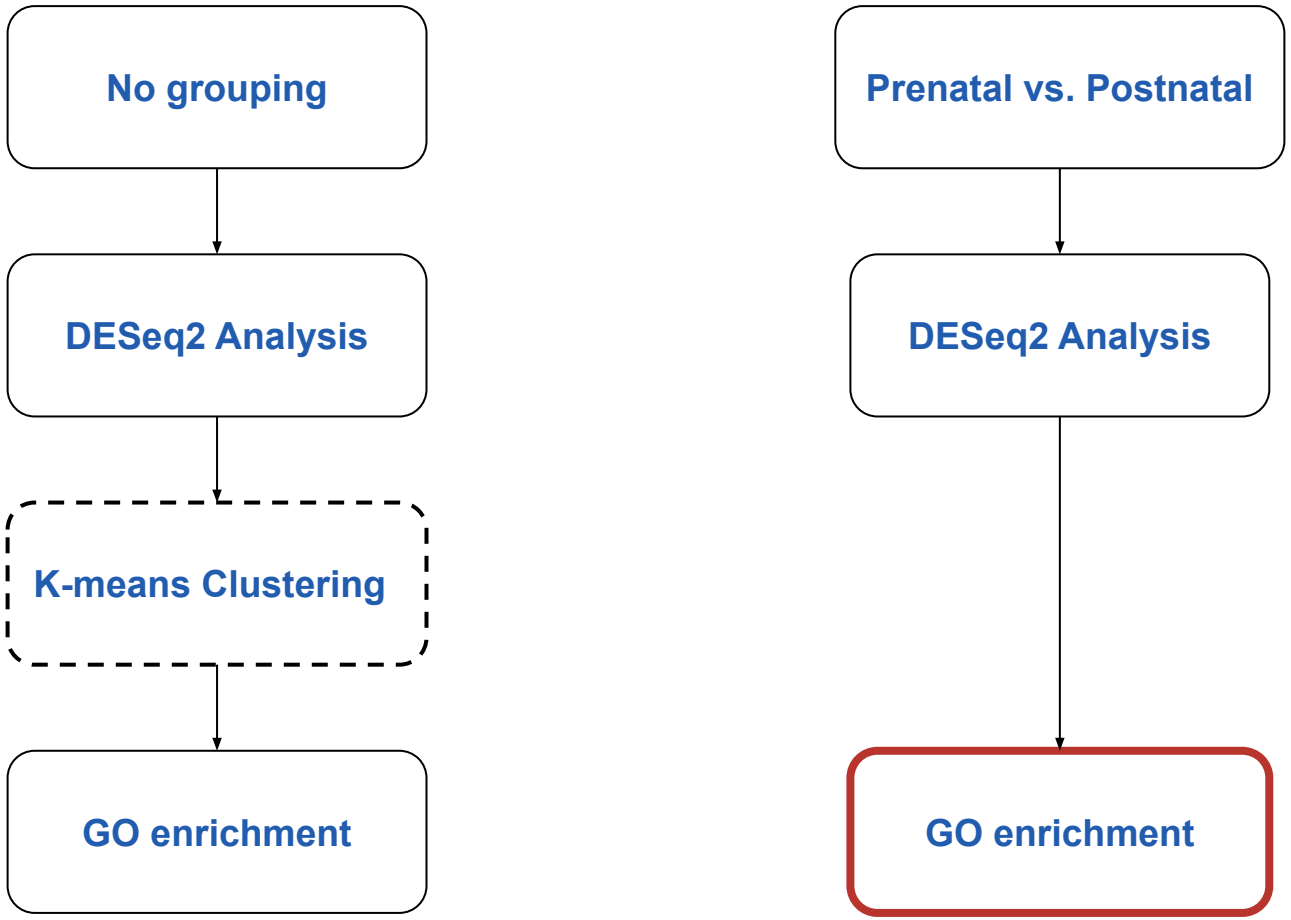
➔ **Myelination !**

Clusters 5


Cluster 5 (227 genes)
Scaled expression
-2
-1
0
1
2
3
4
E60
E82
P0
7M
2Y
4Y
7Y
Age


C
1 (Neurogenesis)
2 (Neuronal differentiation)
3 (Astrogliogenesis)
4 (Synaptogenesis)
5 (Myelination)
W
0.0
0.1
0.2
0.3
0.4
3 4 5 6 7 8 9 10 11 12 13
MFC
OFC
DFC
VFC
M1C
S1C
IPC
A1C
STC
ITC
V1C
Human (H)
Macaque (M)

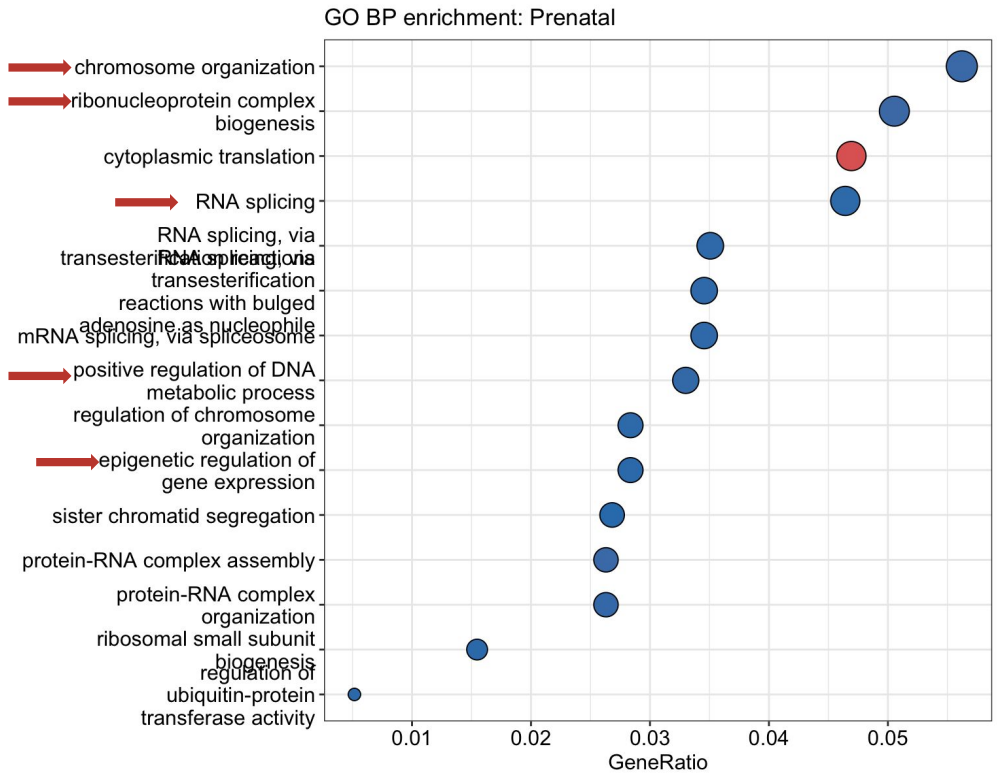
K-means Clustering and Enrichment Analysis

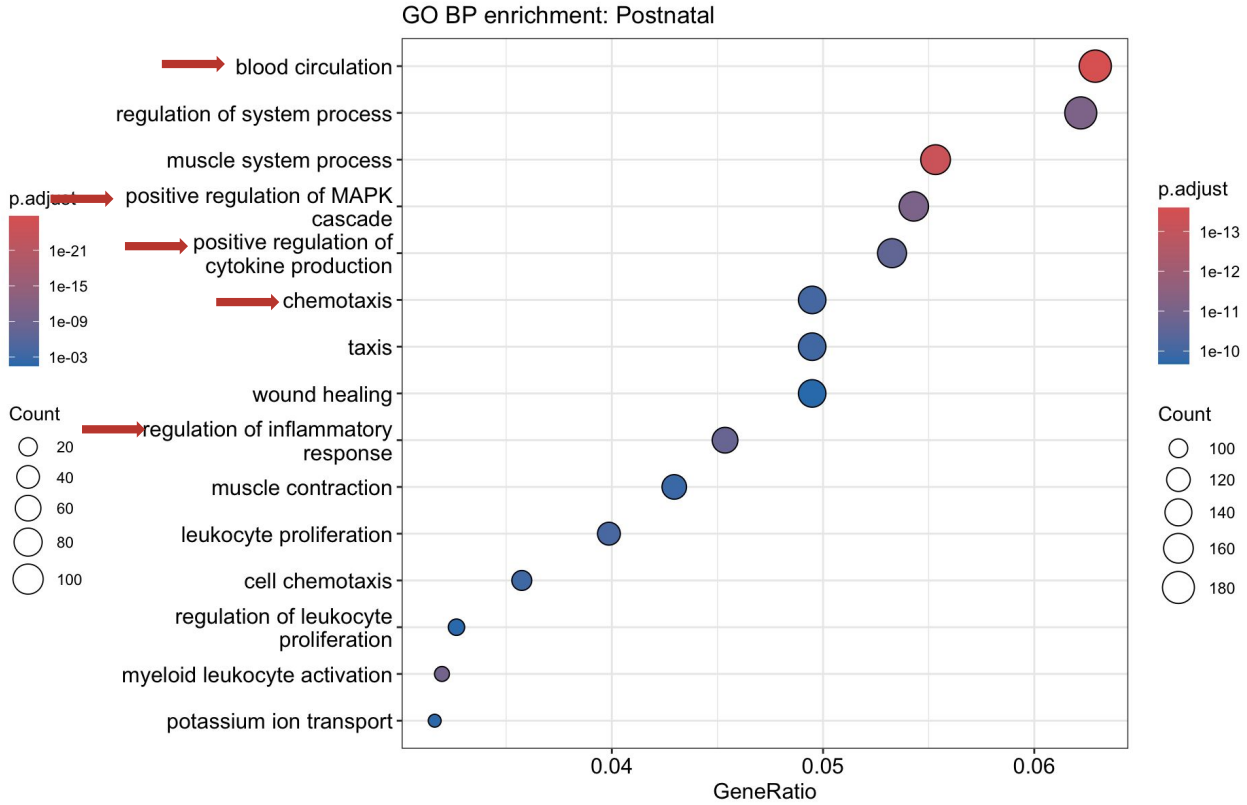


DESeq2 differential expression (Prenatal vs Postnatal)

- ➔ Nuclear and transcriptiona processes
- ➔ Signaling-related processes

Enrichment Analysis





Conclusion

Conclusion

- **Developmental age is a key driver of transcriptomic variation**, with a clear separation between prenatal and postnatal samples

- When treating development **without grouping**, we recovered **specific, stage-resolved biological programs**, including neurogenesis, synaptogenesis, and myelination

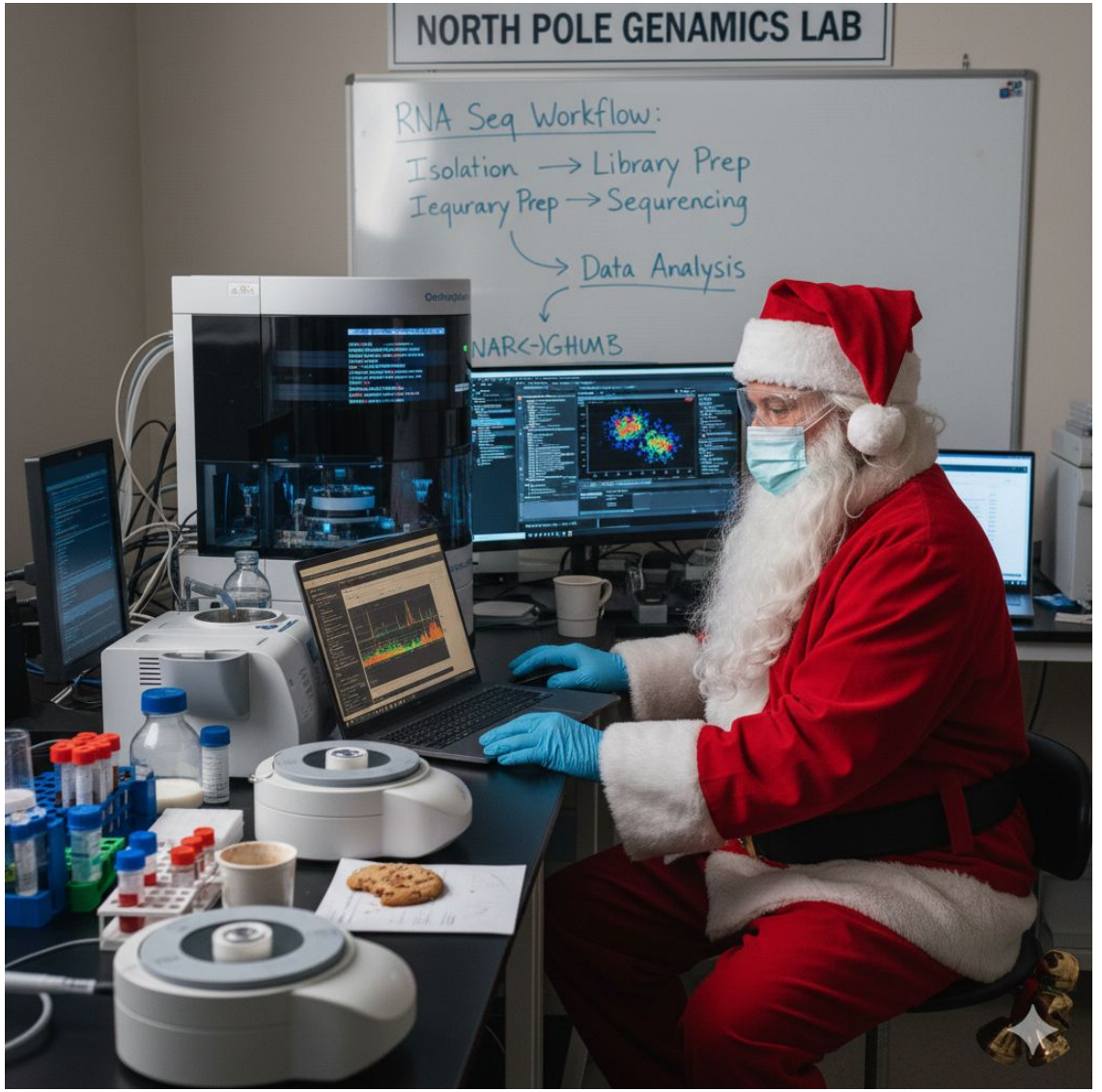
- In contrast, **grouping samples into prenatal versus postnatal stages** captured global developmental differences but masked finer temporal structure

Recommendations

- **Move from bulk RNA-seq to single-cell transcriptomics** to resolve cell-type–specific developmental programs
- **Increase sample size and improve age coverage** to strengthen statistical power and better capture developmental transitions
- **Extend the analysis to cross-species comparisons (e.g., human vs. macaque)** to identify conserved and species specific transcriptional programs during brain development

Thank you

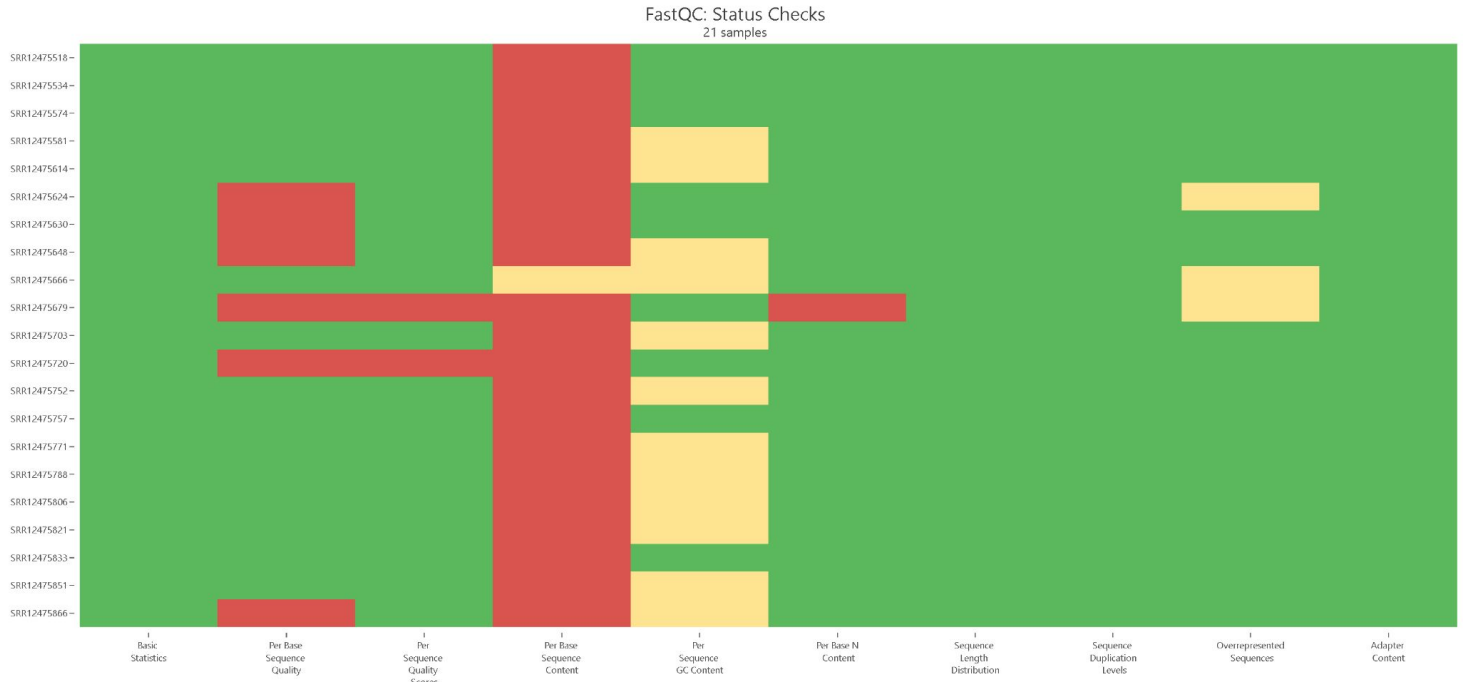
Happy holidays!



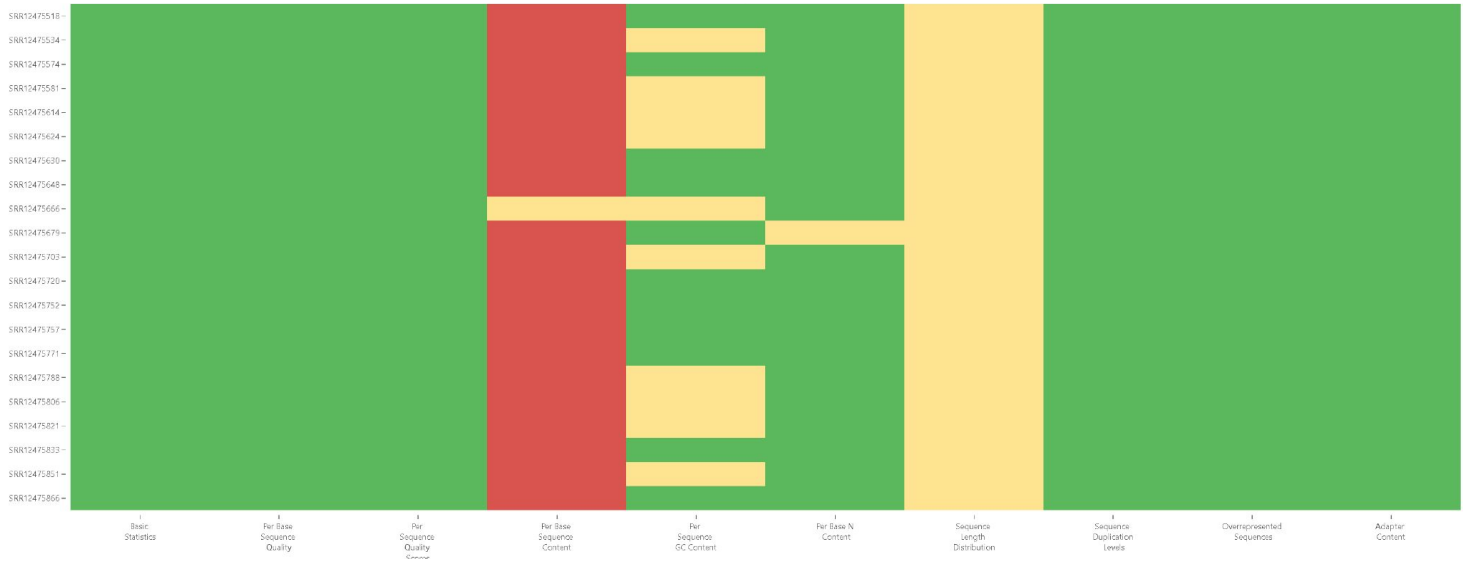
https://gemini.google.com/

Supplementary

Before



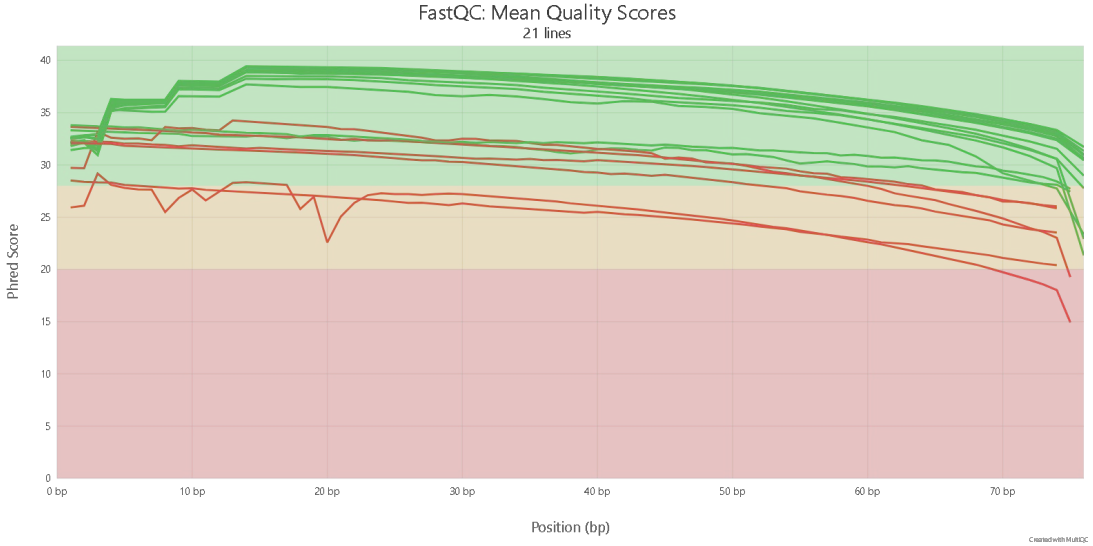
After



MultiQC Outputs for Raw Samples

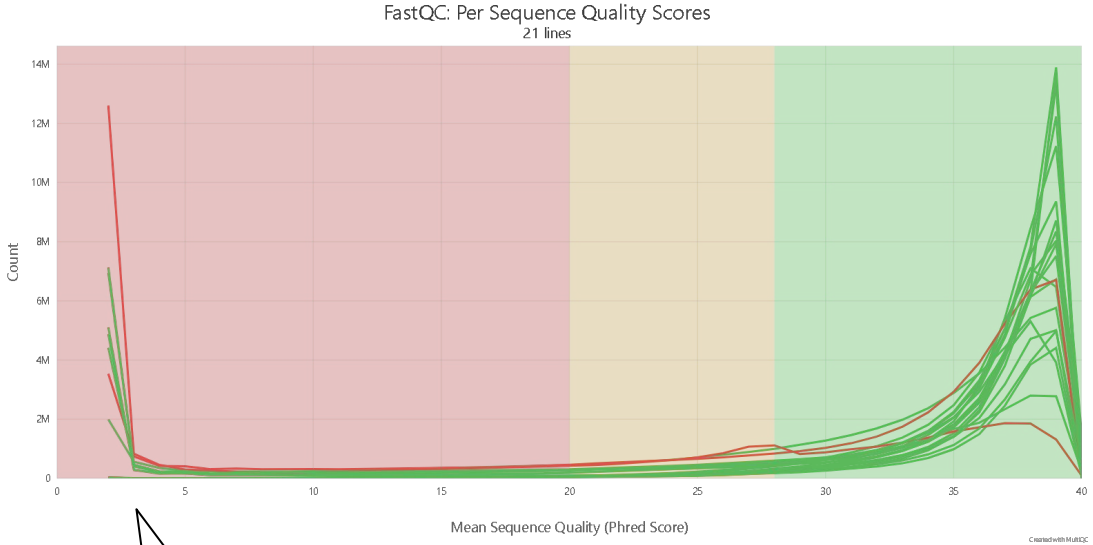
Sequence Quality Histograms

15 6



Per Sequence Quality Scores

19 2



Cluster of low-quality reads

Leads to across-the-board bad scores

A Note on Phred Scores/Encodings

Basic Statistics

Measure	Value
Filename	SRR12475720.fastq
File type	Conventional base calls
Encoding	Illumina 1.5
Total Sequences	58302814
Total Bases	4.3 Gbp
Sequences flagged as poor quality	0
Sequence length	74
%GC	46

Basic Statistics

Measure	Value
Filename	SRR12475752.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	32215873
Total Bases	2.4 Gbp
Sequences flagged as poor quality	0
Sequence length	76
%GC	47

We noticed that our fastq files had heterogenous encodings (this was a super fun problem to identify and debug):

- Illumina 1.5/phred64 (older)
- Illumina 1.9/phred33 (newer)

Illumina 1.5 flags reads/bases that should be ignored with a “B” character. This also corresponds to a phred of 2 in that encoding.

(**seqtk** (https://github.com/lh3/seqtk) was used to convert from phred64 to phred33 for the analysis)

Preprocessing Parameters (fastp)

```
fastp \
-i "$fq" \
-o "$out" \
--html "${outdir}/${base}.html" \
--json "${outdir}/${base}.json" \
-w 8 \
-q 30 \
-u 30 \
--cut_front \
--cut_tail \
--cut_window_size 4 \
--cut_mean_quality 30 \
--length_required 25
```

Read Filtering:

- At least 30% of bases needed to have Phred ≥ 30

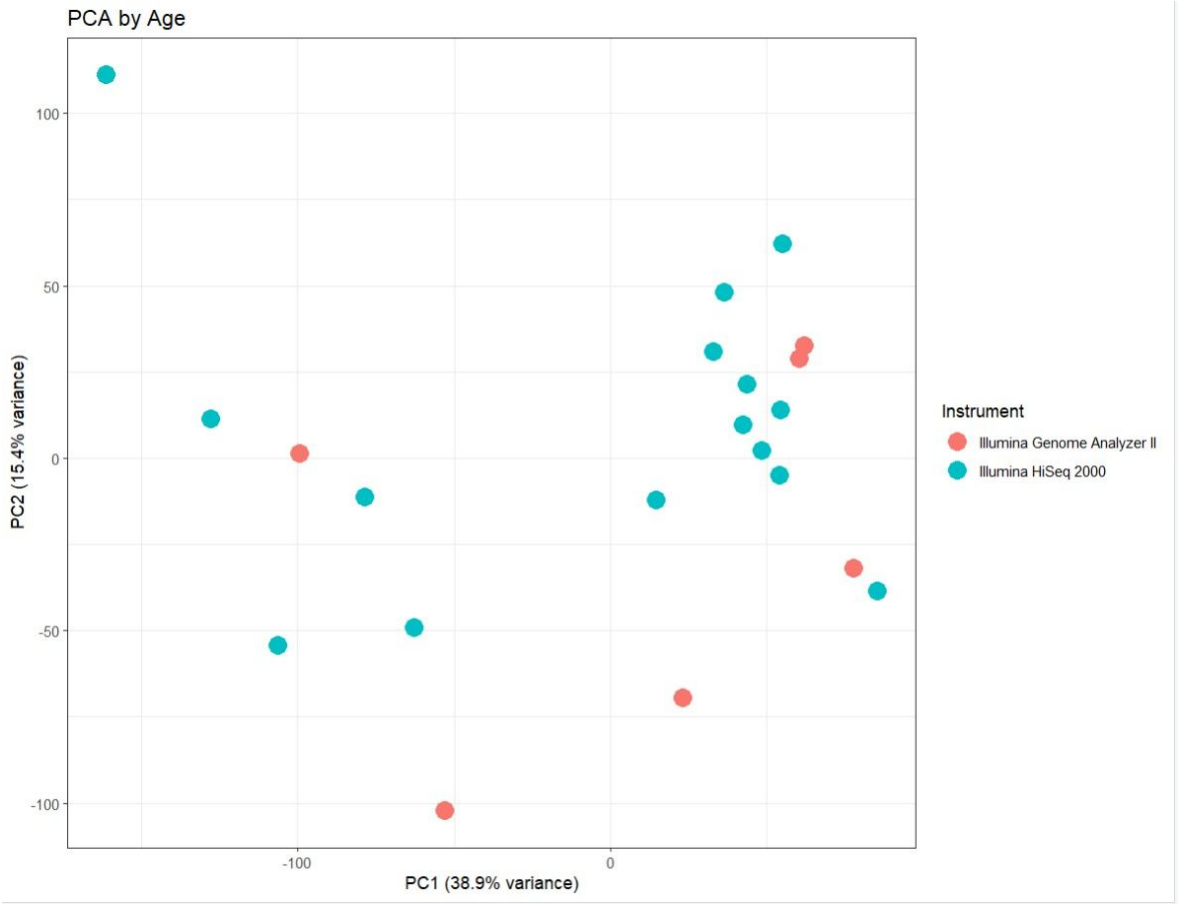
Adapter Trimming:

- Identifies and removes common Illumina adapters by default

Trimming by Quality Score:

- Cut at 5’ and 3’
- Window size of 4 bases and mean quality in the window requires Phred ≥ 30

Batch Effects



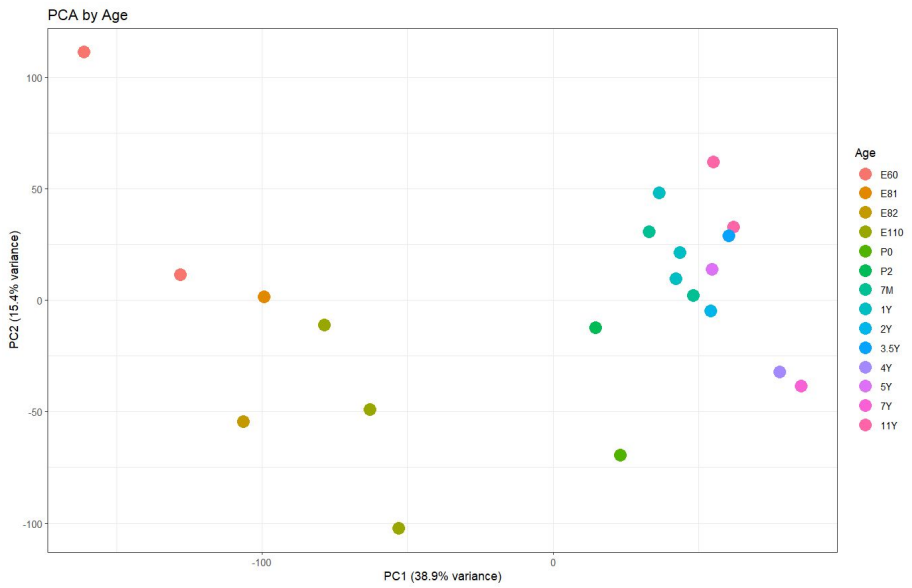
Our sample were sequenced using 2 different instruments

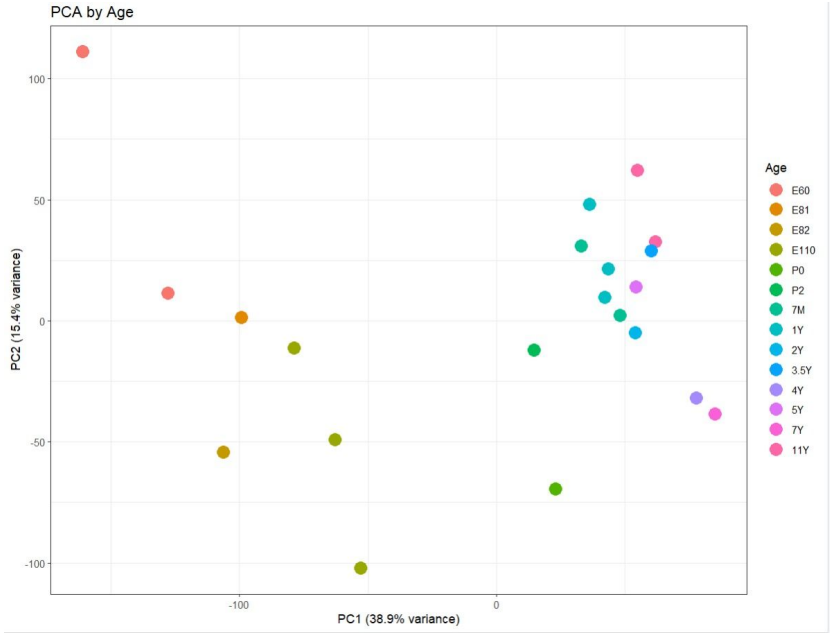
We were worried about potential batch effects

That being said it is quite clear that instrument type does not seem to clearly cluster (compare the mean vector of the instrument groups)

Batch Effect Correction (Combat-Seq)

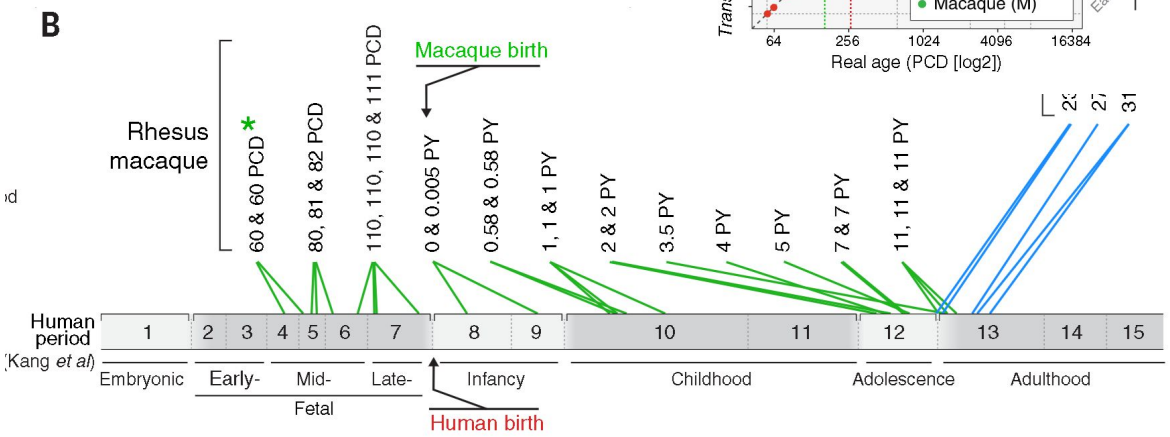
Zhang, Yuqing, Giovanni Parmigiani, and W. Evan Johnson. "ComBat-seq: batch effect adjustment for RNA-seq count data." *NAR genomics and bioinformatics* 2.3 (2020): lqaa078.

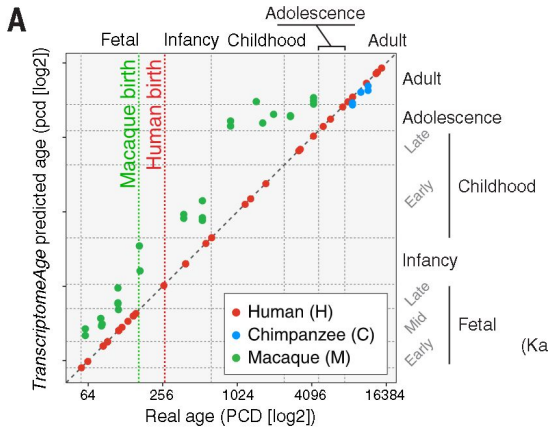




Age Periods defined by Kang et. al

They matched the ages using their own TranscriptomeAge Algorithm





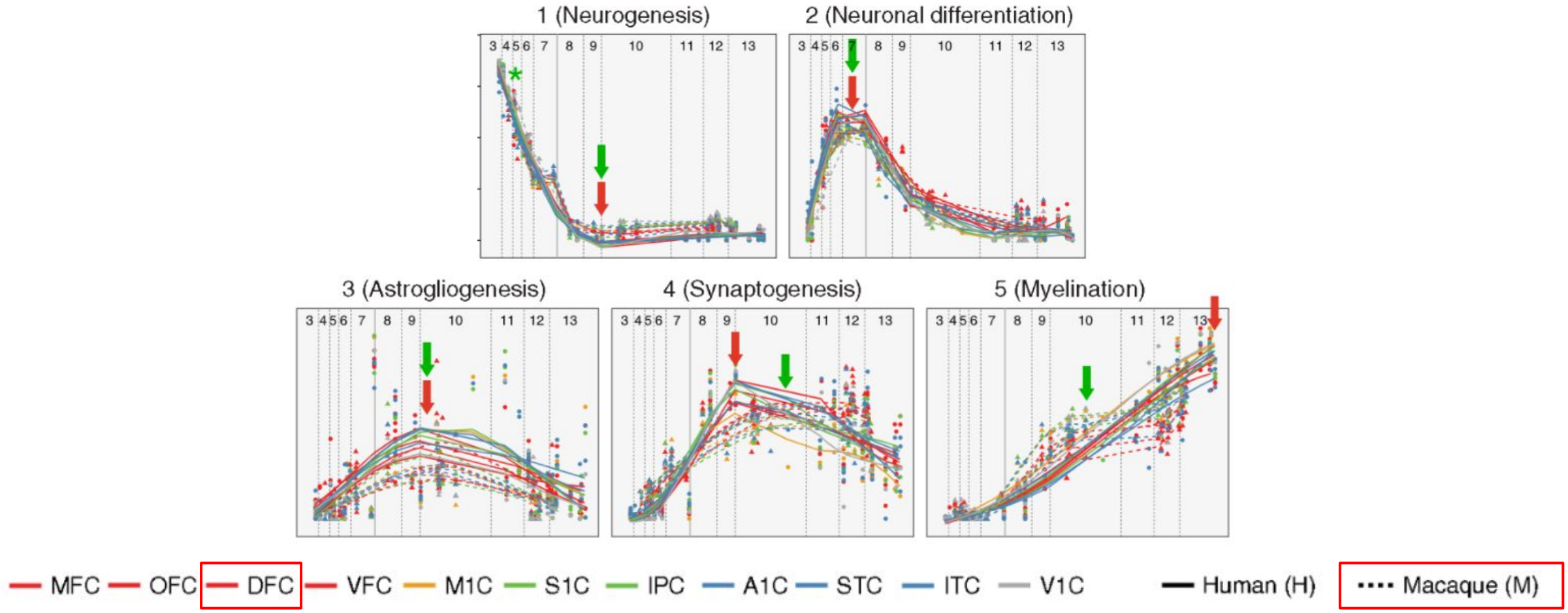
Zhu, Y., et al. "Spatiotemporal transcriptomic divergence across human and macaque brain development. Science 362, eaat8077." 2018,

Period	Description	Age
1	Embryonic	4 PCW ≤ Age < 8 PCW
2	Early fetal	8 PCW ≤ Age < 10 PCW
3	Early fetal	10 PCW ≤ Age < 13 PCW
4	Early mid-fetal	13 PCW ≤ Age < 16 PCW
5	Early mid-fetal	16 PCW ≤ Age < 19 PCW
6	Late mid-fetal	19 PCW ≤ Age < 24 PCW
7	Late fetal	24 PCW ≤ Age <38 PCW
8	Neonatal and early infancy	0 M (birth) ≤ Age <6 M
9	Late infancy	6 M ≤ Age < 12 M
10	Early childhood	1 Y ≤ Age <6 Y
11	Middle and late childhood	6 Y ≤ Age < 12 Y
12	Adolescence	12 Y ≤ Age < 20 Y
13	Young adulthood	20 Y ≤ Age < 40 Y
14	Middle adulthood	40 Y ≤ Age < 60 Y
15	Late adulthood	60 Y ≤ Age

M, postnatal months; PCW, post-conceptional weeks; Y, postnatal years.

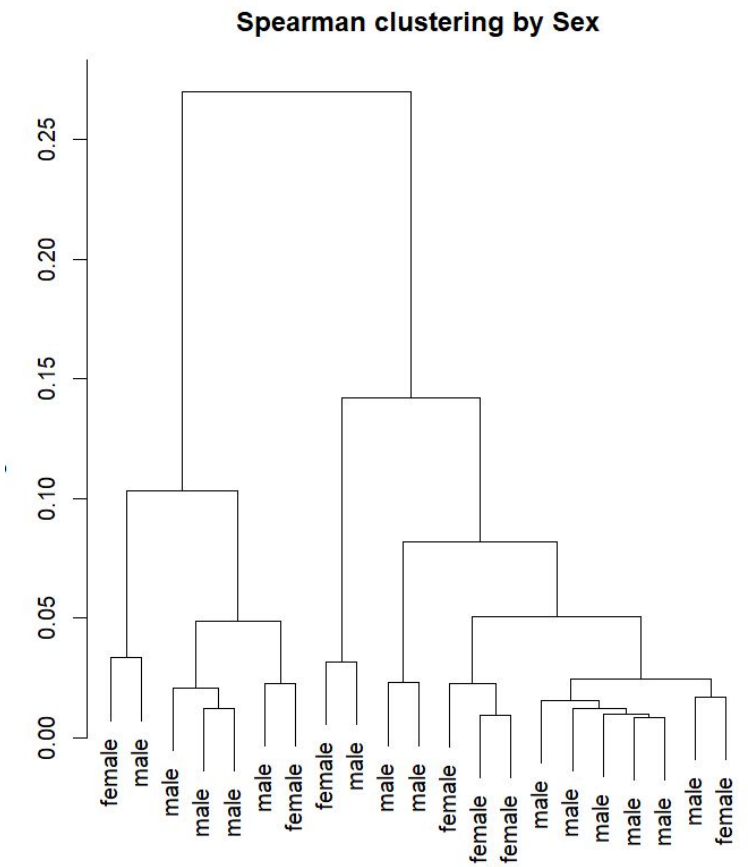
Kang, Hyo Jung, et al. "Spatio-temporal transcriptome of the human brain." *Nature* 478.7370 (2011): 483-489.

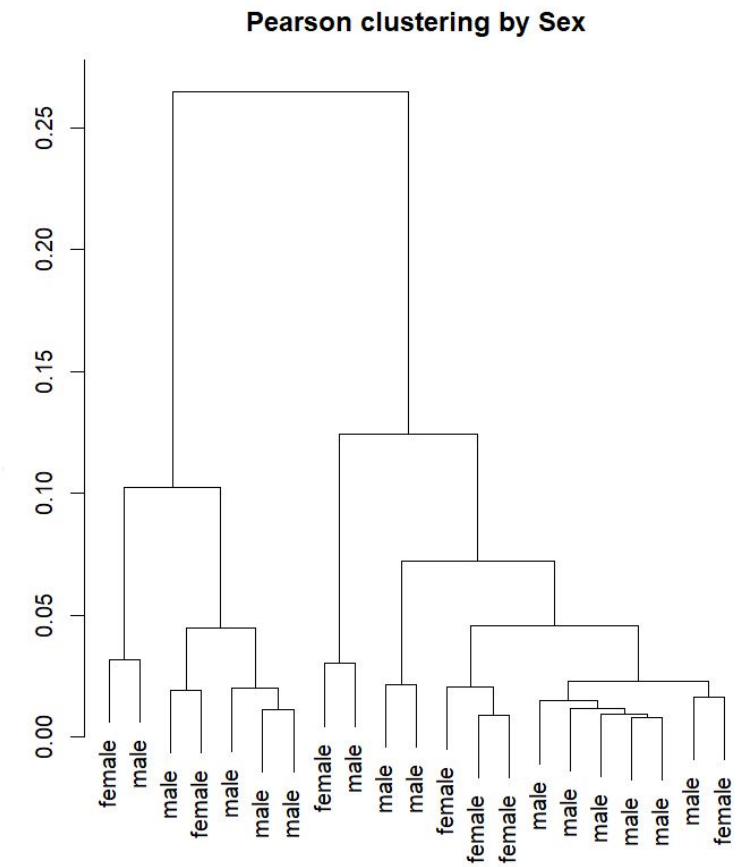
Our Paper: 5 Transcriptomic Profiles

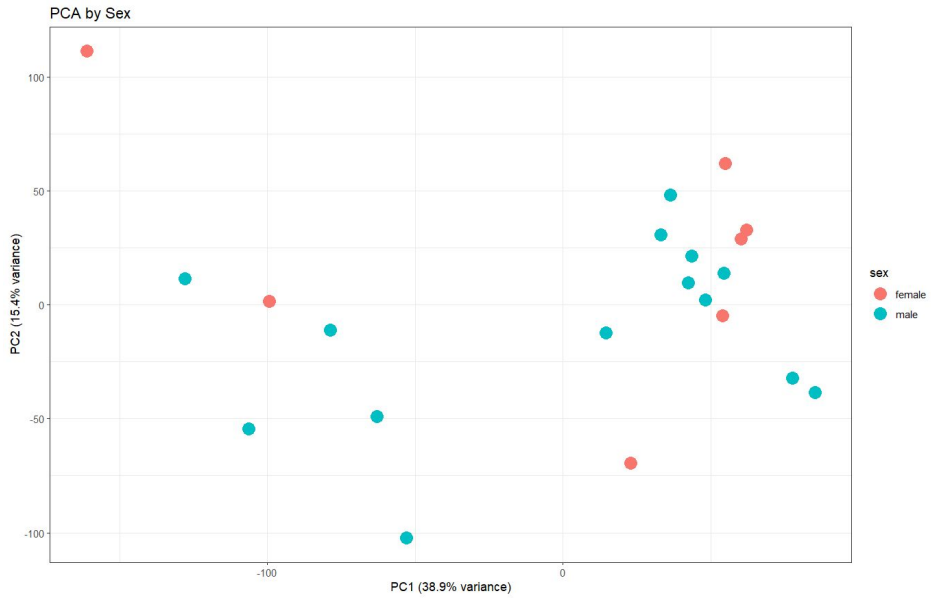


Zhu, Ying, et al. "Spatiotemporal transcriptomic divergence across human and macaque brain development." *Science* 362.6420 (2018): eaat8077.

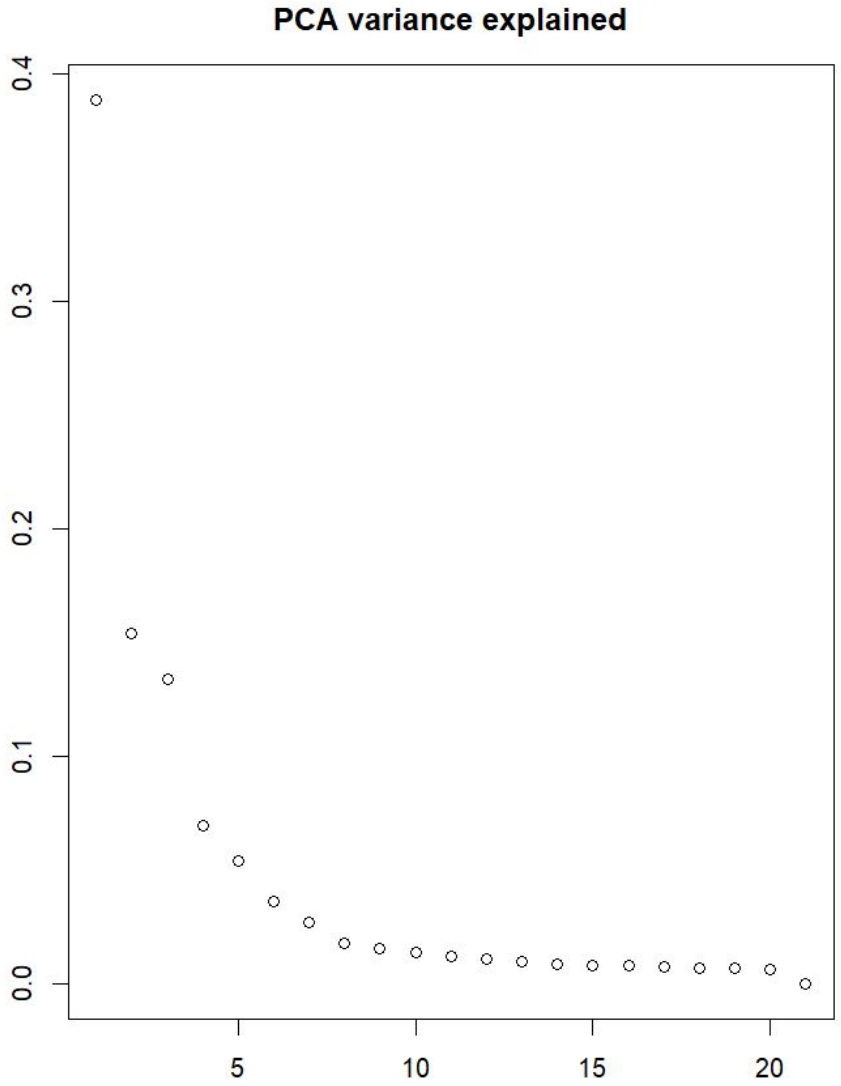
Clustering by sex



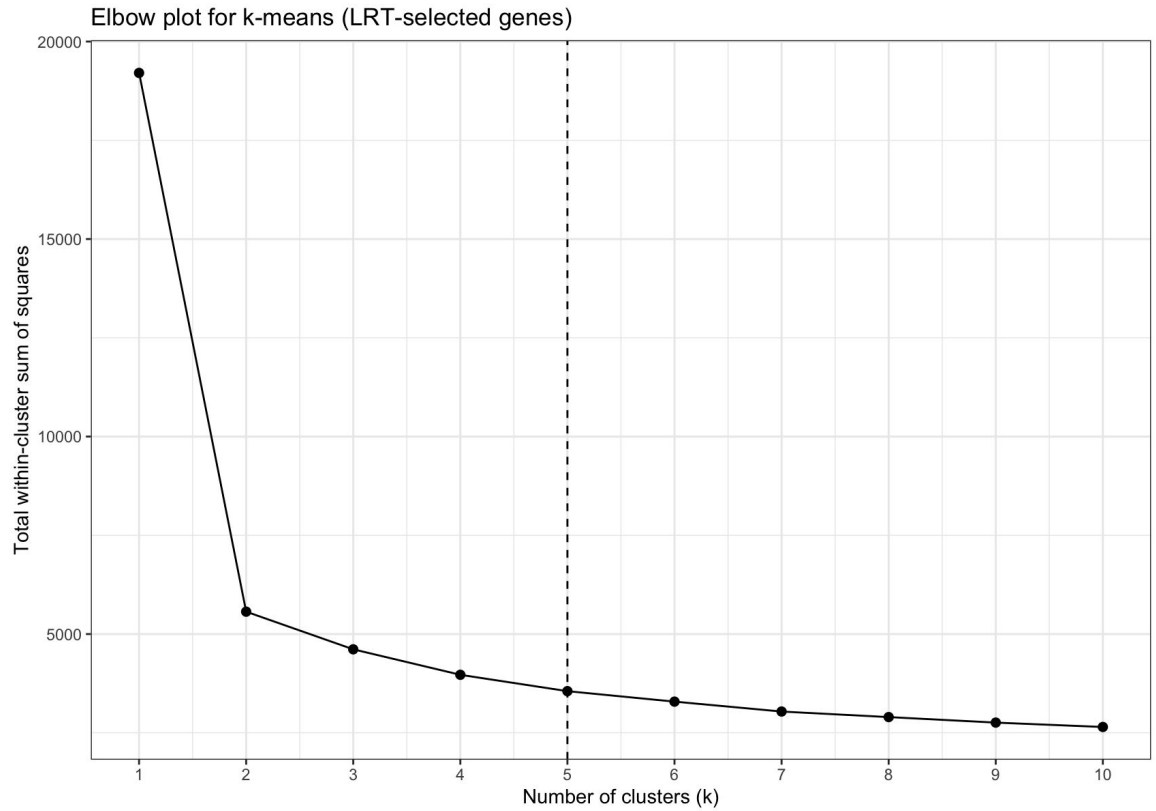




PCA variance



ETH zürich


Elbow plot for k-means (LRT-selected genes)
Total within-cluster sum of squares
20000
15000
10000
5000
1
2
3
4
5
6
7
8
9
10
Number of clusters (k)

ETH zürich

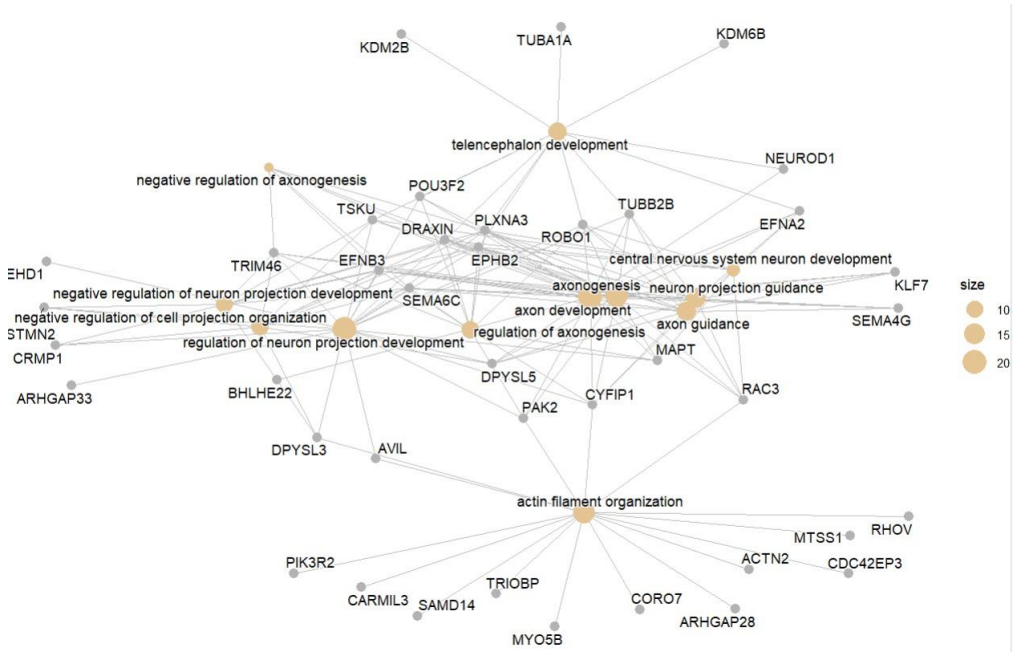
Zhu *et al., 2018*

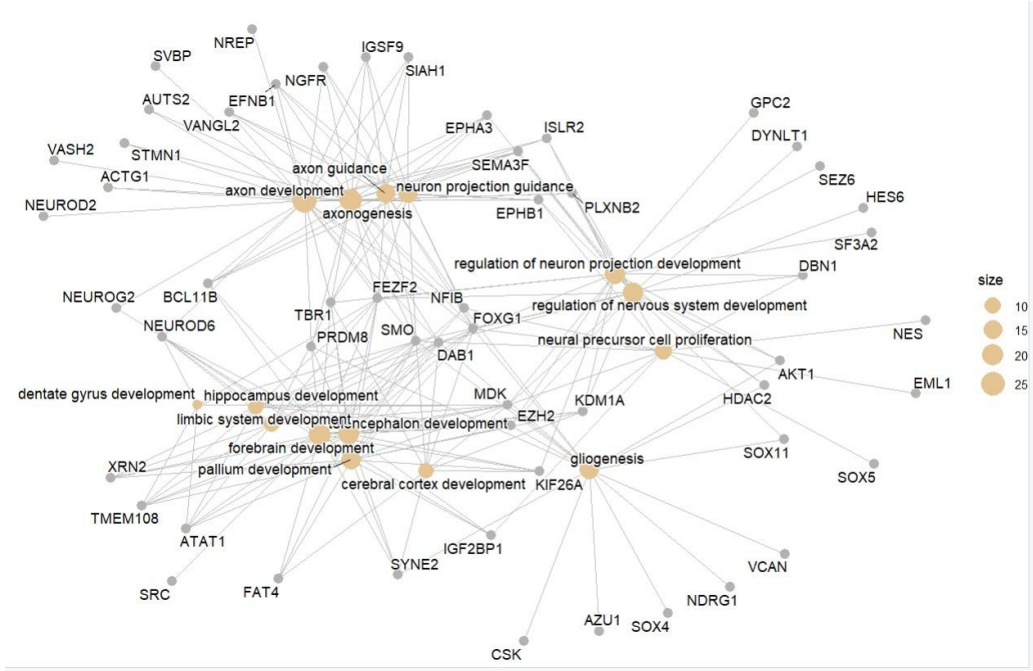
Developmental Stage	Trajectory
Neurogenesis	Very high early -> collapses prenatally
Neuronal Differentiation	High prenatally -> sharp drop at birth
Astrogliogenesis	Peaks late prenatal -> early postnatal
Synaptogenesis	Rises postnatally -> plateaus in childhood
Myelination	Slow long-term rise (infancy -> juvenile)

Our results

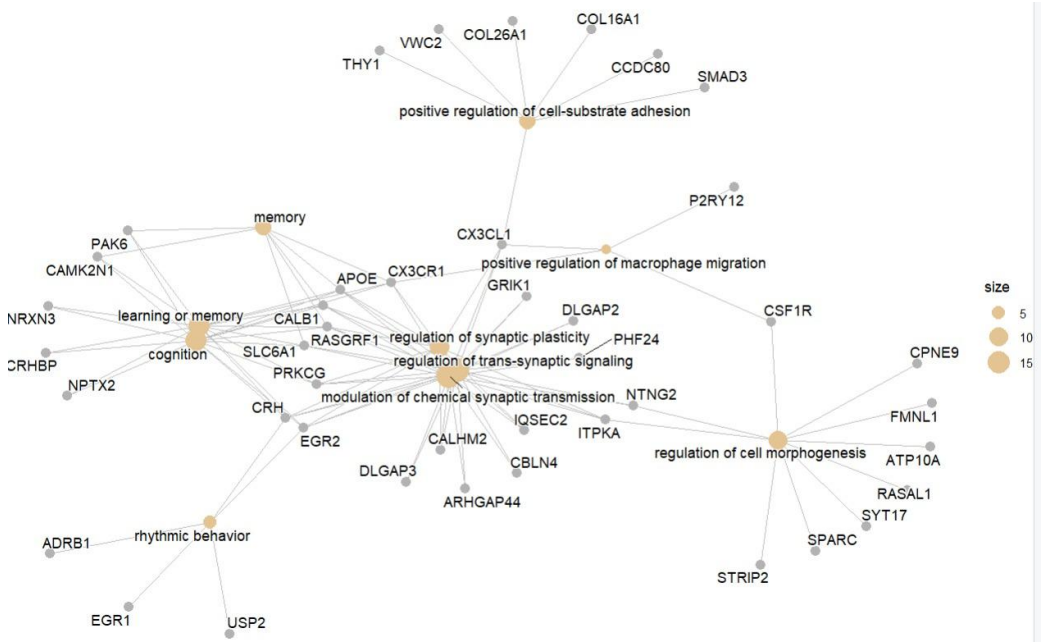
Cluster	Trajectory	GO Enrichment	BP
Cluster 2	High in prenatal -> declines	forebrain development, neuronal precursor proliferation, axonogenesis, gliogenesis	Neurogenesis
Cluster 3	Prenatal peak -> decreases across postnatal	axonogenesis,neuron projection development	Neuronal differentiation Astrogliogenesis
Cluster 1	Low prenatal -> steady rise -> plateaus	synaptic plasticity, learning, memory	Synaptogenesis
Cluster 4	Low prenatal -> increases into infancy -> strong in adulthood	neurotransmitter transport, membrane potential, ion homeostasis	
Cluster 5	Low prenatal -> rises across infancy -> high in adulthood	myelination, oligodendrocyte differentiation, axon ensheathment	Myelination

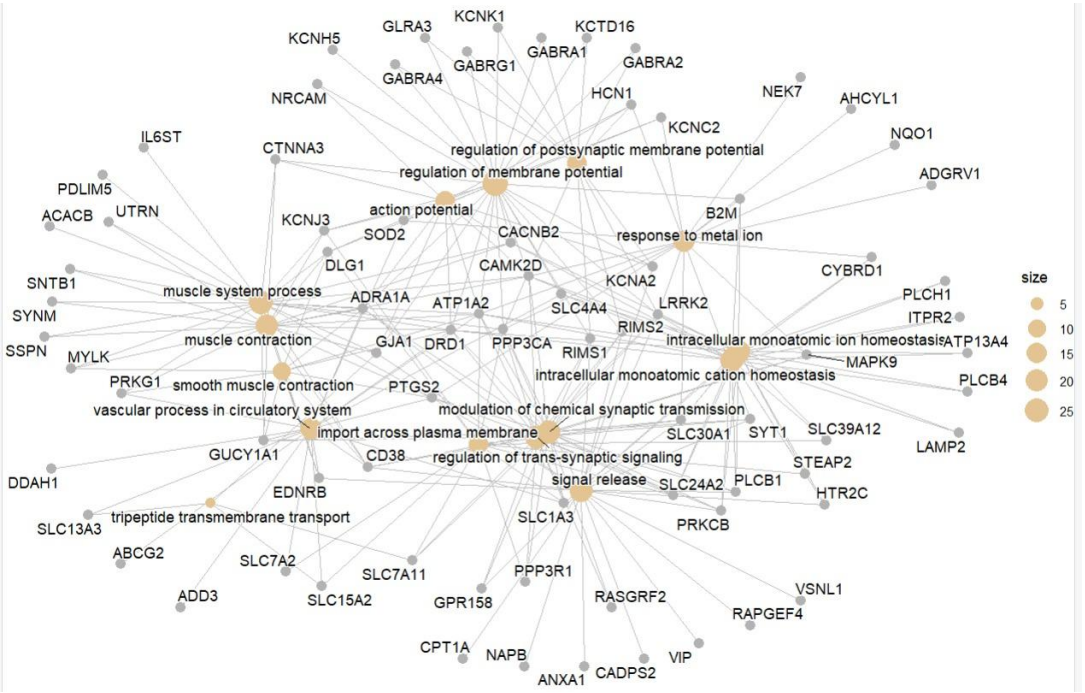
Clusters 2 and 3 (Neurogenesis)





Clusters 1 and 4 (Synaptogenesis)





Clusters 5 (Myelination)

